

**VARIABILITY STUDIES ON *Macrophomina
phaseolina* (TASSI) GOID. CAUSING DRY
ROOT ROT OF CHICKPEA (*Cicer arietinum* L.)
AND ITS MANAGEMENT**

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PALB 1075

**DEPARTMENT OF PLANT PATHOLOGY
UNIVERSITY OF AGRICULTURAL SCIENCES
BENGALURU-560065**

2014

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Thesis submitted to the

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IN

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Affectionately Dedicated

to

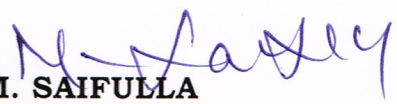
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and friends*

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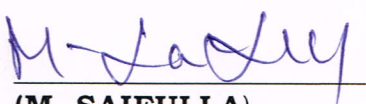
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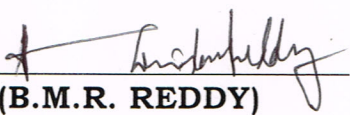
This is to certify that the thesis entitled "**Variability studies on *Macrophomina phaseolina* (Tassi) Goid. causing dry root rot of chickpea (*Cicer arietinum* L.) and its management**" submitted by **Mr. Manjunatha, H., ID. No. PALB 1075** in partial fulfillment of the requirements for the award of the degree of **DOCTOR OF PHILOSOPHY (Agriculture)** in **PLANT PATHOLOGY** to the University of Agricultural Sciences, Bengaluru, is a record of research work done by him during the period of his study in this University under my guidance and supervision and that no part of the thesis has been submitted for the award of any other degree, diploma, associateship, fellowship or any other similar titles.

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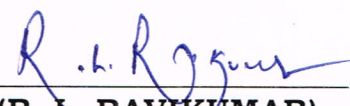

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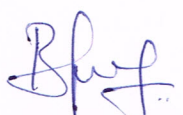
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With Regardful memories...

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19nd September, 2014

MANJUNATHA, H.

**Variability studies on *Macrophomina phaseolina* (Tassi) Goid.
causing dry root rot of chickpea (*Cicer arietinum* L.) and its
management**

Thesis abstract

Chickpea dry root rot samples infected by *Macrophomina phaseolina* were collected from ten major growing states viz., Andhra Pradesh, Himachal Pradesh, Karnataka, Madhya Pradesh, Maharashtra, Punjab, Tamil Nadu, Uttarakhand, Uttar Pradesh and West Bengal of India. *M. phaseolina* isolated from the samples were tested for pathogenicity on susceptible chickpea cultivars viz., H 09-23, L-550 and BG-212. Based on cultural and morphological characters like growth, colony appearance and color, mean diameter and number of microsclerotia production, shape and size of microsclerotia, the isolates were categorized as different groups. With respect to physiological requirements viz., the optimum temperature for the growth of *M. phaseolina* was 30 to 35°C, pH 5-6, besides alternate light and dark periods favored growth and microsclerotial production. Molecular identification with species specific primers (MpKFI and MpKRI) confirmed that all the isolates belonged to *M. phaseolina*. Variability studies through PCR-RAPD, revealed that all the isolates fall under two major clusters A and B. Phylogenetic tree constructed by using ITS sequence formed two clusters A and B, all 31 sequences of *M. phaseolina* were grouped in cluster A with two sub clusters. While, only one isolate KAMP-2 diverged to form single subcluster, remaining all isolates formed another sub cluster. Among the 215 entries screened in sick plot and blotter paper method, none were found resistant. However, only two entries viz., RSG 888 (2) and ICCV 93706 showed moderately resistant reaction. Of the fourteen treatments imposed to manage the dry root rot and root rot complex in chickpea during *Kharif* and *Rabi* 2013-14 seed treatment with tebuconazole (2g/kg) recorded least per cent disease incidence of 11.77 and 7.08 with a yield of 666.57 and 779.04 kg/ha respectively. Further, lowest root rot complex incidence of 26.61 per cent with a highest average yield of 722.81 kg/ha was recorded in both the seasons.

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**ಕಡಲೆ ಬೆಳೆಯ ಒಣ ಬೇರು ಕೊಳೆರೋಗಕಾರಕ ಮ್ಯಾಕ್ರೋಫೋಮಿನ ಫ್ಯಾಸಿಯೋಲಿನದ
ವೈವಿಧ್ಯತೆಯ ಅಧ್ಯಯನ ಮತ್ತು ಅದರ ನಿರ್ವಹಣೆ**

ಮಂಜುನಾಥ, ಹೆಚ್.

ಸಂಶೋಧನಾ ಸಾರಾಂಶ

ಮ್ಯಾಕ್ರೋಫೋಮಿನ ಫ್ಯಾಸಿಯೋಲಿನ ಸೊಂಕಿತ ಕಡಲೆ ಒಣ ಬೇರು ಕೊಳೆರೋಗ ಮಾದರಿಗಳನ್ನು ಭಾರತದ ಹತ್ತು ಪ್ರಮುಖ ಕಡಲೆ ಬೆಳೆಯುವ ರಾಜ್ಯಗಳಾದ ಆಂಧ್ರಪ್ರದೇಶ, ಹಿಮಾಚಲ ಪ್ರದೇಶ, ಕರ್ನಾಟಕ, ಮಧ್ಯಪ್ರದೇಶ, ಮಹಾರಾಷ್ಟ್ರ, ಪಂಜಾಬ್, ತಮಿಳುನಾಡು, ಉತ್ತರಾಖಂಡ, ಉತ್ತರಪ್ರದೇಶ ಮತ್ತು ಪಶ್ಚಿಮ ಬಂಗಾಳಗಳಿಂದ ಸಂಗ್ರಹಿಸಲಾಗಿತ್ತು. ಪ್ರತ್ಯೇಕಿತ ಮ್ಯಾಕ್ರೋಫೋಮಿನ ಫ್ಯಾಸಿಯೋಲಿನದ ರೋಗಕಾರಕ ಪರೀಕ್ಷೆಯನ್ನು ರೋಗಗ್ರಾಹಕ ಕಡಲೆ ತಳಿಗಳಾದ ಹೆಚ್ ೦೯-೨೩, ಎಲ್-೫೫೦ ಮತ್ತು ಬಿಜಿ-೨೧೨ ಗಳಲ್ಲಿ ನಡೆಸಲಾಯಿತು. ಸಾಂಸ್ಕೃತಿಕ ಮತ್ತು ಅಂಗರಚನಾಶಾಸ್ತ್ರೀಯ ಗುಣಗಳಾದ ಬೆಳವಣಿಗೆ, ಕಾಲೋನಿ ಕಾಣಿಸಿಕೊಳ್ಳುವ ರೀತಿ, ಬಣ್ಣ ಮೈಕ್ರೋಸ್ಕ್ಪೀರೋಷಿಯಾದ ಸರಾಸರಿ ವ್ಯಾಸ, ಸಂಖ್ಯೆ, ಆಕಾರ, ಗಾತ್ರ ಮತ್ತು ಉತ್ಪಾದನೆಗೆ ತೆಗೆದುಕೊಳ್ಳುವ ದಿನಗಳ ಆಧಾರದ ಮೇಲೆ ಪ್ರತ್ಯೇಕೀಕರಣಗಳನ್ನು ವಿವಿಧ ಗುಂಪುಗಳಾಗಿ ವಿಂಗಡಿಸಲಾಗಿತ್ತು. ಶಾರೀರಿಕ ಅವಶ್ಯಕತೆಗಳ ಪೈಕಿ ೩೦^o ರಿಂದ ೩೫^o ಸೆಲ್ಸಿಯಸ್ ತಾಪಮಾನ, ೫ ಮತ್ತು ೬ ರಸಸಾರ ಪ್ರಮಾಣ ಮತ್ತು ಪರ್ಯಾಯ ಬೆಳಕು ಹಾಗೂ ಕತ್ತಲೆಯು ಉತ್ತಮ ಬೆಳವಣಿಗೆಗೆ ಮತ್ತು ಮೈಕ್ರೋಸ್ಕ್ಪೀರೋಷಿಯಾದ ಉತ್ಪಾದನೆಗೆ ಸಹಕರಿಸಿವೆ. ಪ್ರತ್ಯೇಕೀಕರಣಗಳನ್ನು ಅಣುಶಾಸ್ತ್ರೀಯ ಆಧಾರದ ಮೇಲೆ ಗುರುತಿಸಲು ಪ್ರಭೇದ ನಿರ್ದಿಷ್ಟ ಪ್ರೈಮರ್‌ಗಳಾದ ಎಂ.ಪಿ.ಕೆ.ಎಫ್-೧ ಮತ್ತು ಎಂ.ಪಿ.ಕೆ.ಆರ್-೧ಗಳನ್ನು ಬಳಸಲಾಗಿ ಎಲ್ಲಾ ಪ್ರತ್ಯೇಕೀಕರಣಗಳು ಮ್ಯಾಕ್ರೋಫೋಮಿನ ಫ್ಯಾಸಿಯೋಲಿನ ಎಂದು ದೃಢೀಕರಿಸಲಾಯಿತು. ಪಿ.ಸಿ.ಆರ್-ಆರ್.ಎ.ಪಿ.ಡಿ. ಮುಖಾಂತರ ಅಣುಶಾಸ್ತ್ರೀಯ ವೈವಿಧ್ಯತೆಯ ಅಧ್ಯಯನ ಮಾಡಿದಾಗ ಎಲ್ಲಾ ಮ್ಯಾಕ್ರೋಫೋಮಿನ ಫ್ಯಾಸಿಯೋಲಿನ ಪ್ರತ್ಯೇಕೀಕರಣಗಳು 'ಎ' ಮತ್ತು 'ಬಿ' ಸಮೂಹಗಳಾಗಿ ವಿಂಗಡಣೆಯಾದವು. ಐ.ಟಿ.ಎಸ್. ಸರಣಿಯ ಆಧಾರದ ಮೇಲೆ ಜೀವವಿಕಾಸ ಚರಿತ್ರೆಯನ್ನು ವಿಶ್ಲೇಷಿಸಿದಾಗ ಎಲ್ಲಾ ಪ್ರತ್ಯೇಕೀಕರಣಗಳು 'ಎ' ಮತ್ತು 'ಬಿ' ಸಮೂಹಗಳಾಗಿ ಗುಂಪುಗೊಂಡವು ಮತ್ತು ಎಲ್ಲಾ ೩೧ ಮ್ಯಾಕ್ರೋಫೋಮಿನ ಫ್ಯಾಸಿಯೋಲಿನದ ಐ.ಟಿ.ಎಸ್. ಸರಣಿಗಳು 'ಎ' ಸಮೂಹದ ಒಂದು ಉಪಸಮೂಹಕ್ಕೆ ಸೇರಿದ್ದು, ಏಕೈಕ ಪ್ರತ್ಯೇಕೀಕರಣವಾದ ಕೆ.ಎ.ಎಂ.ಪಿ-೨ಯು ಮತ್ತೊಂದು ಉಪಸಮೂಹಕ್ಕೆ ಸೇರಿದೆ. ೨೧೫ ಕಡಲೆ ನಮೂದುಗಳನ್ನು ಕ್ಷೇತ್ರ ಷರತ್ತು ಅಡಿಯಲ್ಲಿ ಮತ್ತು ಬ್ಲಾಟರ್ ಕಾಗದ ಅಡಿಯಲ್ಲಿ ಒಣ ಬೇರು ಕೊಳೆರೋಗದ ತೀವ್ರತೆಗೆ ಎರಡು ವರ್ಷ ಒಳಪಡಿಸಿದಾಗ ಯಾವುದೇ ನಮೂದು ನಿರೋಧಕತೆಯನ್ನು ತೋರಿಸಲಿಲ್ಲ, ಆದರೆ ಕೇವಲ ಎರಡು ನಮೂದುಗಳಾದ ಆರ್.ಎಸ್.ಜಿ. ೮೮೮ (೨) ಮತ್ತು ಐ.ಸಿ.ಸಿ.ವಿ. ೯೩೭೦೬ಗಳು ಮಧ್ಯಮ ನಿರೋಧಕ ಪ್ರತಿಕ್ರಿಯೆಯನ್ನು ತೋರಿದವು. ಮುಂಗಾರು ಮತ್ತು ಹಿಂಗಾರು ೨೦೧೩-೧೪ ರಲ್ಲಿ ಅಳವಡಿಸಿದ ೧೪ ಒಣ ಬೇರು ಕೊಳೆರೋಗದ ನಿರ್ವಹಣಾ ಕ್ರಮಗಳಲ್ಲಿ, ಟೆಬುಕೊನಾಜೋಲ್ ಬೀಜೋಪಚಾರ (೨ ಗ್ರಾಂ/ಕೆ.ಜಿ. ಬೀಜ) ಕ್ರಮವು ಅತಿ ಕಡಿಮೆ, ಶೇಕಡಾ ೧೧.೭೭ ಮತ್ತು ೭.೦೮ ರೋಗದ ತೀವ್ರತೆಯನ್ನು ಹೊಂದಿದ್ದು, ಕ್ರಮವಾಗಿ ೬೬೬.೫೭ ಮತ್ತು ೭೭೯.೦೪ ಕಿ.ಗ್ರಾಂ/ಹೆ. ಇಳುವರಿ ನೀಡಿದೆ ಹಾಗೂ ಶೇಕಡಾ ೨೬.೬೧ ಅತಿಕಡಿಮೆ ಬೇರು ಕೊಳೆರೋಗ ಸಂಕೀರ್ಣದೊಂದಿಗೆ, ಎರಡೂ ಹಂಗಾಮಿನಲ್ಲಿ ಸರಾಸರಿ ೭೨೨.೮೧ ಕಿ.ಗ್ರಾಂ/ಹೆ.ಇಳುವರಿ ನೀಡಿದೆ.

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(ಎಂ. ಸೈಫುಲ್ಲಾ)
ಮುಖ್ಯ ಸಲಹೆಗಾರರು

CONTENTS

CHAPTER	TITLE	PAGE No.
I	INTRODUCTION	1-4
II	REVIEW OF LITERATURE	5-31
III	MATERIAL AND METHODS	32-49
IV	EXPERIMENTAL RESULTS	50-102
V	DISCUSSION	103-120
VI	SUMMARY	121-124
VII	REFERENCES	125-156

LIST OF TABLES

Table No.	Title	Page No.
1	OPA primers used in RAPD to study genetic variability among <i>M. phaseolina</i> isolates	41
2	Categorization of chickpea genotypes into different disease reaction categories	45
3	Collection of <i>M. phaseolina</i> isolates from different geographical regions	52
4	Dry root rot incidence caused by <i>M. phaseolina</i> isolates in chickpea cultivars	54
5	a. Growth characters of <i>M. phaseolina</i> isolates	55
	b. Grouping of <i>M. phaseolina</i> isolates based on colony diameter (mm)	57
	c. Grouping of <i>M. phaseolina</i> isolates based on colony color	57
	d. Grouping of <i>M. phaseolina</i> isolates based on colony appearance	57
6	a. Morphology of microsclerotial bodies produced by <i>M. phaseolina</i> isolates	59
	b. Grouping of <i>M. phaseolina</i> isolates based on mean microsclerotial diameter (µm)	60
	c. Size of microsclerotia produced by <i>M. phaseolina</i> isolates	62
7	a. Effect of temperature levels on growth of <i>M. phaseolina</i> isolates	64
	b. Effect of temperature levels on microsclerotial production by <i>M. phaseolina</i> isolates	65
8	Effect of hydrogen ion concentration (pH) levels on growth <i>M. phaseolina</i> isolates	67

Table No.	Title	Page No.
9	a. Effect of light on radial growth of <i>M. phaseolina</i> isolates	69
	b. Effect of light on microsclerotial production by <i>M. phaseolina</i> isolates	70
10	Scorable bands generated with 16 RAPD primers in 20 <i>M. phaseolina</i> isolates	72
11	Similarity coefficients of <i>M. phaseolina</i> isolates based on RAPD analysis	74
12	Nucleotide identities for ITS region sequences of <i>M. phaseolina</i> isolates with other selected sequences from GenBank	76
13	a. Screening of desi entries for dry root rot under field and blotter method (2012-13 <i>Rabi</i>)	78-80
	b. Disease reaction of desi entries for dry root rot under field conditions	81
	c. Disease reaction of desi entries for dry root rot under blotter paper method	83
14	a. Screening of Kabuli entries for dry root rot under field and blotter paper method (2012-13 <i>Rabi</i>)	84-85
	b. Disease reaction of Kabuli entries for dry root rot under field conditions	87
	c. Disease reaction of Kabuli entries for dry root rot under blotter method	88
15	a. Screening of local germplasm and advance breeding entries for dry root rot under field and blotter paper method (2012-13 <i>Rabi</i>)	90
	b. Disease reaction of local germplasm and advance breeding entries for dry root rot under field	91

Table No.	Title	Page No.
	c. Disease reaction of local germplasm and advance breeding entries for dry root rot under blotter paper method	93
16	a. Screening of promising entries for dry root rot under field 2013-14 (<i>Rabi</i>)	94
	b. Disease reaction of promising chickpea entries for dry root rot under field condition	96
17	Management of root rot complex in chickpea during <i>Kharif</i> 2013-14 at GKVK, Bangalore	97
18	Management of root rot complex in chickpea during <i>Rabi</i> 2013-14 at GKVK, Bangalore	98
19	Management of root rot complex in chickpea during 2013-14	100
20	Management of root rot complex in chickpea during <i>Rabi</i> 2013-14, Hassan	102

LIST OF FIGURES

Figure No.	Title	Between Pages
1	Colony diameter of <i>M. phaseolina</i> isolates on potato dextrose agar medium four days after inoculation	57-58
2	Number of microsclerotia produced and days taken for microsclerotial production by <i>M. phaseolina</i> isolates	62-63
3	Effect of temperature levels on growth of <i>M. phaseolina</i> isolates	65-66
4	Effect of temperature levels on microsclerotia production by <i>M. phaseolina</i> isolates	67-68
5	Effect of pH levels on growth of <i>M. phaseolina</i> isolates	67-68
6	Effect of light on radial growth of <i>M. phaseolina</i> isolates	70-71
7	Effect of light on microsclerotia production and days taken by <i>M. Phaseolina</i> isolates	70-71
8	Dendrogram constructed with UPGMA clustering method among 20 isolates of <i>M. phaseolina</i> based on polymorphism	74-75
9	Phylogeny tree showing relationships among the <i>M. phaseolina</i> isolates based on their ITS sequences.	76-77
10	Management of root rot complex in chickpea during the year 2013-14 (pooled data)	100-101

LIST OF PLATES

Plate No.	Title	Between Pages
1	Chickpea plant showing symptoms of dry root rot disease	50-51
2	<i>M. phaseolina</i> on PDA and its morphological characters	52-53
3	Isolates of <i>M. phaseolina</i>	52-53
4	Pathogenicity test of <i>M. phaseolina</i> isolates on chickpea cultivars	54-55
5	Colony color of <i>M. phaseolina</i> isolates	57-58
6	Colony growth pattern of <i>M. phaseolina</i> isolates	57-58
7	Morphology of microsclerotia produced by <i>M. phaseolina</i> isolates	60-61
8	PCR amplification of <i>M. phaseolina</i> with species specific primers MpKF1 and MpKR1	72-73
9	Banding pattern of Random amplified polymorphic DNA (RAPD) obtained from 20 <i>M. phaseolina</i> isolates	72-73
10	PCR amplification of <i>M. phaseolina</i> isolates with ITS 1 and ITS 4 primers	74-75
11	General view of dry root rot screening plot	81-82
12	Screening of chickpea genotypes by blotter paper technique	81-82
13	Dry root rot disease reaction under blotter paper technique (Scale 1-9)	81-82
14	General view of chickpea root rot complex management field at GKVK	102-103
15	General view of chickpea root rot complex management field at Hassan	102-103



Introduction

I. INTRODUCTION

Pulses are major sources of proteins among the vegetarians in India and complement the staple cereals in the diets with proteins, essential amino acids, vitamins (A, B and C) and minerals. They contain 22-24% protein, which is almost twice the protein in wheat and thrice that of rice. Pulses provide significant nutritional and health benefits, and are known to reduce several non-communicable diseases such as colon cancer and cardiovascular diseases (Jukanti *et al*, 2012). India, being a major pulse producing country, they are grown over an area of 10.84 m ha with a production of 6.43 million tonnes with a productivity of 593 kg /ha (Singh, 2013).

Chickpea (*Cicer arietinum* L.) is the oldest and third most important pulse crop after beans (*Phaseolus vulgaris* L.) and peas (*Pisum sativum* L.) and is believed to be originated in south-eastern Turkey (Ladizinsky, 1975). Cicer, which was classified under viciaeae alef., was later reported to belong to the monogeneric tribe, Cicereae. The genus Cicer, includes nine annuals and 34 perennial herbs (Van der Maesen, 1972). Chickpea seeds contain an average of 23% protein, 38-59% carbohydrate, 4.8-5.5% oil, 3% ash, 0.2% calcium. (Hulse, 1991). Every 100 g of whole raw seeds contains 357 calories, 4.5-15.69% moisture and 14.9-24.6 g protein (Duke, 1981). Boiled and roasted seeds contain similar amounts. Sprouting in chickpea is said to increase the proportionate amounts of ascorbic acid, niacin and available iron. The leaves also contain 4-8% protein and the crop meets up to 80% of the nitrogen requirements from symbiotic nitrogen fixation (Duke, 1981).

Chickpea has been known in this country for a long time. It is said to be one of the oldest pulses known and cultivated from ancient times both in Asia and Europe. The important gram growing countries are

India, Pakistan, Ethiopia, Burma and Turkey. India ranks first in the world in respect of production as well as acreage followed by Pakistan. The world's total production of chickpeas hovers around 8.5 million metric tonnes annually and is grown over 10 million hectares of land approximately. The desi type chickpea contribute to around 80% and the Kabuli type around 20% of the total production.

In India, chickpea ranks second in area and third in production and perhaps is the largest producer of chickpea accounting for about 75 per cent of world's production that is grown in an area of about 9.51 million ha producing 8.88 million tonnes with the productivity of 929 kg/ha (Singh, 2013). The major chickpea production area is situated in Andhra Pradesh, Karnataka, Madhya Pradesh, Rajasthan, Uttar Pradesh, Haryana, Maharashtra and Punjab together contributing 90 per cent of the area and 91 per cent of the production in the country (Singh, 2010).

Karnataka is one of the major chickpea producing states in the country where it is grown over an area of 0.8 m ha with a production of 0.38 million tonnes having an average productivity of 473 kg/ha (Singh, 2013).

Among various factors attributing to low productivity of chickpea, biotic and abiotic factors are very important. It was estimated that yield loss due to insects and diseases ranges from 5 to 10% in temperate and 50 to 100% in tropical regions (Van Emden *et al.*, 1988). The chickpea crop is attacked by 172 pathogens (67 fungi, 22 viruses, 3 bacteria, 80 nematodes and mycoplasma) from all over the world (Nene *et al.*, 1996). Among all, only a few of them have the potential to devastate the crops. Some of the serious diseases in order of their importance are wilt, dry root rot, collar rot, colletotrichum blight, alternaria blight, rust and ascochyta blight caused by *Fusarium oxysporum* f. sp. *ciceri*, *Macrophomina phaseolina*, *Sclerotium rolfsii*, *Colletotrichum dematium*,

Alternaria alternata, *Uromyces ciceris-arietini* and *Ascochyta rabiei* respectively (Nene *et al.*, 1984). Among them dry root rot caused by *M. phaseolina* (Tassi) Goid. is becoming a major constraint in the chickpea production and causing 10–20% annual loss in yield (Vishwadhar and Chaudhary, 2001).

M. phaseolina is primarily seed and soil-borne fungal pathogen. In chickpea, infected seeds and microsclerotia surviving in the soil are the major source of primary inoculum. The pathogen also has wide host range. Since 75% cultivation of chickpea in India is under rainfed, the crop faces severe moisture stress at flowering to podding stage which predisposes the crop to dry root rot development.

Prolonged saprophytic survival ability of the pathogen in soil makes chemical control and crop rotation ineffective. Cultivation of resistant varieties is an economical approach for the management of dry root rot of chickpea, but only a few sources with low level of genetic resistance are available.

Management can be made feasible and cost effective by identification of new resistant sources for which development of sick plot is essential which is the only alternative method so as to combat this serious disease, even though chemical control plays an important role in modern agriculture for disease management (Gupta *et al.*, 1997).

Against this background, the present study has been specifically planned to collect isolates from different parts of India and identify the causal organism, proving its pathogenicity and screening for resistant chickpea cultivars and management of dry root rot pathogen by using different seed treatment chemicals and bioagents.

The objectives of the study are as follows:

1. Collection of representative *Macrophomina phaseolina* isolates from different parts of India and their variability studies.
2. Molecular characterization of *M. phaseolina* isolates.
3. Evaluation of chickpea genotypes for identification of host plant resistance.
4. Management of dry root rot and root rot complex in chickpea.



Review of Literature

II. REVIEW OF LITERATURE

Development of resistance or tolerant chickpea varieties to *Macrophomina phaseolina* is challenging since the pathogen is having a wide host range. Prolonged saprophytic survival ability of the pathogen in soil makes chemical control and crop rotation ineffective. Cultivation of resistant varieties is an economical approach for the management of dry root rot of chickpea, but only a few sources with low level of genetic resistance are available. However, commercial cultivars of chickpea with desired level of genetic resistance to dry root rot have not yet been developed (Pande *et al.*, 2004). The information regarding morphological and molecular diversity among the populations of the pathogen and management of disease under field conditions using various components will pave path to manage the disease at best in the farmers field. Hence the related literature pertaining to the present investigation are reviewed and presented here under.

2.1 Current name of the causal organism: *Macrophomina phaseolina*

M. phaseolina (Tassi) Goid. is a soilborne plant pathogen belonging to the phylum Deuteromycetes and class Coelomycetes. It is highly variable, with isolates differing in microsclerotial size and presence or absence of pycnidia. The pycnidial stage is not common on soybean, but is on peanut. Pycnida are initially immersed in host tissue, then erumpent at maturity. They are 100-200 µm in diameter, dark to grayish, becoming black with age, globose or flattened globose, membranous to subcarbonaceous with an inconspicuous or definite truncate ostiole. The pycnida bear simple, rod-shaped conidiophores, 10-15 µm long. Conidia (14-33 x 6-12 µm) are single celled, hyaline, and elliptic or oval (Dhingra and Sinclair, 1978).

Microsclerotia are jet black in color and appear smooth and round to oblong or irregular, across isolates and on different substrates they vary on size and shape. The microsclerotia formed from aggregates of hyphal cells are joined by a melanin material with 50 to 200 individual cells (Dhingra and Sinclair, 1978).

Colonies in culture range in color from white to brown or gray and darken with age. Hyphal branches generally form at right angles to parent hyphae, but branching is also common at acute angles. Aerial mycelium with completely or partially appressed growth may or may not be produced in culture. Some isolates may form concentric growth rings (Dhingra and Sinclair, 1978).

2.2 The causal organism

Haware and Nene (1978) first reported *Rhizoctonia bataticola* as the causal organism of dry root rot of chickpea. Later, appearance of *Rhizoctonia* root rot in pulses was reported by Tripathi and Sharma (1983), Pandey and Singh (1990).

The earlier names of the *R. bataticola* includes *Botryodiplodia phaseoli* (Maubl.), *Dothiorella cajani* (Syd.), *Dothiorella phaseoli* (Maubl.), *Dothiorella philippinensis* (Petr.), *Macrophoma cajani* (Syd.), *Macrophoma corchori* (Sawada), *Macrophoma phaseoli* (Maubl.), *Macrophoma phaseolina* (Tassi), (Muriel *et al.*, 1977) Goidanich (1947) examined the original culture of *Macrophoma phaseolina* (Tassi) collected by Tassi and renamed it as *Macrophomina phaseolina* (Tassi) Goid.

Earlier, it was reported that *Macrophomina phaseolina* was the pycnidial stage of *R. bataticola* (Ashby, 1927; Haigh, 1928 and McRae, 1929). Gowdra Nagamma and Saifulla (2013) identified the causal agent of dry root rot of chickpea as *M. phaseolina* from CABI. Recently the application of separate names to different stages in the life cycle of

pleomorphic fungi is abandoned in favour of one name for each fungus especially species (www.iapt-taxon.org). So, now there is only one name for casual organism of dry root rot of chickpea is *M. phaseolina*.

2.3 Host Range of dry root rot pathogen and distribution of disease

The pathogen infects over 500 plant species and has a wide geographic distribution. Major cultivated hosts include: *Arachis hypogaea* (peanut), *Beta vulgaris* (beet), *Brassica oleracea* (cabbage), *Capsicum annuum* (pepper), *Cicer arietinum* (chickpea), *Citrus* spp., *Corchorus* sp., *Cucumis* sp., *Fargaria* sp., *Glycine max* (soybean), *Gossypium* sp., *Helianthus annuus* (sunflower), *Ipomoea batatas* (sweet potato), *Medicago sativa* (alfalfa), *Phaseolus* sp., *Pinus* sp., *Prunus* sp., *Sesamum indicum* (sesame), *Solanum tuberosum* (potato), *Sorghum bicolor* (sorghum), *Vigna unguiculata* (bean) and *Zea mays* (corn) (Reuveni *et al.*, 1983).

The dry root rot disease caused by *M. phaseolina* is one of the most destructive pathogens in arid regions and has wide host range (Waller, 1976). In India, it was reported from Andhra Pradesh, Bihar, Gujarat, Karnataka, Maharashtra, Madhya Pradesh, Rajasthan, Tamil Nadu, Uttar Pradesh and West Bengal (Uppal, 1931; Likhite and Kulkarni, 1934; Patel *et al.*, 1949; Venkataram, 1950; Thirumalachar, 1955; Chattopadhyay and Bhattacharya, 1968; Deshpande *et al.*, 1969 and Phillip *et al.*, 1969). Dry root rot due to *R. bataticola* has been reported in Austria, Ethiopia, India, and Turkey (Nene, 1978), on chilli (Mathur and Gurjar, 1999; Bunker and Mathur, 2001) and on urdbean (John *et al.*, 2008).

Raut and Ingle (1989) reported that the plant species of Leguminosae, Pedaliaceae and Compositae encourage the growth and survival of conidia, sclerotia and pycnidiospores of *R. bataticola*. Singh *et*

al. (1990b) found that all the commonly cultivated crops in the semi-arid tropics were found to be infected. Su *et al.* (2001) noted that *M. phaseolina* readily infects soybean, corn, sorghum and cotton.

2.4 Economic importance of dry root rot

Chickpea is susceptible to many diseases (Nene *et al.*, 1984) and among them dry root rot (DRR) is one of the major limitations in chickpea production causing 10 to 20 per cent annual loss (Dhar and Chaudhary, 2001). With the drastic reduction of area in northern India and its quantum increase in hot, drier rainfed central and southern India, the dry root rot is emerging as the most destructive constraint to chickpea cultivation. The disease is more prevalent between flowering and podding stages especially during hot (temperatures of 30 to 35°C) and dry (deficit soil moisture) environmental conditions (Sharma *et al.*, 2012).

Sharma *et al.* (1983) recorded appearance of *Rhizoctonia* root rot in pulses. Periodic surveys on wilt and root rot incidence of chickpea by Sharma, *et al.* (1983) in northern region of Madhya Pradesh for two consecutive years showed on incidence of wilt (15.90-37.99 per cent) and root rot (3.58-20.63 per cent) respectively in the districts surveyed. The highest wilt and root rot incidence was 27.79 and 14.17 percent respectively and the lowest was 20.81 and 8.76 per cent incidence.

Later, Pandey and Singh (1990) conducted a survey of chickpea diseases with reference to dry root rot in Allahabad region and reported that chickpea was less infected by *Rhizoctonia*, *Fusarium oxysporum* f. sp. *ciceri*, and 83 per cent of villages showed less than 25 percent disease intensity and remaining ones recorded 26-50 per cent disease intensity. Dollar (1996) conducted surveys in the main chickpea growing areas in Turkey and noted the most common root rot complex pathogens which

included 49 per cent of the isolates were includes *Rhizoctonia*, *Fusarium* and *Sclerotium*.

Singh and Agarwal (2002) conducted a survey in the Vindhya plateau zone of Madhya Pradesh (Bhopal, Raisen, Sagar, Seshore and Vidish) to evaluate the prevalence of dry root rot in chickpea and reported that the incidence of *R. bataticola* ranged from 8 to 20 per cent.

Prajapati *et al.* (2003) reported that the incidence of dry root rot of chickpea caused by *R. bataticola* was highest in bold-seeded cultivars. Singh and Sirohi (2003) evaluated the incidence of chickpea diseases in the Himachal Pradesh and reported that the incidence of dry root rot was highest in Una (4.86 %) and Sirmour (3.04 %) districts.

Veerendra Kumar (2004) noticed an incidence of 9.97 per cent dry root rot of chickpea in Gulbarga followed by Raichur (5.16 per cent) and Bidar (4.28 per cent). The highest incidence was 12.24 per cent in Chittapur of Gulbarga and least was in one of the locations of Raichur (3.21 per cent). Gurha and Trivedi (2008) conducted a survey in all the chickpea growing districts of Karnataka and reported that *R. bataticola* was found to be the predominant pathogen which infected 60.0 per cent plants in the fields of Gulbarga.

2.5 Symptomatology of dry root rot

The disease generally appears around flowering and podding time as scattered dried plants in field, where the leaves and stems of affected plants remain straw-colored. The roots dry and the tap root becomes brittle, shows shredding of bark and is broken easily. The disease appears suddenly when the ambient temperatures are between 25^o and 30^o C. The fungus lacks fruiting bodies and spores. The mycelium is light-brown and thick in which black sclerotia are formed (Haware, 1990).

Infected plants, at first do not show above ground symptoms. However, in advanced stages, leaves turn yellow and wilt, but remain attached to the plant. When the epidermis is removed sclerotia appear as grayish block on the tissue. In the absence of the plants, it survived over many seasons as small black sclerotia in diseased root and stem debris or in soil after decay of the plant material (Smith, 1969). Thus, the primary source of inoculum is sclerotia in the soil.

Ilyas and Sinclair (1974) observed intra-xylem sclerotia of *R. bataticola* that totally plugged the xylem vessels of soybean infected plants. Later, Singh and Mehrotra (1982a) in their histopathological studies reported that, roots inoculated by the pathogen showed only discoloration but no necrosis.

Prashanthi (1994) reported that, safflower plants infected by *R. bataticola* were characterized by gradual yellowing and drying of leaves followed by loss of vigour and premature death. The bark of such plants was easily peeled off along with extensive shredding of affected bark.

Singh *et al.* (1990a) observed that roots inoculated with *R. bataticola* (*M. phaseolina*) showed disintegration of cortical tissues while mycelium and sclerotial bodies plugging the xylem were seen in longitudinal sections. The sudden collapse of infected plants at the pod-filling stage was attributed due to this plugging of the xylem vessels.

2.6 Collection and isolation of *M. phaseolina* isolates

M. phaseolina was isolated from different plant parts *viz.*, diseased seeds and seedlings (Rajeevpant and Mukhopadhyay, 2001; Bagri *et al.*, 2004; Vinod Kumar *et al.*, 2007), roots (Sajeena *et al.*, 2004; Gowdra Nagamma and Saifulla, 2013), stem (Pan and Bhagat, 2008) and leaves (Sarkar and Pradhan, 1999).

Goidanich and Camici (1947) collected isolates of *M. phaseoli* from bean and Lavender in three widely separated parts of Italy. Similarly, Dhingra and Sinclair (1973b) collected nine isolates of *M. phaseoli* from various parts of United States. Shekhar *et al.* (2006a) obtained seven isolates of *M. phaseolina* incitant of charcoal rot of maize from different agro-ecological zones of India.

Aghakhani and Dubey (2009b) collected 23 isolates of *M. phaseolina* from chickpea plants showing characteristic dry root rot symptoms from different chickpea growing states of India and isolated the associated fungus was isolated from the roots of diseased plants on PDA medium. Manjunatha and Naik (2011) collected thirty isolates of *R. bataticola* causing dry root rot of chickpea collected from three major chickpea growing regions from Karnataka.

Sharma *et al.* (2012) collected fifty isolates of *R. bataticola* in different years (2004 to 2010) from different chickpea growing regions of India covering eight states including Uttar Pradesh, Rajasthan, Tamil Nadu, Andhra Pradesh, Madhya Pradesh, Delhi, Himachal Pradesh and Uttarakhand and the pathogen was isolated by plating surface sterilized (0.8% sodium hypochlorite for 2 min) pieces from dry root rot infected roots of the chickpea plants on potato dextrose agar (PDA) medium.

2.7 Maintenance of the pathogen

Potato dextrose agar (PDA) was found to be the best supporting medium for *R. bataticola* (Vinod Kumar *et al.*, 2007; Konde *et al.*, 2008). The pathogen *R. bataticola* was stored in an Agar Papa dextrose culture medium (ADP, Difco) in a bioclimatic chamber at 4°C (Valiente *et al.*, 2008).

2.8 Pathogenicity test

Prameela Devi and Singh (1998) reported 10 per cent inoculum density causing 71 per cent seedling mortality in greengram and blackgram. Kataria *et al.* (2007) assessed nine isolates of *R. bataticola*, the incitant of dry root rot of chickpea and revealed that all nine isolates were found pathogenic on C-235 chickpea cultivar in both sterilized and unsterilized soil. The root rot incidence varied from 26 to 100 per cent in sterilized soil and 20 to 80 per cent in an unsterilized soil.

Jaiman and Jain (2008) conducted pathogenicity test for *M. phaseolina* in cluster bean seeds and observed that seed inoculation with *M. phaseolina* resulted in pre-(33 %) and post emergence mortality and loss of seed germination (59 %) and vigour index (856) in comparison to control (93 %).

Aghakhani and Dubey (2009a) evaluated the pathogenicity and virulence of twenty-three isolates of *R. bataticola* against 10 chickpea cultivars namely ICC 12441, ICC 11332, ICC 11224, ICC 12450, Pusa 362, BGD 112, Pusa 1103, Pusa 212, Pusa 1088 and BCP 17 under sick pot conditions. Based on the dry root rot incidence the study revealed that the isolates were highly variable in their virulence. Cultivar Pusa 212 showed maximum dry root rot incidence (31.7-100 %) against the isolates tested followed by ICC 12441 and ICC 11332. whereas, root rot incidence was minimum on cultivar Pusa 1103. Among the isolates, Rb1 from Bangalore proved to be highly virulent and caused 16.6-100 percent (76.4 % mean) root rot in different cultivars. Faridkot (Rb5) and Badanapur (Rb11) isolates were comparatively less virulent.

2.9 Mass multiplication of *M. phaseolina*

The pathogen *R. bataticola* was mass multiplied on sand maize medium (Sajeena *et al.*, 2004; Kataria *et al.*, 2007). Sterilized sorghum

grains were used for mass culturing of *R. bataticola* (Rajeswari *et al.*, 1999; Durai, 2004).

2.10 Variability studies of *M.phaseolina*

Owing to the highly polykaryotic characteristics of the mycelium, *M. phaseolina* appears to be highly variable in nature. This may at least, in part, help to explain, the differences in pathogenicity and cultural characters as reported by several workers (Uppal *et al.*, 1936; Dhingra and Sinclair, 1973a; Ghosh and Sen, 1973; Anilkumar and Sastry, 1980; Gupta and Kolte, 1982; Hooda and Grover, 1982; Byadgi and Hegde, 1985; Arca and Yildizn, 1989; Than *et al.*, 1991; Ali and Dennis, 1992 and Shekhar *et al.*, 2006a).

2.10.1 Cultural variability in *M. phaseolina*

Dhingra and Sinclair (1973a, 1973b) noticed variation in growth rate among the isolates from the same plant and between isolates from different plants on different media. Byadagi and Hegde (1985) grouped isolates of bean, chickpea and cowpea as virulent, sorghum and soybean isolates as intermediate and *Glyricidia* isolate as mild, based on, morphology, cultural characters and pathogenic behaviour. Arca and Yildizn (1989) reported that, out of 362 isolates of Aegean region, 333 exhibited variation in cultural characteristics, such as colony color, mycelial growth, colony radius and the number of sclerotia. Groups of isolates were established based on radius and the number of sclerotia.

Ali and Dennis (1992) isolated *M. phaseolina* from field peas in South Australia and reported that 21 isolates can be differentiated into 15 pathotypes. Shekhar *et al.* (2006a) on the basis of colony color, divided seven isolates into four groups *viz.*, grayish white, blackish gray, dark black in centre periphery creamish and cottony white color.

2.10.2 Morphological variability in *M. phaseolina*

Uppal *et al.* (1936) reported variation in growth rate and colony character for two sorghum strains of *M. phaseolina* from two different localities *viz.*, Broach farm and Mohol farm in Maharashtra. Cultures of Broach strain showed concentric rings on PDA, while that of Mohol strain, produced aerial mycelium.

Vasudeva (1937) reported that *R. bataticola* grew equally well on Richard's agar, cotton root extract and synthetic agar but the sclerotia formation was favoured by the latter two media and was poor on Brown's agar. Thomas (1938) observed that the pycnidia of *M. phaseoli* were developed in cultures of several solid media.

Goidanich and Camici (1947) observed similar cultural and morphological characters among the strains of *M. phaseoli* isolated from bean and Lavender. Khare *et al.* (1970) reported variability in growth pattern and rate of growth among six isolates from root, stem, pod, leaf and seeds of urd bean and soil, when grown on different media. Further, they also observed that the soil isolate had the least amount of growth in almost all media while, the pod isolate showed maximum growth.

Dhingra and Sinclair (1973a) noticed variation in growth rate among the isolates from the same plant and between isolates from different soybean plants on different media. The stem isolates produced fluffy growth; root isolates showed partially fluffy growth and all other isolates produced appressed growth. Seed and petiole isolates produced concentric alternating zones of sclerotia which was more pronounced for the seed isolate.

They also collected nine isolates of *M. phaseoli* from different parts of United States and reported variation in growth rate and colony characters. Sobti and Sharma (1992) also observed no correlation

between radial growth and virulence or virulence and pycnidial formation in isolates of *R. bataticola* from groundnut in Rajasthan.

Jain *et al.* (1973) isolated *M. phaseolina* from urd bean plant parts *viz.*, from root, stem, leaf, pod and seeds and one isolate from the soil, where urd bean crop was taken and taken growth studies on different media. The isolates differed from each other in the growth pattern either in the diameter of colony or nature of aerial mycelium or the type of mycelial growth.

Waseer *et al.* (1990) isolated *M. phaseolina* from soybean and reported that the pathogen grew best on potato dextrose agar or potato dextrose broth at 35°C. However, least growth occurred on either media at 20°C or 40°C.

2.10.3 Morphology of microsclerotial bodies

Haigh (1928) classified 27 strains of *M. phaseolina* into three groups A, B and C, where C group included those strains having a mean sclerotial diameter of 120 µm or less, the B group included those with about 200 µm, while A comprised those, having > 200 µm of mean sclerotial diameter. On the basis of differences in size and number of sclerotia, Hildebrand *et al.* (1945) differentiated two strains of *M. phaseolina* isolated from soybean (the Ontario strain), having mean sclerotial diameter of 90.4 × 75.8 µm on host plant where as 99.9 × 89.4 µm in culture. From cotton (the Texas strain) whose, sclerotia diameter in culture measured 85.4 × 73.0 µm and were produced in greater number.

Goidanich and Camici (1947) recorded sclerotial diameter of 65 to 75 µm on host and 100 to 115 µm in cultures with bean isolate and 115 to 135 µm on culture with broad bean (*Vicia faba* L.) isolate. Bean, egg plant, potato, chilli, tomato, Bengal gram and pumpkin isolates produced

smooth and round sclerotia whereas, those of cotton, tobacco and sesame isolates were more or less irregular (Reichert and Hellinger, 1947). Similarly, studies with eight *M. phaseolina* isolates (B4, B6, B43, B45, B46, B50, B61 and B68) revealed that the sclerotia of B4 and B43 were smaller, less compact and of irregular shape as compared with the other strains (Anon., 1955).

Sclerotial diameter of 65.6 to 114 μm , 45.2 to 81.6 μm , 98.1 to 119.2 μm , 40.2 to 85.0 μm , 52.6 to 80.1 μm , 110 to 114 μm and 89.1 to 116.3 μm was recorded in respect to justica (S1), soybean (S2), Bengal gram (S3), jute (S7), kochia (S8), groundnut (S9) and red gram (S11) isolates, respectively by Ghosh and Sen (1973).

Dhingra and Sinclair (1973b) observed irregular shaped sclerotia in the root isolates and round to elliptical sclerotia in other isolates involved in their study. Raut and Bhombe (1976) reported that stem isolate of sorghum produced more sclerotial bodies than that of the leaf isolate on Elliott's medium at 30°C.

Adam and Stokes (1942) observed that mean diameter of sclerotia varied from 63- 165 μm on Dox's agar and 50-18 μm on PDA. Anil Kumar and Sastry (1980) reported that the ability of the isolates to produce sclerotia changed with growth media. Sunflower and horse gram isolates produced maximum number of sclerotia on Rose Bengal Agar (RBA) and Nutrient Agar (NA) respectively. The brinjal isolate produced least sclerotia on NA and no sclerotia on RBA, while, the sesamum and bean isolates did not develop sclerotia on NA. Further, they observed that the sclerotial initiation was quick in sunflower, on all the media tried, while the brinjal and cowpea isolates took maximum duration for sclerotial initiation (ten days).

Smits and Noguera (1988) reported that the sclerotial formation began with branching and intertwining of adjacent hyphal filaments, an increase in size of associated cells and compaction of the sclerotial mass. They also observed that the external cells of peripheral hyphae collapsed, but internally, sclerotia were uniformly reticulate.

Raut and Ingle (1989) reported differences in sclerotial size of *M. phaseolina* isolated from 15 crops including sorghum, soybean and safflower. Waseer *et al.* (1990) isolated *M. phaseolina* from soybean and reported that, sclerotial size on PDA differed from that on other media.

Dhar and Sarbhoy (1993) isolated *M. phaseolina* from the pod and seed of soybean causing ashy stem blight and reported the smallest known sclerotia in the world (mean $32 \times 24 \mu\text{m}$). Lokesha and Benagi (2004) recorded maximum sclerotial production on PDA (123.3/microscopic field of 10x). The ICRISAT isolate recorded maximum sclerotial production (71.2/microscopic field) followed by Bidar isolate (68.68/microscopic field).

Vishwadar and Sarabhay (1993) studied the variation in isolates of *R. bataticola* isolated from soybean plants collected from 11 locations at IARI. About 44 isolates were isolated from different parts of the plants and it was observed that the isolates differed with reference to colour of the colony, mycelial growth, shape and size of sclerotia. Morphological and cultural variations in isolates of *R. bataticola* isolated from different crops were also reported by other workers.

Shekhar *et al.* (2006a) observed that on the basis of sclerotial morphology, two groups of isolates could be formed, the one with oblong shape having irregular edges and the other being round with regular edges.

According to Shekhar *et al.* (2006a) Hyderabad isolate produced highest number of sclerotia (180.3 sclerotia/9 mm disc and 52.0/microscopic 10X field) and bigger size (95.7 μm) whereas, Coimbatore isolate produced minimum number of sclerotia (169) that too with smaller size (66.9 μm).

Species of *M. phaseolina* were classified based on differences in micro-sclerotial size, cultural characteristics, chlorate sensitivity, pycnidia formation and pathogenecity (Cloud and Rupe, 1988; Mihail, 1992; Karunanithi *et al.*, 1999; Suriachandraselvan and Seetharam, 2000).

2.10.4 Effect of temperature on growth of *M. phaseolina*

The growth of two isolates of *R. bataticola* from sorghum was maximum on Elliotts medium at 35°C (Raut and Bhombe., 1976). Later, Singh and Mehrotra (1980) also reported similar findings.

Uppal *et al.* (1936) studied the optimum temperature for growth of two strains of *M. phaseolina* and revealed that Broach strain grew best between 30 and 35°C while Mohol strain between 30 and 37°C. At higher temperature, the radial growth of the two strains fell sharply, they also noted very slow growth at lower temperature.

Khare *et al.* (1970) studied variation among the isolates of *R. bataticola* from urd bean plants found differences in growth pattern and sclerotial size at 5 and 25°C. All the six isolates differed from their growth pattern and rate of growth at different temperature levels.

Waseer *et al.* (1990) isolated *M. phaseolina* from soybean and reported that the pathogen grew best on PD agar and PD juice at 35°C while least growth occurred at 20 or 40°C.

Bainade *et al.* (2006) subjected various levels of temperature for culturing *M. phaseolina* causing leaf spot of mungbean, observed that 38°C temperature favoured the maximum growth and sclerotial production.

Chowdary and Govindaiah (2007) studied the growth and sclerotial production of *M. phaseolina* (incitant of root rot of mulberry) at different temperature *in vitro* and found that the growth was maximum (90 mm) between 30 and 35°C and minimum (40 mm) at 20°C. Similarly, number of sclerotia/9 mm disc was found maximum (162.3 sclerotia) at 40°C which gradually reduced to 150.3 at 35°C, followed by 120.0 sclerotia at 30°C and 8.66 sclerotia at 25°C. It failed to produce any sclerotia at 20°C.

The fungus, *R. bataticola* grew well on a wide range of temperature (25-36°C) and pH (5.5-7.5), with the optimum being 32°C and 6.5 pH respectively for mycelial growth and sclerotia formation (Mishra and Sinha, 1982).

The effect of different temperatures was evaluated for growth and sclerotia production of *Rhizoctonia*. The maximum growth of the fungus was 90 mm of radial growth at 30 and 35°C followed by 40°C (87.3 mm) (Veerendra Kumar, 2004).

Csondes *et al.* (2007) studied the effect of temperature on the growing patterns of 35 *M. phaseolina* isolates, collected from different districts of Hungary. The isolates were grown at 10, 15, 20, 25, 30, 35 and 40°C. All the isolates grew well at favorable temperature regime at 25 to 35°C with microsclerotial colonies reaching the maximum (90 mm) at the 5th day. Mycelia growth of the pathogen was very low at 10, 15 and 40°C, and did not form microsclerotia.

2.10.5 Effect of pH on growth of *M. phaseolina*

Uppal *et al.* (1936), studied variability in *M. phaseolina* revealed that both Mohol and Broach strain can tolerate wide range of pH while the optimum range lies between pH 3.4 and 6.4, indicating that acidic reaction favours the growth of the isolates. Sclerotia formation was numerous in acidic condition compared to alkaline condition.

Khare *et al.* (1970) observed variation among *R. bataticola* from urdbean plant parts *in vitro* found differences in growth pattern and sclerotial size at pH 6.5. All the six isolates varied in their growth pattern and growth rate at different pH levels.

Shanmugam and Govindswamy (1973) studied the physiologic aspects of *M. phaseolina* causing groundnut root rot and found that, the pathogen grew best on Richards' medium at optimum pH of 5.0.

Chowdary and Govindaiah (2007) observed that the growth of *M. phaseolina* (causal agent of root rot of mulberry) was maximum (90.0 mm) at pH 7.0, followed by 59.0 mm at pH 8.0. At different levels of pH (6.0, 6.5 and 7.5), growth of pathogen varied (62.3, 84.3 and 87.6 mm respectively). Similarly, maximum number of sclerotia (120)/9 mm mycelial disc was recorded at pH 7.0 and the same was minimum (16.3) at pH 6.0. The growth rate of pathogen and number of sclerotia varied at different levels of pH.

According to Ghosh and Sen (1973) the optimum pH for the growth of *M. phaseolina* isolates was 6.8. Bainade *et al.* (2006) observed that the pH levels of 7.0 (neutral) recorded 89.66 mm growth followed by pH 8.0 (61.33 mm growth) were favourable for *M. phaseolina* on mungbean.

2.10.6 Effect of light on growth of *M. phaseolina*

Muthukrishnan *et al.* (1995) noticed that continuous light or light alternated with dark favored the growth of *Rhizoctonia* from pulses *in vitro* on Czapeck's Dox liquid medium. Veerendra Kumar (2004) showed the effect of alternate light and darkness on good growth and sclerotia formation as compared to continuous light and continuous darkness.

Manjunatha and Naik (2011) recorded maximum (89.00 mm) colony diameter in Raichur isolate (RR1) when cultures were exposed to alternate cycles of light and darkness followed by Bidar isolate (88.00 mm) and Gulbarga isolate (84.50 mm). While, continuous darkness favoured less growth of all three isolates compared to alternate light and darkness. Alternate cycles of light and darkness supported the maximum mean (73.28) sclerotia production of *R. bataticola* followed by continuous light (71.89) and continuous darkness (68.64).

2.11 Molecular characterization of *M. phaseolina* isolates

With the advent of new molecular techniques, the use of molecular markers as a tool in comprehensive genetic analysis of plant pathogens has become important. Ever, the first introduction of RFLP markers in genetic mapping (Botstein *et al.*, 1980), molecular markers have opened a new era for pathogen genetics. Presently, an array of genetic markers are available to plant pathologists to study variability of pathogens.

Different molecular methods have been used for differentiating *M. phaseolina* populations including Restriction Fragment Length Polymorphism (RFLP) of rDNA-ITS regions (Aghakhani and Dubey, 2009b; Almeida *et al.*, 2003; Purkayastha *et al.*, 2006; Su *et al.*, 2001), Random Amplified Polymorphic DNA (RAPD) (Aboshosha *et al.*, 2007; Aghakhani and Dubey, 2009b; Almeida *et al.*, 2003; Almeida *et al.*, 2008; Babu *et al.*, 2010; Das *et al.*, 2006; Jana *et al.*, 2003; Omar *et al.*, 2007;

Purkayastha *et al.*, 2008, Rajkumar and Kuruvinaashetti, 2007; Su *et al.*, 2001; Zade *et al.*, 2009), Amplified Fragment Length Polymorphism (AFLP) (Brooker *et al.*, 2008; Mayek-Perez *et al.*, 2001; Reyes-Franco *et al.*, 2006; Vandemark *et al.*, 2000), Universal Rice Primer PCR (URP-PCR) (Jana *et al.*, 2005b), Inter simple sequence repeats (ISSR) (Jana *et al.*, 2005a; Purkayastha *et al.*, 2008), Repetitive Sequence-Based Polymerase Chain Reaction (Rep-PCR) (Purkayastha *et al.*, 2008) and SSR (Baird *et al.*, 2010). Inter simple sequence repeat (ISSR) markers are more powerful tools which can be utilized to access the variation in the flanking regions of microsatellite loci that are dispersed throughout all genomes (Zietkiewicz *et al.*, 1994).

2.11.1 Molecular identification of the fungus

The identification of *M. phaseolina* isolates is usually based on morphological criteria, but due to wide variations in the phenotype of the isolates these criteria are often not reliable. Although useful in phytopathology, these tests are subjective (Singh *et al.*, 1990b; Wrather *et al.*, 1998; Kendig *et al.*, 2000 and Mayek-Perez *et al.*, 1997).

Babu *et al.* (2007) designed two species-specific primers MpKFI and MpKRI from the specific nucleotide areas and one oligonucleotide probe MpKH1 (19 mer) from the conserved region, adjacent to ribosomal 5.8 S gene. The designed primers yielded single amplified product of 350 bp with all the *M. phaseolina* isolates tested. The specificity of the primers was tested on representative species of common soil borne microbes. The primer pair was found to be specific for *M. phaseolina* as none of the other soil microbes tested could yield any amplification product under identical conditions of amplification.

Mahdizadeh *et al.* (2011) confirmed that fifty two *M. phaseolina* isolates were recovered from 24 host plant species through the 14 Iranian provinces using species-specific primers MpKFI and MpKRI.

Asran-Amal (2012) employed the microsatellite primers (ATG)5, (TAGG)4 and a primer derived from the intergenic spacer regions (T3B) for identification of *M. phaseolina*. The three primers tested amplified species-specific microsatellites that unambiguously distinguished *M. phaseolina* from other fungal species tested.

2.11.2 Random Amplified Polymorphic DNA (RAPD)

RAPD involves the use of single short random oligonucleotide sequence (called random primers) defined cyclic amplification of DNA, which expose the polymorphism, distributed throughout the genome. The amplified fragments are called random amplified polymorphic DNA's (Williams *et al.*, 1990).

Su *et al.* (2001) investigated host specialization in *M. phaseolina*. The fungus was isolated from soybean, corn, sorghum and cotton root tissue and soil from fields cropped continuously to these species for 15 years in St. Joseph, L. A. These isolates were examined by RFLP and RAPD analysis. No variations were observed among isolates in restriction pattern of DNA fragments amplified by PCR, covering internal transcribed spacer (ITS) region 5.8 rRNA and part of 25S RNA but variations were observed when ten random (RAPD) primers were used to amplify the total DNA of isolates.

Tarakanta *et al.* (2003) showed that single RAPD primer A13 could be used to identify and discriminate several isolates of *M. phaseolina* and *Fusarium* sp. obtained from 20 hosts including soybean, cotton, chickpea and safflower.

Rajkumar (2004) observed that ten isolates of *M. phaseolina* representing Dharwad and Bijapur regions proved pathogenic to sorghum; molecular profiling using RAPD markers indicated the genetic differences among isolates and species specific finger print to *M.*

phaseolina was identified. Isolates were grouped using molecular and virulence data.

Shekhar *et al.* (2006b) analysed seven isolates of *M. phaseolina* incitant charcoal rot of maize through RAPD marker for genetic diversity. The UPGMA cluster analysis for 706 loci score permitted identification of three main clusters. Similarity matrix and Jaccard' similarity co-efficient between the isolates indicated that the maximum genetic variation was among isolates of Arabhavi and Coimbatore with 70.8 per cent followed by Ludhiana and Coimbatore with 69.5 per cent. The most closely related isolates were Hyderabad and Delhi with an affinity percentage of 65.5 followed by Udaipur and Bangalore isolates with 62.9 per cent similarity.

2.11.3 Internal Transcribed Spacer (ITS)

Ribosomal DNA (rDNA) in eukaryotes typically present in several hundred tandemly repeated multiple copies. They are one of the most extensively used molecular markers. The region is a part of the rDNA cistron which consists of 18S, ITS1, 5.8S, ITS2, and 28S sequences. Both the ITS1 and ITS2 are non coding regions that lie between the 18S and 5.8S (ITS1) and between the 5.8S and 28S (ITS2) coding regions. The entire rDNA unit is transcribed by RNA polymerase I. as a single 45S precursor molecule (Atkins and Clark, 2004; Brunner *et al.*, 2007).

Multiple copies make it easy to amplify rDNA even from small amounts of samples. Because of each of the two internal transcribed spacer (ITS1 and ITS2) are flanked by rDNA highly conserved coding regions, primers can be designed for the amplification of the first internal transcribed spacer (ITS1) and the second transcribed spacer (ITS2).

Aghakhani and Dubey (2009b) determined genetic diversity among six isolates of *R. bataticola* representing five different ITS-RFLP groups with respect of their nucleotide sequences of ITS1, 5.8S and ITSII regions.

The ITS of Karnataka isolate (Rb1) was largest (670 bp), whereas it was smallest (499 bp) in Haryana (Rb21) isolate. After sequencing the ITS sequences of the isolates were deposited to Gen-Bank nucleotide database. The phylogeny tree constructed from the nucleotide sequence, similarity of these six isolates along with 27 other ITS sequences of fungi which showed 96–100% similarity during blast analysis clustered them into three major groups. The isolates Rb16 (Jharkhand), Rb21 (Haryana) and Rb26 (Madhya Pradesh) clustered in one group and showed 96–100% ITS nucleotide sequences similarity with the nucleotide sequences of *M. phaseolina* (*R. bataticola*) available in the GenBank.

2.12 Evaluation of chickpea genotypes for identification of host plant resistance

Nene *et al.* (1981) developed pot culture techniques for greenhouse screening and a paper towel technique for laboratory screening.

Singh and Mehrotra (1982b) reported BG-203, G-543 and Hare Chhole cultivars as resistant to *R. bataticola* when grown in infested soil.

Nene *et al.* (1989) evaluated 225 wilt and root rot resistant *Cicer arietinum* genotypes for broad-based resistance to root rots caused by *R. bataticola*, *R. solani* and *F. solani* at 24 locations in 11 countries during 1978-82. Genotypes ICC 2862, ICC 9023, ICC 9032, ICC 10803, ICC 11550 and ICC 11551 were found resistant to wilt and dry root rot.

Reddy *et al.* (1990) evaluated chickpea lines to dry root rot at ICRI SAT found 100 per cent mortality of JG 62 in all the four seasons. However, the lines Avrodhi, ICC48, and GCP6, had a mortality of <20 per cent.

Baker and Ahmed (1991) tested the resistance of 90 chickpea genotypes in field infested with wilt and dry root rot pathogens. They found that ICC 12263 to be most resistant.

Hunde *et al.* (1992) evaluated chickpea lines to dry root rot at ICRISAT center and plant genetic resource centre, Ethiopia and found 100 per cent mortality in JG 62. However ICC 14400, ICC 12241 and ICC 12445 had <20% mortality in two years of testing while seven lines *viz.*, ICCL 84204, ICCL 85221, ICCL 88001, ICCL 84215, ICC 12884, ICC 12205 and ICC 12408 showed <20% mortality in the second year.

Chaturvedi and Dua (2009) have reported 25 resistant cultivars including KPG-59, Radhey and K-50 against dry root rot.

Jayanti and Bhatt (1993) tested pre-germinated seeds of 21 *Cicer arietinum* varieties in *R. bataticola* contaminated soil and observed seed rot within 24 h in NEC 874 and BG 234 varieties whereas, Bold 2375, BG 209, JG 62, JG 315, ICC 3357 and JG 1133 varieties developed necrotic lesions up to 3 to 5 cells deep on the hypocotyl region within 7 days. Whereas, RSG 44, AGC 677, NEC 41, GL2 69, JG12 19, JG 74, ICC 8983 and ICC 5003 developed only superficial necrosis along the hypocotyl region. Whereas, BGM 416, FBG 416, ICC 1376 and ICC 11314 were found resistant.

Gupta (1995) screened a set of 300 diverse chickpea lines for resistance to *R. bataticola* during 1990-92 in a multiple disease sick plot. Seven lines *viz.*, ICCV 89402, ICCV 90201, ICCV 90254, BG 209, H 88-2, GF 88426 and RSG 180 showed resistant reaction while susceptible line JG 62 recorded 80-100 per cent disease.

Oad *et al.* (1995) found two varieties *viz.*, Sanyasi and C 612, as resistant to *R. bataticola* and *R. solani* with better seed quality. Mayek-

Perez *et al.* (2002) demonstrated that BAT 477 expressed resistance to *M. phaseolina* under both greenhouse and field conditions.

Gangwar *et al.* (2002) screened thirty-five chickpea cultivars for resistance to dry root rot caused by *R. bataticola* in a field experiment conducted in Uttar Pradesh, India during 1998-2001. Disease intensity was recorded 30 days after sowing up to maturity at 15-day intervals. All cultivars exhibited disease symptoms. Ten cultivars *viz.*, IICC 2644, IICC 10384, IICC 10630, IICC 112244, IICC 11332, ICCL 81002, IICC 81010, ICC 12263, IICC 12441 and ICCV were found resistant, whereas, five cultivars *viz.*, GCP 9504, Phule G 96020, Phule G 96105, Phule G 96313 and GL 91059 were found moderately resistant to dry root rot.

Out of 47 lines screened against dry root rot of chickpea under blotter paper method, three genotypes showed resistant, 22 moderately resistant, 19 susceptible and three highly susceptible (Pande *et al.*, 2004).

Om Gupta and Babbar (2006) screened 233 desi and Kabuli chickpea genotypes along with 38 promising lines with susceptible check JG 62 against collar rot, dry root rot and vascular wilt in multiple disease sick plot at Jabalpur, Madhya Pradesh, India. Genotypes H 99-264, PG 9425-5, PG 9425-9 (desi) and HK 00297, PG 97-313 (Kabuli) exhibited resistance against all the 3 pathogens. WCG 97-29, WCG 97-37, IPC 97-72 and H 00-256 (desi) showed less than 10% mortality for dry root rot. GF 88240, MPJG 98-9032 and MPJG 98-11551 (desi) and HK 94-134 (Kabuli) showed resistance to wilt and dry root rot diseases.

Jayalakshmi *et al.* (2008) conducted pot culture experiment for screening of chickpea promising entries against dry root rot. Out of 12 genotypes tested against dry root rot disease, four found resistant, two moderately susceptible, three susceptible and three highly susceptible.

Al Shareef *et al.* (2009) observed that out of the forty chickpea genotypes screened for resistance to wilt and root-rot diseases in the sick-plot. Thirty five genotypes were found resistant while, the other five genotypes moderately susceptible.

Nagamma and Saifulla (2012) screened sixty four Kabuli genotypes against dry root rot, out of which six genotypes *viz.*, Phule G 04305, IPCK 07-62, RVSSG 12, HK 08-212, Phule G 09305 and AKG 2002-1K showed resistant reaction, nine genotypes *viz.*, HK 94-134, GNG 1888, HK 06-152, GNG 2112, HK 06-171, Phule G 09316, CSJK 74, JGK 13 and CSJK 70 showed moderately resistant reaction.

Saifulla *et al.* (2011a) screened 196 entries against dry root rot and collar rot diseases in sick plot of which 21 entries *viz.*, GNG 1861, Phule G 07112, BGD 1056(R), RSG 931, Phule G 07101, Vijay, RSG 888 ,GNG 0825, PG 064, GNG 1947, JG 1307, GBC6 (AVT-1), GNG 2002, GNG 1936, CSJ 313, JG 18, HIR 60 and JG 11, ICCV 07107 and ICCV 08323 resistant and 36 entries moderately resistant to dry root rot disease.

Saifulla *et al.* (2011b) screened 16 chickpea genotypes for collar rot, dry root rot, wilt and rust diseases at GKVK, Bengaluru during the year 2010-11. For dry root rot HIR 60, BGD 256, A 1, JG 11, BBG 1, BBG 2, JG 62, and L 550 were found resistant to moderately resistant, whereas HIR 60, BGD 256, A1, BBG 1 and BBG 2 showed multiple disease resistance to collar rot, dry root rot, wilt and rust diseases.

Om Gupta *et al.* (2012) screened 170 chickpea genotypes for dry root rot at JNKVV, Jabalpur during the year 2007-10. 68 genotypes exhibited resistant reaction (<10% mortality), out of which 26 are the promising lines namely JG 1-14, 2-125, 2-4-110, 14-11, 14-10, 2001-13, 2001-13, 2001-18, 2001-80, 2001-115, 2002-20, 2003-95, 2003-14-16, 2004-110, 210, 9605, 1-9, 99-115, 2001-04, 2003-14-2, JG

2000-07, JSC 37, MPJG 89-11551, MPJG 89-9023, CSJ 592 and Rajas. These lines were further evaluated for their performance in sick field for three consecutive years and six lines viz., JG 2000-07, JSC 37, MPJG 89-11551, MPJG 89-9023, CSJ 592 and Rajas were found to be resistant exhibiting <10 per cent mortality, however 14 lines showed moderately resistance reaction.

2.13 Management of dry root rot and root rot complex in chickpea

Vyas (1987) confirmed that the soybean dry root rot could be reduced by combination of thiram + carbendazim (2:1) followed by seed treatment with fungicide tolerant strain of *Trichoderma* sp. or *Bacillus subtilis* in *Macrophomina* sick soils. Simultaneous application of *T. viride* or *T. harzianum* with carbendazim treatment was effective in reducing dry root rot of soybean (Vyas, 1994).

Sankar and Jeyarajan (1995) conducted a field experiment to manage root rot of sesame caused by *M. phaseolina* by seed treatment with antagonists *T. harzianum* and *T. viride* that significantly reduced the root rot incidence of 10.1 and 12.8 per cent respectively, compared to 60 per cent in control that were superior to carbendazim seed treatment.

Rajeswari (1997) reported that dry root rot of mungbean caused by *M. phaseolina* was effectively controlled by integration of *T. harzianum* as soil application (5g/kg) + seed treatment (10^8 cfu/ml) with sub-lethal doses of carbendazim (0.02%) significantly reduced the dry root rot incidence (95.3 %) over soil treatment (91.5%) and seed treatment alone.

Patel and Anahosur (2001) enumerated rhizosphere microflora of chickpea under irrigated conditions and found that carbendazim as seed treatment and soil drench coupled with soil application of *T. harzianum* reduced maximum fungal population of *Fusarium* sp. and *M. phaseolina*. FYM also reduced pathogen population and stimulated all other

microflora more than other treatment there by showing antagonism against *Fusarium* sp. and *M. phaseolina*.

Venkateswar Rao (1998) reported that an integration of soil solarization (six weeks), *T. viride* (native) and mancozeb (0.3%) effectively controlled the stem and root rot of sesame to the tune of 82.2 per cent caused by *M. phaseolina* under field conditions.

Jayasree *et al.* (2000) investigated that the *Pseudomonas fluorescens* strain pf1 effectively inhibited the mycelia growth of *M. phaseolina*, the pathogen causing dry root rot in blackgram and sesame. Among various methods of pf1 application, seed treatment + soil application recorded maximum yield in blackgram (1238 kg/ha) and sesame (1200 kg/ha) followed by seed treatment + soil drenching of carbendazim, which recorded grain yield to the extent of 1205 kg/ha in blackgram and 820 kg/ha in sesame.

Gaikwad *et al.* (2002) studied the management of charcoal rot of sorghum by using three biocontrol agents *viz.*, *T. viride*, *T. harzianum* and *P. fluorescens* and the fungicides thiram and carbendazim. They have found that seed treatment with biocontrol agents and fungicides were highly effective against *M. phaseolina*.

Sindhan *et al.* (2002) reported satisfactory disease control when *P. fluorescens* was used as seed treatment along with carbendazim against *M. phaseolina* in chickpea.

Khan and Gangopadhyay (2008) evaluated the combined seed treatment of *P. fluorescens* and fungicides against the dry root rot of chickpea and revealed that *P. fluorescens* isolate PFBC-26 was effective against the pathogen when used in combination with carbendazim.

Many of the workers reported that soil application of *Trichoderma* sp. and *P. fluorescens* effectively controlled the diseases caused by soil borne plant pathogens and also a synergistic effect in the growth of plants was observed (Suriachandraselvan, 1997; Manoranjitham *et al.*, 2000; Bharati *et al.*, 2004 and Poddar *et al.*, 2004) in several crops.

Konde *et al.* (2008) studied the efficacy of fungicides against *R. bataticola*, *in-vitro* study revealed that carbendazim + thiram (0.1 + 0.2 %), penconazole (0.1%) and thiophanate- M (0.1 %) completely inhibited (100 %) the growth of pathogen.

Nagamani *et al.* (2011) reported that seed treatment with carbendazim @ 2 g/kg of seed+ *T. viride* @ 4 g/kg of seed +soil application of FYM fortified with *T. viride* recorded least dry root rot incidence (4.33 %) compare to the control (35.03 %). It also recorded highest growth parameters like shoot length (50.2 cm), root length (7.20 cm) and also highest yield (893 kg/ha). Where as in control least shoot length (26.5 cm.), root length (3.5 cm) and yield (660 kg/ha) was recorded.

Kanwal *et al.* (2012) evaluated the potential of tebuconazole at different concentrations viz., 335, 70, 105 and 140 ppm against three problematic soil-borne fungal phytopathogens in chickpea which are responsible for root rot complex in chickpea namely *M. phaseolina*, *F. oxysporum* f. sp. *ciceri* and *S. rolfsii*. At all the concentrations of tebuconazole completely arrested the growth of all the three fungal species.



Material and Methods

III. MATERIAL AND METHODS

The laboratory and field experiments were carried out during 2011-2014, at All India Co-ordinated Research Project on Chickpea, University of Agricultural Sciences, Gandhi Krishi Vignan Kendra (GKVK) Bengaluru located in Eastern dry zone (Zone-5), geographically the place is located at 12° 58' latitude north and 77° 35' longitude east. The details of the Materials used and Methodology adopted during the course of investigation are described in this chapter.

3.1 General laboratory procedure

3.1.1 Glassware cleaning

For all the laboratory experimental studies, Corning and Borosil glasswares were used. The glasswares were kept in cleaning solution for 24 h, containing 60 g of potassium dichromate ($K_2Cr_2O_7$), 60 ml of concentrated sulphuric acid (H_2SO_4) in 1000 ml of water. They were washed with vim powder followed by washing in running tap water and then rinsed in distilled water before use.

3.1.2 Sterilization

All glassware, solid and liquid media were subjected to sterilization by autoclaving at 1.1 kg/ cm² (121.6°C) for 20 min. For isolation of the fungus seeds and plant tissues were surface sterilized in 1: 1000 sodium hypochlorite solution followed by three changes in sterile water. All cultural studies were conducted in aseptic conditions under laminar flow. The tip of inoculation needle and forceps were sterilized over flame.

3.2. Collection of representative *Macrophomina phaseolina* isolates from different parts of India

3.2.1 Collection of diseased specimens

The chickpea plants showing typical symptom of dry root rot were collected from different states *viz.*, Karnataka, Andhra Pradesh, Tamil Nadu, Maharashtra, Madhya Pradesh, Punjab, Uttar Pradesh, Uttarakhand, West Bengal and Himachal Pradesh during *Rabi* 2012-13. Totally twenty infected chickpea samples were brought to the laboratory and washed thoroughly in running water. The plants were dried in shade for 3-4 days and preserved for further studies.

Isolation of the fungus was done by following standard tissue isolation method under aseptic condition. The infected tissues of the root were cut into small bits of size 1-2 mm and surface sterilized in 1:1000 sodium hypochlorite solution for a minute and then washed thrice in sterile distilled water to remove the traces of sodium hypochlorite before transferring them to sterile potato dextrose agar [PDA] slants under aseptic conditions and incubated at $27 \pm 1^{\circ}$ C.

3.2.2 Hyphal tip culture

Hyphal tip isolation was done on two per cent water agar plates. Dilute mycelial suspensions (10^{-6} ml⁻¹) of the fungus were prepared in sterile distilled water. One ml of such (10^{-6} ml⁻¹) suspension was spread uniformly on water agar plates, incubated for 24 hrs and observed for mycelial growth initiation under microscope. Such part was marked with marker and that mycelial bit was transferred on PDA plates under aseptic conditions. The PDA plates were incubated at a temperature of $27 \pm 1^{\circ}$ C and observed for fungal growth. No sectoring was observed in any of the plates after isolating them by hyphal tip culture technique.

This method was followed for maintaining pure culture since the fungus is known to be highly heterozygous.

3.2.3 Maintenance of culture

The fungus was subcultured on PDA containing Petriplates and allowed to grow at $27\pm 1^{\circ}\text{C}$ for four days. Such plates are stored in refrigerator at 4°C and maintained by sub culturing at 20 to 25 days interval. This culture was used throughout the study.

3.2.4 Identification of the fungus

The morphology, cultural characters, formation of sclerotia, and branching of mycelium were the principal characters considered for identification of pure culture isolates of *Macrophomina phaseolina*. The pure culture of the pathogen was maintained on PDA slants.

3.2.5 Mass multiplication of *M. phaseolina*

Giant culture was prepared by soaking chickpea seeds in tap water for 12 h and strained. About 500 g of seeds were taken in 1000 ml conical flasks and were sterilized at 15 lb pressure, at temperature of 121.6°C for 20 min. These flasks were inoculated with 7-day old culture of *M. phaseolina* isolates and incubated at a temperature of $28 \pm 1^{\circ}\text{C}$ for 30 days. The flasks were shaken at alternate days to get uniform growth of the culture. The giant culture so obtained was used for proving pathogenicity of these isolates.

3.2.6. Pathogenicity test

The pathogenicity of 20 *M. phaseolina* isolates was determined by modified sick pot method on susceptible chickpea cultivars H 09-23, L-550 and BG-212 (Nene *et al.* 1981). Surface sterilized aluminum trays (45 cm x 30 cm) with 0.1% formalin were filled with soil sterilized by using 0.1% formalin for a week under air tight polythene cover and

inoculated 5 days prior to sowing with 8 day old 500 g of inoculum multiplied on chickpea seed. Ten seeds of chickpea varieties were sown in each pot in two replications and observed for symptoms development after 15-30 days after sowing. Dry root rot incidence was recorded after 60 days of sowing at the time of maturity of the crop on the basis of the number of diseased plants out of the total number of plants. Re-isolations were made from the roots of diseased plants on PDA medium, and isolated fungus was confirmed to its identity.

3.2.7 Variability studies of *M. phaseolina* isolates

The isolates collected from different parts of Karnataka, Andhra Pradesh, Tamil Nadu, Maharashtra, Madhya Pradesh, Punjab, Uttar Pradesh, Bihar, Uttarakhand, West Bengal and Himachal Pradesh were used to study cultural, morphological and physiological variation among isolates on potato dextrose agar potato dextrose broth.

3.2.7.1 Cultural variability

The cultural characters viz., colony diameter, colony color and colony appearance were recorded.

Colony diameter was recorded by measuring the radial growth of the mycelium in mm after four days of incubation at $28 \pm 1^{\circ}$ C. Based on the mean colony diameter of three replications, isolates were categorized into three groups. Group A (80.00-90.00 mm), Group B (70.00-80.00 mm) and Group C (60.00-70.00 mm).

Based on colony color the isolates categorized into four groups viz., Group I-Grayish white, Group II- Blackish gray, Group III-Cottony white and Group IV-Deep black.

Based on colony appearance twenty isolates were categorized into three groups viz., fluffy, partially appressed and appressed.

3.2.7.2 Morphological variability

Twenty *M. phaseolina* isolates collected from different locations of India were studied for their morphological variability on PDA at a temperature of $28 \pm 1^{\circ}\text{C}$ under light and darkness. Morphological characters like number of microsclerotia per microscopic field, diameter, shape and days taken for microsclerotia production were recorded.

The number of microsclerotia present per microscopic field was recorded by using a 0.1 ml of suspension containing microsclerotia obtained by crushing five mm disc of *M. phaseolina* in 1 ml of sterile distilled water by taking mean number of three microscopic fields using binocular stereomicroscope at 10x magnification.

The diameter of microsclerotia was measured by using motic image plus 2.0 software under binocular stereomicroscope at 10x magnification. Based on mean diameter of three microsclerotial bodies, the isolates were categorized into four groups viz., group I (30 -50 μm), group II (50- 100 μm), group III (100-150 μm) and group IV (>150 μm) and shape of microsclerotia observed under binocular stereomicroscope based on that the isolates were categorized into two groups like round and oblong.

3.2.7.3 Physiological studies

3.2.7.3.1 Effect of different temperature levels on the growth of *M. phaseolina* isolates

Growth of each isolate was tested at 15, 20, 25, 30, 35 and 40°C . About 25 ml of PDA was poured into a sterilized Petriplates, the five mm discs of seven day old culture of all the isolates of *M. phaseolina* were inoculated to a Petriplate containing solidified medium. Each isolate was replicated thrice and incubated at selected temperatures for four days in BOD and the mycelial growth was measured in terms of colony diameter

in mm and number of sclerotial bodies produced were recorded per microscopic field by using a 0.1 ml of suspension containing microsclerotia obtained by crushing five mm disc of *M. phaseolina* in 1 ml of sterile distilled water by taking mean of three microscopic fields using binocular stereomicroscope at 10x magnification.

3.2.7.3.2 Effect of different pH levels on the growth of *M. phaseolina* isolates

Isolates of *M. phaseolina* were grown on the potato dextrose broth in selected pH levels of 4.0, 5.0, 6.0, 7.0, 8.0 and 9.0. The pH levels were adjusted by adding 1N alkali (NaOH) or acid (HCl). A seven day old, five mm mycelial disc of the isolates was inoculated separately into conical flasks containing 100 ml medium at different pH levels. Three replications were maintained. From the flasks after incubation at $27 \pm 1^{\circ}\text{C}$ for 7 days, the mycelial mat was harvested, washed and dried in hot air oven at 40°C for 48 h and the growth of the isolates recorded in terms of dry weight of mycelia and analyzed statistically.

3.2.7.3.3 Effect of light on growth of *M. phaseolina* isolates

Effect of light on growth and microsclerotia production by *M. phaseolina* was studied by exposing the pure culture on PDA to different light periods viz., continuous light, continuous dark and alternate cycle of 12 h complete light and 12 h complete darkness. Observations on radial growth (mm) and microsclerotia production/ microscopic field was recorded after four days of inoculation by using a 0.1 ml of suspension containing microsclerotia obtained by crushing five mm disc of *M. phaseolina* in 1 ml of sterile distilled water by a mean of three microscopic fields using binocular stereomicroscope at 10x magnification.

3.3 Molecular characterization of *M. phaseolina* isolates

3.3.1 Isolation of DNA from *M. phaseolina* isolates

The genomic DNA was extracted from *M. phaseolina* isolates by following CTAB extraction method (Doyle and Doyle, 1987).

1. Each fungal culture was inoculated into potato dextrose broth (PDB) and grown at 28° C for 7 days with sufficient shaking and aeration.
2. Approximately two grams of mycelium was harvested, frozen in liquid nitrogen and crushed in liquid nitrogen in a pre-cooled pestle and mortar to make fine powder.
3. The ground powder was transferred to a 2 ml sterile polypropylene tube containing 1 ml of extraction buffer (2% w/v CTAB, 1.4 M NaCl, 0.1 M TrisHCl, pH 8.0, 0.03% βMercapto ethanol, 0.1% PVP) pre-warmed to 65°C.
4. The samples were incubated at 65°C for 60 minutes with intermittent mixing of tubes at every 15 minutes.
5. After bringing back tubes to room temperature, equal volume (1 ml) of chloroform: Isoamyl alcohol (CIA) (24:1 v/v) was added to each tube containing sample and extraction buffer and mixed by inverting (no vortex).
6. Tubes were then centrifuged at 8500 rpm for 30 minutes.
7. Upper aqueous phase was carefully transferred to new sterile tube (CIA step was repeated once again if the upper layer did not look clear). Then equal volume and pre-chilled isopropanol was added and mixed gently.
8. Tubes were kept for 1-2 h at -20°C/overnight and centrifuged at 8500 rpm for 15 minutes to pellet the DNA.

9. The supernatant was decanted and the pellet was washed with 1 ml of 70 per cent ethanol and then air dried completely for 1-2 hour (there should no trace of alcohol).
10. The DNA pellet was suspended in 200-400 μ l of T₁₀E₁ (10 Mm Tris-HCl; 1 mM EDTA maintained at pH8) in sterile centrifuge tube. When DNA was fully dissolved, 1 μ l of RNase (stock 10 mg/ml) per 100 μ l volume of sample was added (Final RNase concentration 100 μ g/ml) and incubated at 37°C for one hour or at room temperature for overnight.
11. Finally the samples were stored at -20°C.

3.3.2 DNA quality and quantity estimation

To assess the quality of DNA, samples were run on 1 per cent agarose gel in 1x TAE buffer (For 250 ml 50x TAE – 60.5 gm of Trisbase, 14.25 ml glacial acetic acid and 25 ml of 0.5 M EDTA) and stained with 5 μ l ethidium bromide (10 mg/ml stock) per 100 ml of gel and checked for shearing of DNA and contamination of RNA.

3.3.3 PCR (Polymerase Chain Reaction)

Requirements:

1. Template: Purified genomic DNA extracts (20-30 ng/ μ l) of isolates were used as template DNA.
2. Buffers: 10x assay buffer from M/s Bangalore Black biotech Pvt. Ltd. Bangalore, India was used.
3. dNTP: Individual dNTPs, *viz.*, dATP, dGTP, dCTP and dTTP were obtained from M/s Bangalore Black biotech Pvt. Ltd., Bangalore.
4. Thermal cycler: BIO-RAD thermal cycler was used for cyclic amplification of DNA.

5. Taq: Taq DNA polymerase (3 U/ μ l) was obtained from M/s Bangalore Genei Pvt. Ltd., Bangalore
6. Primers: Primers were synthesized from Sigma Pvt. Ltd., Bangalore, India.

3.3.4 Molecular identification of *M. phaseolina* isolates

Molecular identification of isolates was done by using species specific primers MpKFI (5'-CCGCCAGAGGACTATCAAAC-3') and MpKRI (5'-CGTCCGAAGCGAGGTGTATT-3') in a PCR. Reactions were carried out in a final volume of 15 μ l containing 1.2 μ l of template DNA (30 ng / μ l), 0.3 μ l of Taq polymerase (3 U/ μ l), 1.5 μ l Taq buffer containing MgCl₂ (10x), 0.5 μ l dNTP's (2.5mM) each, 0.6 μ l (10 pmol) each of forward and reverse primers and 10.3 μ l of molecular biology water.

Amplification conditions (PCR protocol):

Initial denaturation	: 95°C for 5 min followed by 40 cycles of
Denaturation	: 94°C for 45 sec
Annealing	: 54°C for 1 min
Extension	: 72°C for 2 min
Final extension	: 72°C for 8 min
Final hold	: 4°C for ∞

3.3.5 Random Amplified Polymorphic DNA marker analyses

PCR procedure was carried out with 16 RAPD primers (Table 1.) with a total reaction volume of 15 μ l containing 1.2 μ l of template DNA (30 ng / μ l), 0.3 μ l of Taq polymerase (3 U/ μ l), 1.5 μ l Taq buffer containing MgCl₂(10x), 0.5 μ l dNTP's (2.5 mM) each, 2 μ l of primer (10 pmol) and 9.5 μ l of molecular biology water.

Table 1. OPA primers used in RAPD to study genetic variability among *M. phaseolina* isolates

Sl. No.	Primer No.	Primer Name	Sequence of primer (5' to 3')
1	PMBR1	OPA-01	CAGGCCCTTC
2	PMBR2	OPA -02	TGCCGAGCTG
3	PMBR3	OPA -03	AGTCAGCCAC
4	PMBR4	OPA -04	AATCGGGCTG
5	PMBR5	OPA -05	AGGGGTCTTG
6	PMBR7	OPA -07	GAAACGGGTG
7	PMBR8	OPA -08	GTGACGTAGG
8	PMBR9	OPA -09	GGGTAACGCC
9	PMBR10	OPA -10	GTGATCGCAG
10	PMBR12	OPA -12	TCGGCGATAG
11	PMBR13	OPA -13	CAGCACCCAC
12	PMBR14	OPA -14	TCTGTGCTGG
13	PMBR15	OPA -15	TTCCGAACCC
14	PMBR16	OPA -16	AGCCAGCGAA
15	PMBR17	OPA -17	GACCGCTTGT
16	PMBR20	OPA -20	GTTGCGATCC

Amplification conditions (PCR protocol):

Initial denaturation	: 95°C for 2 min followed by 40 cycles of
Denaturation	: 94°C for 1 min
Annealing	: 36°C for 1 min
Extension	: 72°C for 2 min
Final extension	: 72°C for 8 min
Final hold	: 4°C for ∞

3.3.5.1 Scoring for RAPD primers

Distinct band or fragments (for RAPD markers) were given identification numbers according to their position on the gel and were scored visually on the basis of their presence (1) or absence (0), separately for each isolate and primer. The scores obtained using all the primers, in the RAPD analysis were then pooled to create a single data matrix to estimate polymorphic loci. The binary data was analyzed using standard procedure in NTSYS-PC (version 2.1; Exeter Biological Software, Setauket, NY) software package (Rhoft, 2000). Clustering analysis was performed with the Unweighted Pair-Group Method using Arithmetic averages (UPGMA) in the SAHN module of NTSYS-PC.

3.3.6 The ITS region amplification

The ITS regions and the 5.8S rDNA of 20 *M. phaseolina* isolates representing various RAPD groups were amplified with a set of primers ITS1 (5' TCCGTAGGTGAACCTGCGG 3') and ITS4 (5' TCCTCCGCTTA TTGATATGC 3') as described by White *et al.* (1990). The PCR-amplification reaction was optimized with various concentrations of MgCl₂, dNTPs, primer and template DNA. The amplification was carried out in a BIO-RAD thermal cycler in a 30 µl PCR reaction consisting of 3µl PCR buffer with MgCl₂ (10X), 0.6µl of (3U/µl) Taq polymerase (Bangalore Genei, India), 1 µl (2.5mM) dNTPs each, 1µl (10 pmol) of each

primer and 2.4 µl (30-50 ng) of genomic DNA. Initial denaturation of 95°C for 5 min was followed by 40 cycles of 94°C for 45 sec, 58°C for 1 min (primer annealing), 72°C for 2 min (primer extension). A final extension of 72°C for 8 min was incorporated into the program, followed by cooling to 4°C until recovery of the samples. PCR product (5 µl) was mixed with loading buffer (2 µl) containing 0.25% bromophenol blue, 40 % w/v sucrose in water and then loaded in 1.5% Agarose gel with 0.1 % ethidium bromide for examination with horizontal electrophoresis. Nucleotide base pairs of the amplicons were determined on the basis of its migration and conformation relative to the molecular size marker 100 kb ladder (Bangalore Genei, India).

3.3.7 The ITS sequencing

The ITS amplified PCR product produced bands (~550–600 bp) of 20 isolates on agarose gel were sent to sequencing to Genotypic Technology Pvt. Ltd. Bangalore.

M. phaseolina rDNA gene sequences obtained from NCBI Genebank of various host plants viz., soybean, maize, sweet potato, sorghum, sugarcane, jatropha, mungbean, banana and golden sapphire and sequence of *Rhizoctonia solani* and *Athelia rolfsii* was obtained from sedum and crotalaria respectively were used for comparison with the rDNA gene sequence of *M. phaseolina* isolate of chickpea. Sequences were aligned following the Clustal W algorithm included in the Megalign module (DNASTAR Inc.) Multiple alignment parameters used were gap penalty = 10 and gap length penalty = 10. Both of these values are aimed to prevent lengthy or excessive numbers of gaps. The default parameters were used for the pair wise alignment. The use of Clustal W determines that, once a gap is inserted, it can only be removed by editing. Therefore, final alignment adjustments were made manually in order to remove artificial gaps. Phylogenetic analyses were completed using the MEGA

package (version 6.01; Institute of Molecular Evolutionary Genetics, University Park, PA) (Tamura *et al.*, 2007). Neither gaps (due to insertion-deletion events) nor equivocal sites were considered phylogenetically informative. Hence, complete deletion prevented the use of any of these sites in further analyses. Phylogenetic inference was performed by the neighbour joining method. Bootstrap tests with 1,000 replications were conducted to examine the reliability of the interior branches and the validity of the trees obtained (Felsenstein, 1985). An additional standard error test was performed with the data set using the same characters in order to evaluate the statistical confidence of the inferred phylogeny.

The evolutionary distances were computed using the Maximum Composite Likelihood method and are in the units of the number of base substitutions per site. All positions containing gaps and missing data were eliminated from the dataset (Complete deletion option). Phylogenetic analyses were conducted in MEGA 6.

3.4 Evaluation of chickpea genotypes for identification of host plant resistance

The chickpea genotypes were evaluated by two methods *viz.*, field and blotter paper method for the identification of resistant sources for dry root rot.

3.4.1 Screening under field

A total of 215 chickpea genotypes including one hundred and thirteen desi and sixty one Kabuli types supplied by IIPR Kanpur, forty one entries of local germplasms and advance breeding material (which include CWRRN entries) from UAS, GKVK and ICRISAT, Patancheru, Hyderabad, were screened at field in two replications for host plant resistance at Zonal Agricultural Research Station, G.K.V.K., Bengaluru,

during *Rabi* season of 2012-2013. The promising entries includes two resistant and twenty one moderately resistant entries found during the year 2012-13 were again screened during *Rabi* season of 2013-2014. Each genotype was sown in five meter row length. After every five test entries one line of susceptible check L-550 was sown. At the time of sowing, giant culture of *M. phaseolina* of Bengaluru isolate (KAMP-1) was incorporated to the soil to improve sickness of the soil. Observations on per cent dry root rot incidence were recorded from pod formation to maturity stage.

The following formula was used to calculate dry root rot disease incidence

$$\text{Per cent disease incidence} = \frac{\text{Number of plants infected}}{\text{Total number of plants}} \times 100$$

After estimating disease incidence, the entries were categorized into different disease reactions as per AICRP scale mentioned below.

Table 2. Categorization of chickpea genotypes into different disease reaction categories

Sl. No.	Disease Reaction category	Dry root rot incidence (%)
1.	Resistant	0 – 10
2.	Moderately resistant	11 – 20
3.	Moderately susceptible	21 – 30
4.	Susceptible	31 – 50
5.	Highly susceptible	51 – 100

3.4.2 Blotter paper technique

Two hundred and fifteen genotypes of desi and Kabuli types supplied by IIPR Kanpur, ten genotypes of local germplasm and advanced breeding material entries from UAS, GKVK and thirty one chickpea entries supplied by ICRISAT Patancheru, Hyderabad were also screened under blotter paper method for dry root rot (Nene *et al.*, 1981).

The *M. phaseolina* cultured in sterilized 250 ml of PDB was poured into 500 ml conical flasks. The flasks were then inoculated with mycelial discs and incubated for seven days. The mycelial mat from the flasks was removed and macerated in a warring blender along with 100 ml of sterilized distilled water for a minute. The inoculum was later collected in a beaker. In the mean time, the chickpea seedlings were raised in plastic pots containing sterilized sand mixed soil. One week old seedlings were uprooted and the roots were immersed in sterile water in order to remove the adhered soil particles. The seedlings were then immersed completely in the inoculum in a beaker for a minute. The seedlings particularly the root portion were then placed side by side on a blotter paper (45 cm x 25 cm with one fold); in such a way that only the cotyledons and roots are covered and the green tops of seedlings remained outside and then blotter paper was folded. The folded blotter papers were then placed in trays, and kept in an incubator at 35^o C for eight days provided with 12-h artificial light, 80 per cent RH and the blotters were moistened with sterile water every day. At the end of the incubation period, the seedlings were examined for the extent of root damage. One folder blotter paper had seedlings of one test line and along with susceptible check. Folded blotter were kept one above the others, in heaps of ten in a tray. One of these ten blotters had the seedlings of susceptible check. At the end of the incubation period (8 days), the seedlings were examined for the extent of root damage, and scored for the disease. Dry root rot severity was scored on a 1-9 rating scale on the 8th day after inoculation (Nene *et*

al., 1981). Ten seedlings of each entry was considered as one replication, Based on the disease score the genotypes were grouped into different disease reaction categories as mentioned below.

Categorization of chickpea genotypes into different disease ratings on 1-9 scale and categorized as below.

Rating	Disease reaction categories	Symptoms
1	Resistant	No infection on roots
3	Moderately resistant	Very few small lesions on roots
5	Moderately susceptible	Lesions on roots clear but small, new roots free from infection
7	Susceptible	Lesions on roots many, new roots generally free from lesions
9	Highly susceptible	Roots infected and completely discolored

3.5 Management of dry root rot and root rot complex in chickpea

Due to non-availability of sick plot only for dry root rot at field level an integrated approach for the management of dry root rot and root rot complex in chickpea was taken during *Kharif* 2013-14 and *Rabi* 2013-14 at ZARS, GKVK, UAS, Bengaluru and only during *Rabi*, 2013-14 at College of Agriculture, Hassan, which consisted of wilt, collar rot and dry root rot caused by *F. oxysporum* f.sp. *ciceri*, *Sclerotium rolfsii* and *M. phaseolina* respectively. The observations on these three diseases was recorded as per disease incidence individually by uprooting the plants which contributed to root rot complex and yield parameters were recorded from each treatment.

3.5.1 Experimental details

Details	During the Year 2013-14	
Location	ZARS, GKVK	College of Agriculture, Hassan
Season	<i>Kharif and Rabi</i>	<i>Rabi</i>
Variety	JG-11	JG-11
Design	RCBD	RCBD
Plot size	3 X 2 sq.m	3 X 2 sq.m
Spacing	30 x 10 cm	30 x 10 cm
Treatments	14	14
Replications	3	3

3.5.2 TREATMENT DETAILS

- T1- Seed treatment with *Trichoderma viride* @ 5 g / kg of seeds
- T2- Seed treatment with *T. viride* @ 10 g / kg of seeds
- T3- Seed treatment with *Pseudomonas florescence* @ 5 g/ kg of seed
- T4- Seed treatment with *P. florescence* @ 10 g/ kg of seed
- T5- Soil application of *T. viride* @ 5 kg in 50 kg of FYM
- T6- Soil application of *T. viride* @10 kg in 50 kg of FYM
- T7- Seed treatment with carbendazim @ 2 g + thiram @ 1 g/kg of seed
- T8- Seed treatment with carbendazim @ 2 g + captan @ 1 g/kg of seeds
- T9- Seed treatment with (carbendazim + mancozeb) @ 2 g/ kg of seed)
- T10- Seed treatment with tebuconazole (Raxil) @ 2 g/kg of seeds
- T11- Seed treatment with tebuconazole (Raxil) @ 2 g + carbendazim 1 g/ kg of seed.
- T12- Seed treatment with carboxin 37.5 % + thiram 37.5 % (vitavax power) @ 2 g/ kg of seed

T13- Seed treatment with carboxin 37.5 % + thiram 37.5 % (vitavax power) @ 2g + *T. viride* @ 5 g/ kg of seed.

T14- control

3.6 Statistical analysis

The experimental data collected were analyzed statistically for its significant or non-significant difference by the normal statistical procedure adopted for completely randomized design and randomized block design and interpretation of data was carried out in accordance with Walter (1997). The level of significance used in 'F' and 'T' test was $P=0.05$ and $P=0.01$. Critical differences were calculated wherever 'F' test was significant. The values of per cent disease index was subjected to arcsine transformation.



Experimental Results

IV. EXPERIMENTAL RESULTS

The experiments were carried out on various aspects of dry root rot of chickpea caused by *Macrophomina phaseolina* with reference to collection of disease specimens, isolation of the pathogen and pathogenicity test. Variability studies among the isolates with respect to cultural, morphological, physiological aspects and molecular characterization of isolates that includes identification by using species specific primers and molecular variability studies by using PCR-RAPD, ITS sequence. Screening of chickpea entries for disease resistance and management of root rot complex of chickpea by imposing different treatments under field condition. The results obtained on these aspects are presented under here.

4.1 Symptomatology

Visual observations on dry root rot of chickpea plants were recorded at various stages of crop growth in sick plot at ZARS, GKVK, Bengaluru. Symptoms appeared around flowering to podding stage. Affected plants showed various types of symptoms *viz.*, yellowing, drooping, drying and shedding of leaves, the root system of infected plant showed dark and extensive rotting. Roots became brittle and bark was peeled off easily, lateral roots were destroyed. The diseased plants which could be very easily pulled out from the soil showed discoloration of roots by presence of black colored microsclerotia (Plate1).

4.2 Growth of *M. phaseolina* isolates on Potato Dextrose Agar (PDA)

Growth and microsclerotial bodies production studied on PDA indicated that the mycelium was pale white in color in the initial stages of the growth but, later turned to dark brown to black as and when microsclerotial bodies started to form. Hypha showed right angled



A. Dry root rot infected field



B. Closer view of infected plant in field



C. Typically infected plants showing destruction of root system

Plate 1. Chickpea plant showing symptoms of dry root rot disease

branching. The microsclerotia varied in shape and size with growth pattern of scattered to clustered (Plate 2).

4.3 Collection of representative *M. phaseolina* isolates from different parts of India

Twenty dry root rot infected chickpea samples were collected from major chickpea growing states of India *viz.*, Andhra Pradesh, Himachal Pradesh, Karnataka, Madhya Pradesh, Maharashtra, Punjab, Tamil Nadu, Uttarakhand, Uttar Pradesh and West Bengal during *Rabi* 2012-13. *M. phaseolina* was isolated from infected samples by following standard tissue isolation procedure. The pure culture obtained was sub-cultured in Petriplate and slants containing potato dextrose agar and stored in refrigerator for use in further studies. Five isolates collected from Karnataka were designated as KAMP-1, KAMP-2, KAMP-3, KAMP-4 and KAMP-5, four isolates collected from Andhra Pradesh were designated as APMP-6, APMP-7, APMP-8 and APMP-9, two isolates from Tamil Nadu were designated and as TNMP-10 and TNMP-11, one isolate each from Maharashtra designated as MHMP-12, Madhya Pradesh isolate was designated as MPMP-13, Punjab isolate was designated as PUMP-14, three isolates from Uttar Pradesh were designated as UPMP-15, UPMP-16 and UPMP-17, one isolate from Uttarakhand was designated as UKMP-18, West Bengal isolate was designated as WBMP-19 and Himachal Pradesh isolate was designated as HPMP-20 (Table 3 and Plate 3).

4.4 Pathogenicity test

Pathogenicity test was conducted by following artificial inoculation of soil with 20 isolates of *M. phaseolina* separately. The giant culture was prepared by using chickpea seed as a substrate. Ten seeds of each susceptible cultivars *viz.*, H 09-23, L-550 and BG-212 were sown in trays filled with soil to which the giant culture of *M. phaseolina* was added. The

Table 3. List of isolates collected from different geographical regions

Sl. No.	State	District	Place	Designation
1	Karnataka	Bangalore north	GKVK	KAMP-1
2	Karnataka	Chitradurga	Hiriyur	KAMP-2
3	Karnataka	Hubli-Dharwad	Dharwad	KAMP-3
4	Karnataka	Gulbarga	Jevargi	KAMP-4
5	Karnataka	Raichur	Sarjapur	KAMP-5
6	Andhra Pradesh	Karnool	Karnool	APMP-6
7	Andhra Pradesh	Karnool	Nandyal	APMP-7
8	Andhra Pradesh	Ranga Reddy	ICRISAT	APMP-8
9	Andhra Pradesh	Warangal	RARS, Warangal	APMP-9
10	Tamil Nadu	Coimbatore	TNAU (Pappanaicken Pudur)	TNMP-10
11	Tamil Nadu	Coimbatore	Periyanakulampalyam	TNMP-11
12	Maharashtra	Jalna	Badnapur	MHMP-12
13	Madhya Pradesh	Ujjain	Rajwas	MPMP-13
14	Punjab	Ludhiana	PAU, Ludhiana	PUMP-14
15	Uttar Pradesh	Kanpur	IIPR, Kanpur	UPMP-15
16	Uttar Pradesh	Shahjahanpur	Sujalpur	UPMP-16
17	Uttar Pradesh	Bihar	Mokama	UPMP-17
18	Uttarakhand	Udham Singh Nagar	CRC, Pantnagar	UKMP-18
19	West Bengal	Burdwan	Durgapur	WBMP-19
20	Himachal Pradesh	Sirmour	Dhaulikaun	HPMP-20



Colony



Initial stage of microsclerotia formation



Microsclerotia



Right angle branching of mycelium

Plate 2. *M. phaseolina* on PDA medium and its morphological characters



Plate 3. Isolates of *M. phaseolina*

dry root rot symptoms appeared at 15- 30 days after sowing. Symptoms consisted of seed rot, yellowing, drooping and drying of leaves, the root system of infected plant turned dark and extensive rotting destroyed the lateral roots. Symptoms due to dry root rot affected plants in trays pots inoculated with *M. phaseolina* were found similar to that of the plants drying in the field due to the pathogen attack. Cultures obtained from isolation of the pathogen from infected root were compared with original culture of *M. phaseolina* and was found to be similar with regards to all morphological characters of the pathogen on PDA. Out of twenty isolates, only five isolates KAMP-2, KAMP-3, KAMP-4, TNMP-10 and PUMP-14 caused cent per cent dry root rot incidence on all the three cultivars; four isolates *viz.*, KAMP-1, MHMP-12, UPMP-16 and UPMP-17 caused > 90 per cent incidence; six isolates *viz.*, KAMP-5, APMP-6, APMP-7, MPMP-13, UKMP-18 and WBMP-19 caused > 80 per cent incidence and the remaining three isolates APMP-8, APMP-9 and TNMP-11 caused > 70 per cent dry root rot incidence (Table 4 and Plate 4).

4.5 Studies on Variability of *M. phaseolina*

4.5.1 Cultural studies of *M. phaseolina*

The cultural characters of twenty *M. phaseolina* isolates were studied on potato dextrose agar medium as described in material and methods. The observations on colony growth measured in terms of colony diameter in mm and growth characters *viz.*, appressed, partially appressed, fluffy growth and colony color were recorded (Table 5a). The variation was observed in some of the isolates with respect to growth pattern and mycelial color. These characters were considered to assess the existence of variation in the pathogen.

The growth of twenty isolates on PDA were recorded after 4 days of inoculation. Among the isolates studied six isolates *viz.*, APMP-7, APMP-8, APMP-9, MPMP-13, PUMP-14 and HPMP-20 showed maximum growth

Table 4. Dry root rot incidence caused by *M. phaseolina* isolates in susceptible chickpea cultivars

Sl. No.	Isolates	Dry root rot disease incidence (%) on cultivars			Mean
		H 09-23	L-550	BG-212	
1	KAMP-1	70	100	100	90.00
2	KAMP-2	100	100	100	100.00
3	KAMP-3	100	100	100	100.00
4	KAMP-4	100	100	100	100.00
5	KAMP-5	60	100	100	86.67
6	APMP-6	40	100	100	80.00
7	APMP-7	80	80	80	80.00
8	APMP-8	30	100	100	76.67
9	APMP-9	60	70	100	76.67
10	TNMP-10	100	100	100	100.00
11	TNMP-11	40	90	90	73.33
12	MHMP-12	70	100	100	90.00
13	MPMP-13	50	100	100	83.33
14	PUMP-14	100	100	100	100.00
15	UPMP-15	40	90	100	76.67
16	UPMP-16	80	100	90	90.00
17	UPMP-17	90	90	100	93.33
18	UKMP-18	80	90	90	86.67
19	WBMP-19	60	100	100	86.67
20	HPMP-20	60	80	90	76.67
Mean		70.50	94.50	97.00	87.33

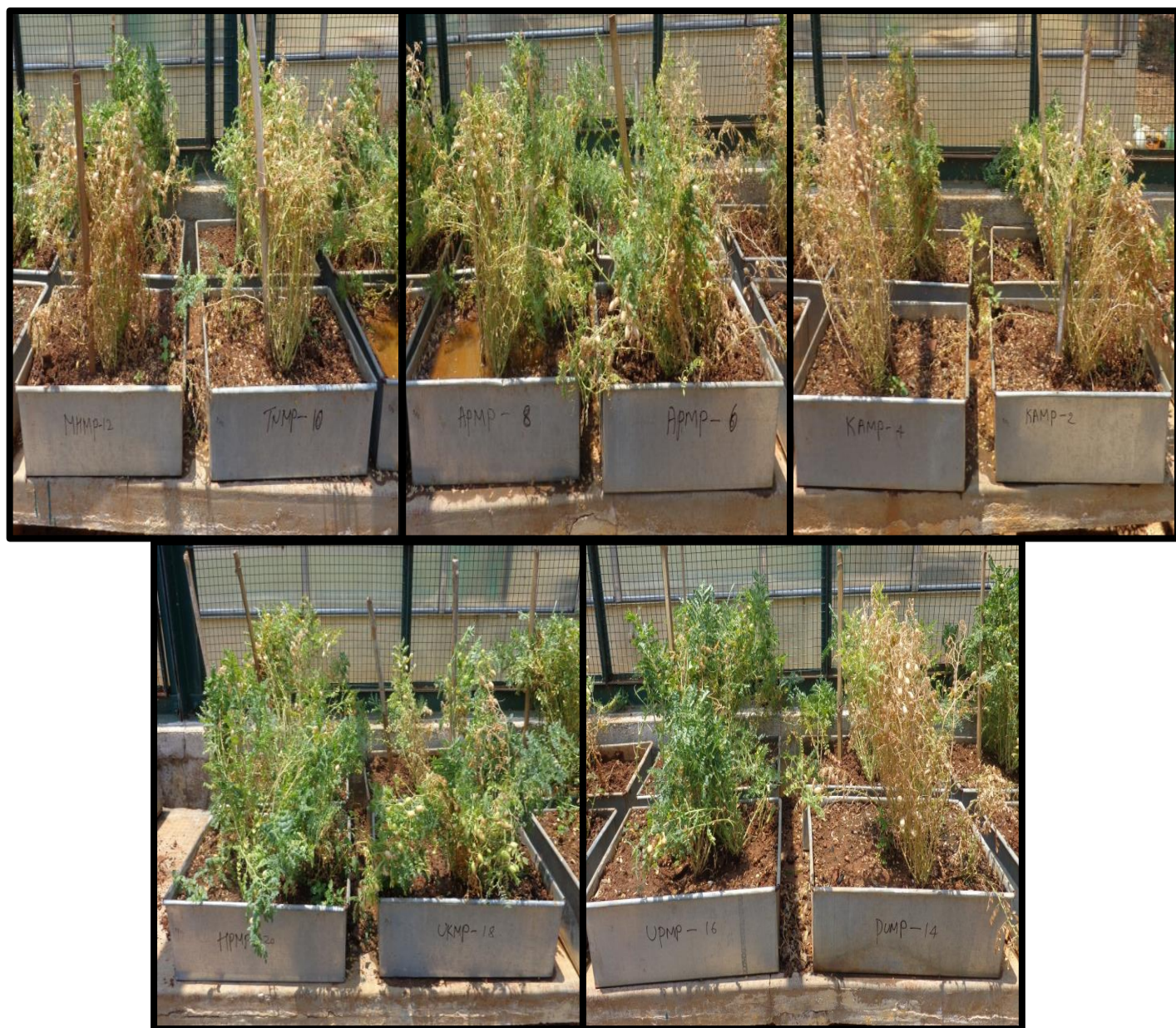


Plate 4. Pathogenicity test of *M. phaseolina* isolates on chickpea cultivars

Table 5a. Growth characters of *M. phaseolina* isolates

Sl. No.	Isolates	Mean colony diameter (mm)	Colony color	Colony appearance
1	KAMP-1	83.83	Blackish gray	Fluffy
2	KAMP-2	88.33	Blackish gray	Partially appressed
3	KAMP-3	89.00	Blackish gray	Fluffy
4	KAMP-4	74.16	Deep black	Appressed
5	KAMP-5	82.00	Deep black	Appressed
6	APMP-6	86.99	Grayish white	Fluffy
7	APMP-7	90.00	Grayish white	Fluffy
8	APMP-8	90.00	Deep black	Appressed
9	APMP-9	90.00	Blackish gray	Fluffy
10	TNMP-10	83.16	Grayish white	Partially appressed
11	TNMP-11	67.33	Grayish white	Fluffy
12	MHMP-12	85.00	Blackish gray	Fluffy
13	MPMP-13	90.00	Blackish gray	Partially appressed
14	PUMP-14	90.00	Cottony white	Appressed
15	UPMP-15	71.50	Grayish white	Partially appressed
16	UPMP-16	74.00	Cottony white	Appressed
17	UPMP-17	84.83	Blackish gray	Fluffy
18	UKMP-18	85.50	Blackish gray	Partially appressed
19	WBMP-19	79.66	Blackish gray	Partially appressed
20	HPMP-20	90.00	Grayish white	Partially appressed
	S.Em $\pm$	4.43		
	CD @ 1%	16.70		

(90 mm) they are significantly superior over the isolates TNMP-11 and UPMP-15 had minimum growth of 67.33 & 71.50 mm respectively and on par with remaining isolates.

Based on the mean colony diameter, isolates were categorized into three groups. Fifteen isolates *viz.*, KAMP-1, KAMP-2, KAMP-3, KAMP-5, APMP-6, APMP-7, APMP-8, APMP-9, TNMP-10, MHMP-12, MPMP-13, PUMP-14, UPMP-17, UKMP-18 and HPMP-20 were placed in group A with mean colony diameter ranging between 80-90 mm; four isolates *viz.*, KAMP-4, UPMP-15, UPMP-16 and WBMP-19 were categorized under group B with mean colony diameter ranging from 70-80 mm and the remaining one isolate TNMP-11 was kept under group C with mean colony diameter of 60-70 mm and there was no significant difference observed in between the isolates with respect growth (Table 5b and Fig1).

Based on colony color the twenty isolates were categorized into four groups. Group I comprised six isolates *viz.*, APMP-6, APMP-7, TNMP-10, TNMP-11, UPMP-15 and HPMP-20 with grayish white colony, group II comprised nine isolates *viz.*, KAMP-1, KAMP-2, KAMP-3, APMP-9, MHMP-12, MPMP-13, UPMP-17, UKMP-18 and WBMP-19 with blackish gray colony, group III comprised two isolates *viz.*, PUMP-14 and UPMP-16 with cottony white colony and group IV comprised three isolates KAMP-4, KAMP-5 and APMP-8 with deep black colony (Table 5c and Plate 5).

Based on colony appearance the twenty isolates were categorized into three groups *viz.*, fluffy, partially appressed and appressed. Group I of eight isolates *viz.*, KAMP-1, KAMP-3, APMP-6, APMP-7, APMP-9, TNMP-11, MHMP-12 and UPMP-17 showed fluffy growth appearance, group II of seven isolates *viz.*, KAMP-2, TNMP-10, MPMP-13, UPMP-15, UKMP-18, WBMP-19 and HPMP-20 produced partially appressed growth

Table 5b. Grouping of *M. phaseolina* isolates based on colony diameter (mm)

Group	Range (mm)	Isolates	No. of isolates
A	80.00-90.00	KAMP-1, KAMP-2, KAMP-3, KAMP-5, APMP-6, APMP-7, APMP-8, APMP-9 TNMP-10, MHMP-12, MPMP-13, PUMP-14, UPMP-17, UKMP-18 and HPMP-20	15
B	70.00-80.00	KAMP-4, UPMP-15, UPMP-16 and WBMP-19	4
C	60.00-70.00	TNMP-11	1

Table 5c. Grouping of *M. phaseolina* isolates based on colony color

Group	Color	Isolates	No. of isolates
I	Grayish white	APMP-6, APMP-7, TNMP-10, TNMP-11, UPMP-15 and HPMP-20	6
II	Blackish gray	KAMP-1, KAMP-2, KAMP-3, APMP-9, MHMP-12, MPMP-13, UPMP-17, UKMP-18 and WBMP-19	9
III	Cottony white	PUMP-14 and UPMP-16	2
IV	Deep black	KAMP-4, KAMP-5 and APMP-8	3

Table 5d. Grouping of *M. phaseolina* isolates based on colony appearance

Group	Colony appearance	Isolates	No. of isolates
I	Fluffy	KAMP-1, KAMP-3, APMP-6, APMP-7, APMP-9, TNMP-11, MHMP-12 and UPMP-17	8
II	Partially appressed	KAMP-2, TNMP-10, MPMP-13, UPMP-15, UKMP-18, WBMP-19 and HPMP-20	7
III	Appressed	KAMP-4, KAMP-5, APMP-8, PUMP-14 and UPMP-16	5

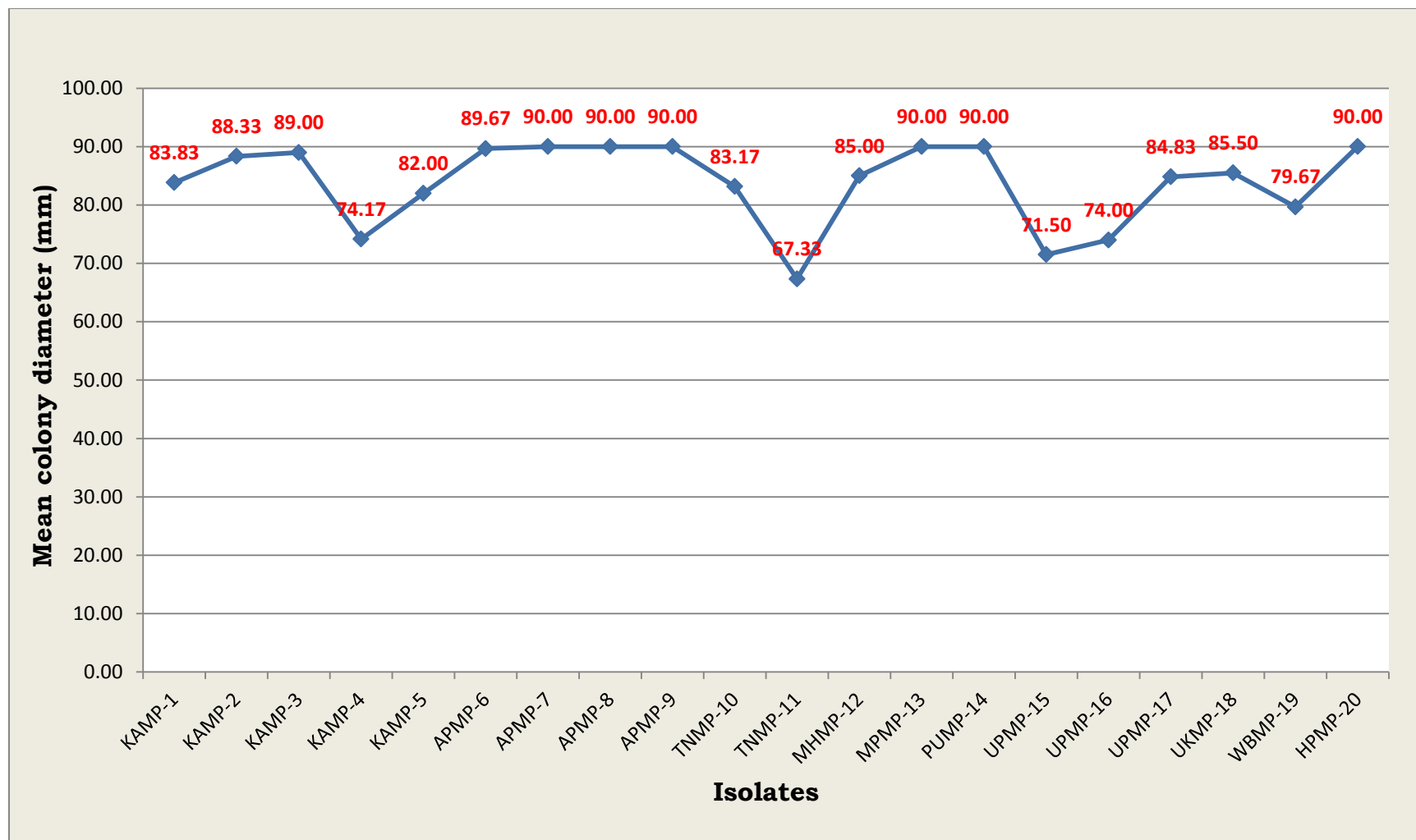


Fig. 1. Colony diameter of *M. phaseolina* isolates on potato dextrose agar medium four days after inoculation



Plate 5. Colony color of *M. phaseolina* isolates



Plate 6. Colony growth pattern of *M. phaseolina* isolates

and group III of five isolates viz., KAMP-4, KAMP-5, APMP-8, PUMP-14 and UPMP-16 produced appressed growth (Table 5d and Plate 6).

4.5.2 Morphological variability of *M. phaseolina* isolates

The data pertaining to morphological variability of twenty isolates observed on PDA is presented in the Table 6a.

The morphological characters like size and shape of microsclerotia, number of microsclerotia per microscopic field, days taken to produce microsclerotia were taken into consideration to assess the existence of variation in the pathogen.

Wide range of variation was noticed among the twenty isolates of *M. phaseolina* with respect to diameter of microsclerotia which varied from 160.73 μm (KAMP-4) which is at par with MHMP-12 KAMP-1, APMP-6, APMP-7, APMP-8, TNMP-10 and which is significantly superior over the isolate TNMP-11 which is having a diameter of 36.98 μm and the remaining isolates KAMP-2, KAMP-3, KAMP-5, APMP-9, MPMP-13, PUMP-14, UPMP-15, UPMP-16, UPMP-17, UKMP-18 and WBMP-19 were on par with each other.

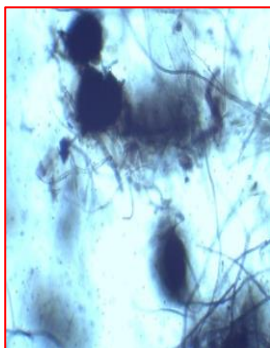
Based on mean microsclerotial diameter, all twenty isolates were categorized into four groups. Group I consisted four isolates viz., TNMP-11, UPMP-15, UPMP-16 and HPMP-20 having mean microsclerotial diameter ranging from 30 -50 μm , group II consisted eleven isolates viz., KAMP-1, KAMP-2, KAMP-3, KAMP-5, APMP-9, TNMP-10, MPMP-13, PUMP-14, UPMP-17, UKMP-18 and WBMP-19 having mean microsclerotial diameter ranging from 50- 100 μm , group III consisted four isolates viz., APMP-6, APMP-7, APMP-8 and MHMP-12 having mean microsclerotial diameter ranging from 100-150 μm and the remaining one isolate KAMP-4 fell under the group IV having the mean microsclerotial diameter > 150 μm (Table 6b and Plate 7).

Table 6a. Morphology of microsclerotial bodies produced by *M. phaseolina* isolates

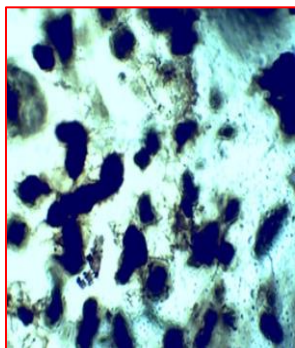
Sl. No.	Isolates	Number/ microscopic field	Diameter (μm)	Shape	Days taken for micro-sclerotia formation
1	KAMP-1	45.00	85.82	Round	2
2	KAMP-2	50.33	56.95	Oblong	2
3	KAMP-3	44.33	69.38	Oblong	2
4	KAMP-4	69.00	160.73	Oblong	2
5	KAMP-5	64.33	80.38	Oblong	2
6	APMP-6	66.33	107.90	Oblong	2
7	APMP-7	31.00	107.40	Round	2
8	APMP-8	60.66	111.47	Oblong	2
9	APMP-9	54.66	82.27	Oblong	3
10	TNMP-10	47.00	90.20	Oblong	3
11	TNMP-11	32.66	36.98	Round	2
12	MHMP-12	77.00	125.95	Oblong	2
13	MPMP-13	37.66	65.33	Round	2
14	PUMP-14	29.66	68.90	Oblong	3
15	UPMP-15	45.33	48.32	Oblong	2
16	UPMP-16	23.00	48.23	Round	2
17	UPMP-17	44.00	59.97	Round	2
18	UKMP-18	45.66	63.68	Round	2
19	WBMP-19	36.66	78.05	Round	3
20	HPMP-20	25.33	42.02	Round	2
S.Em $\pm$		10.21	20.65		
CD @ 1 %		38.50	77.84		

Table 6b. Grouping of *M.phaseolina* isolates based on mean microsclerotial diameter

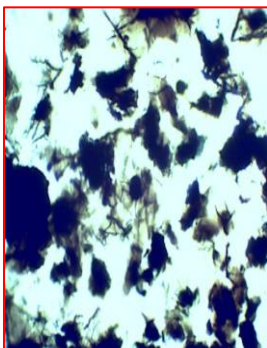
Group	Range (µm)	Isolates	Number of isolates
I	30-50	TNMP-11, UPMP-15, UPMP-16 and HPMP-20	4
II	50-100	KAMP-1, KAMP-2, KAMP-3, KAMP-5, APMP-9, TNMP-10, MPMP-13, PUMP-14, UPMP-17, UKMP-18 and WBMP-19	11
III	100-150	APMP-6, APMP-7, APMP-8 and MHMP-12	4
IV	>150	KAMP-4	1



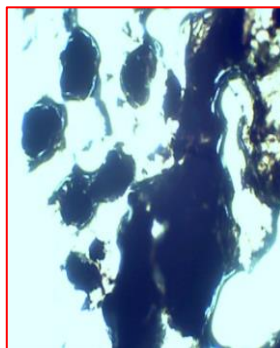
KAMP-1



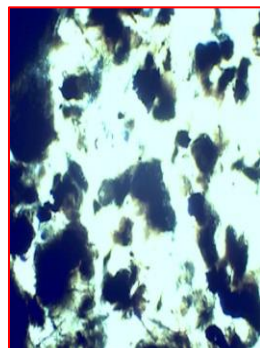
KAMP-2



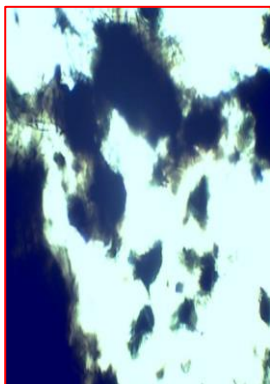
KAMP-3



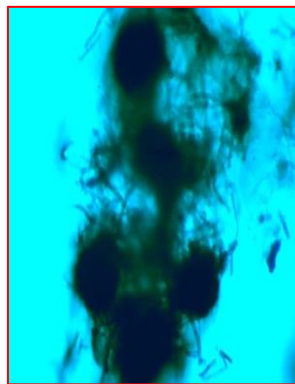
KAMP-4



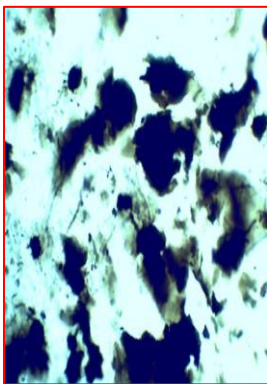
KAMP-5



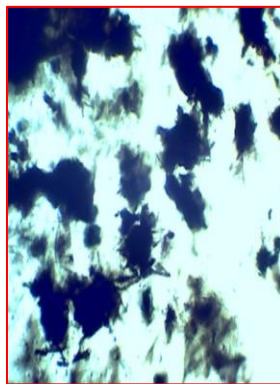
APMP-6



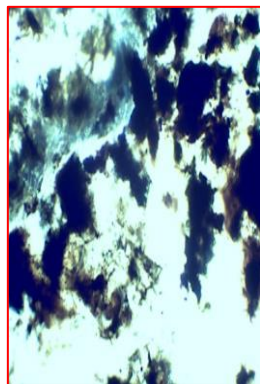
APMP-7



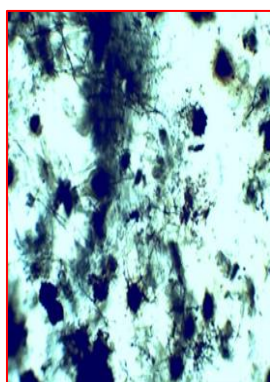
APMP-8



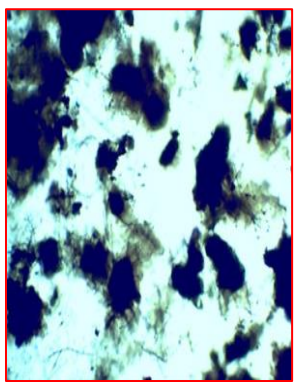
APMP-9



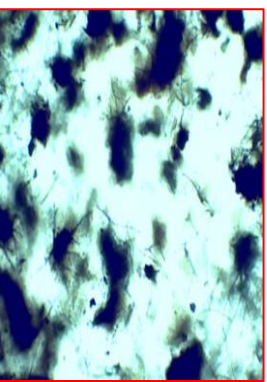
TNMP-10



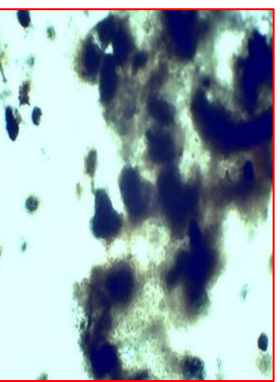
TNMP-11



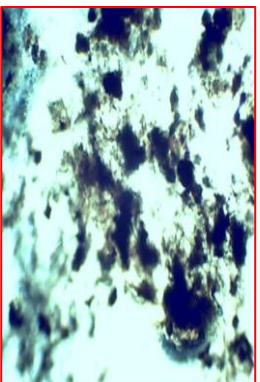
MHMP-12



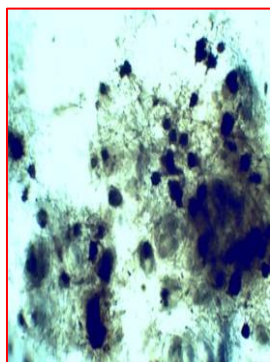
MPMP-13



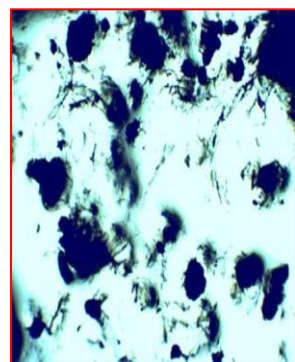
PUMP-14



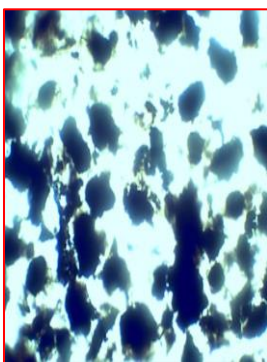
UPMP-15



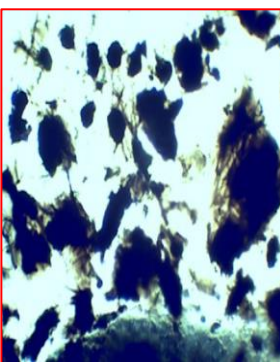
UPMP-16



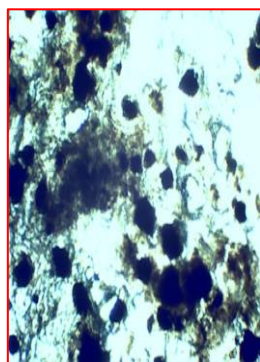
BHMP-17



UKMP-18



WBMP-19



HMMP-20

Based on shape of microsclerotia all the isolates were categorized into two groups. Nine isolates KAMP-1, APMP-7, TNMP-11, MPMP-13, UPMP-16, UPMP-17, UKMP-18, WBMP-19 and HPMP-20 produced round shape microsclerotia while, remaining eleven isolates *viz.*, KAMP-2, KAMP-3, KAMP-4, KAMP-5, APMP-6, APMP-8, APMP-9, TNMP-10, MHMP-12, PUMP-14 and UPMP-15 produced oblong shape microsclerotia in the culture (Table 6a).

The number of microsclerotial bodies produced per microscopic field was minimum (25.33) by isolate UPMP-16 which is significantly different over the isolate MHMP-12 produced maximum microsclerotial bodies (77) under 10x magnification and the isolates PUMP-14, APMP-7, WBMP-19, UPMP-17, KAMP-3, KAMP-1, UKMP-18, UPMP-15, KAMP-2, APMP-8, APMP-9 and TNMP-10 were on par with each other. All the isolates took two days to form microsclerotial bodies in the culture except four isolates *viz.*, APMP-9, TNMP-10, PUMP-14 and WBMP-19 which took three days in culture after inoculation (Table 6a and Fig 2).

The observations on length and breadth of microsclerotia are presented in Table 6c. The minimum length and breadth of microsclerotia 37.53 x36.43µm was observed in isolate TNMP-11, while the maximum length and breadth of about 183.63x137.83 µm was noticed in KAMP-4 isolate.

4.5.3 Physiological studies of *M. phaseolina* isolates

4.5.3.1 Effect of temperature levels on growth and microsclerotial bodies production of *M. phaseolina* isolates

Temperature influences the growth rate, metabolism and morphological characters of the fungi. The fungus can thrive under wide range of temperatures and there is minimum, optimum and maximum

Table 6c. Size of microsclerotia produced by *M. phaseolina* isolates

Sl. No.	Isolates	Measurement	
		Range (LXB) (µm)	Average (LXB) (µm)
1	KAMP-1	85.6-95 x 65.9- 102.6	90.1 x81.53
2	KAMP-2	49.7-90 x 33.6-45.6	74.66 x39.23
3	KAMP-3	70 -90.9 X55.1-58.2	81.96 x56.8
4	KAMP-4	134.7-214.4 X 95.4- 162.4	183.63 x137.83
5	KAMP-5	56.7- 135.6 X 47.2- 90.3	97.33 x63.43
6	APMP-6	44.5-215 X 65.4- 104.5	126.56 x89.23
7	APMP-7	78.2- 173.5 X 80.5-144.2	110.73 x104.06
8	APMP-8	100- 151.1 X 87.2- 105.2	128.43 x94.5
9	APMP-9	58.7- 110.6 X 43.2- 91.4	90.66 x73.86
10	TNMP-10	87.7- 144.8 X 59.5- 67.2	116 x64.4
11	TNMP-11	32.9-41.1 X 27.7- 48.4	37.53 x36.43
12	MHMP-12	98.47 -175.6 X 93.8-118.6	144.22 x107.66
13	MPMP-13	45.6 -106.9 X 40.2- 60.5	78.1 x52.56
14	PUMP-14	63 -80.8 X 48.7-91.2	70.96 x66.83
15	UPMP-15	53.9- 59.8 X 38.1- 42.4	56.76 x39.86
16	UPMP-16	43.3- 55.3 X 40.6- 52.1	49.63 x46.83
17	UPMP-17	49.6- 85.6 X43.2 – 57.3	68.9 x51.03
18	UKMP-18	69.9-81.6 X34.5- 72.3	75.06 x52.3
19	WBMP-19	84.2- 90 X 58.4- 76.9	87.86 x68.23
20	HPMP-20	44.1- 53 X 30.9- 43.7	48 x36.03

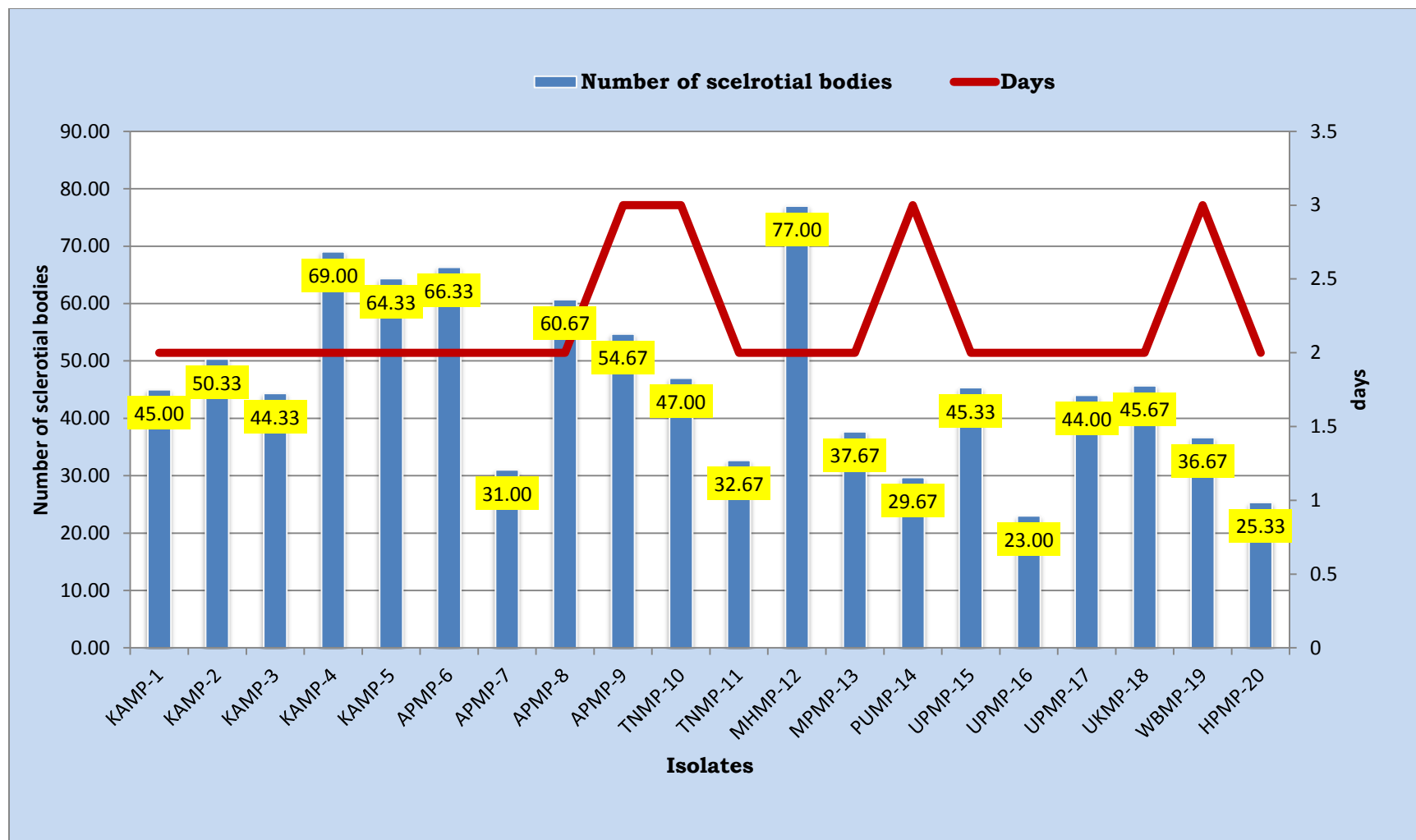


Fig. 2. Number of microsclerotia produced and days taken for microsclerotial production by *M. phaseolina* isolates

temperature requirements for each species and even for a particular strain.

The variations in growth among the isolates at different temperature regimes were found highly significant (Table 7a). KAMP-3 showed mean maximum growth with colony diameter of 84.22 mm at all temperature evaluated which was significantly superior over other isolates. The isolates MHMP-12 & WBMP19 are on par having a growth of 81.86 & 80.94 mm respectively and were significantly superior over other isolates, whereas, least growth of that with 60.31 mm was recorded in UPMP-15.

Among different temperature regimes evaluated, the temperature 35^o C is on par i.e., contributing the same effect with 30^o C and 30^o C is highly significant with the remaining temperature. Similarly other temperature are significantly differ with the other. 30^oC and 35^oC supported better growth in all the isolates (90mm) whereas, at temperature of 40^oC poor growth (32.37mm) was observed. The growth of all isolates increased substantially from 15^oC to 30^oC. However the temperatures of 30^oC and 35^oC favored the growth of *M. phaseolina* isolates. Most of the isolates differed significantly in their relative growth between 15-25^oC indicating that each isolate had an ability to withstand specific temperature upto 25^oC (Fig 3) and the effect of temperature on growth of isolates and interaction were found to be highly significant.

The variations in microsclerotial bodies production among the isolates at different temperature regimes was found significant (Table 7b). MHMP-12 produced more microsclerotial bodies with a mean of 27.72 per microscopic field across the temperatures evaluated followed by KAMP-3 with mean microsclerotial bodies production of 27.44 and KAMP-3 was on par with KAMP-5, KAMP-4 and MPMP-13 were significantly different with other isolates, whereas, least microsclerotial

Table 7a. Effect of temperature levels on growth of *M. phaseolina* isolates

Sl. No.	Isolates	Temperature (°C) and mean radial growth (mm)						Mean
		15	20	25	30	35	40	
1	KAMP-1	61.83	85.83	90.00	90.00	90.00	29.83	74.58
2	KAMP-2	56.17	87.33	90.00	90.00	90.00	63.00	79.42
3	KAMP-3	77.33	86.33	90.00	90.00	90.00	71.67	84.22
4	KAMP-4	34.50	78.33	79.33	90.00	90.00	17.00	64.86
5	KAMP-5	41.50	86.23	87.67	90.00	90.00	33.67	71.51
6	APMP-6	36.00	85.17	74.33	90.00	90.00	26.67	67.03
7	APMP-7	49.00	87.00	86.00	90.00	90.00	28.67	71.78
8	APMP-8	46.13	78.00	86.67	90.00	90.00	10.67	66.91
9	APMP-9	25.00	87.00	90.00	90.00	90.00	34.33	69.39
10	TNMP-10	36.67	85.17	55.97	90.00	90.00	10.33	61.36
11	TNMP-11	34.33	81.67	62.17	90.00	90.00	12.17	61.72
12	MHMP-12	79.33	90.00	90.00	90.00	90.00	51.83	81.86
13	MPMP-13	78.00	87.33	86.33	90.00	90.00	33.00	77.44
14	PUMP-14	52.33	87.00	90.00	90.00	90.00	24.33	72.28
15	UPMP-15	34.17	63.67	72.33	90.00	90.00	11.67	60.31
16	UPMP-16	55.17	83.33	86.33	90.00	90.00	23.50	71.39
17	UPMP-17	56.30	88.33	90.00	90.00	90.00	40.00	75.77
18	UKMP-18	62.50	90.00	87.70	90.00	90.00	22.67	73.81
19	WBMP-19	82.30	90.00	90.00	90.00	90.00	43.33	80.94
20	HPMP-20	63.83	81.67	72.67	90.00	90.00	59.00	76.19
MEAN		53.12	84.47	82.88	90.00	90.00	32.37	72.14
		Isolates (A)		Temperature (B)		Interaction (AXB)		
S.Em ±		0.83		0.45		2.03		
CD @ 1 %		2.16		1.18		5.29		

Table 7b. Effect of temperature levels on microsclerotial production by *M. phaseolina* isolates

Sl. No.	Isolates	Temperature (°C) and number of microsclerotial bodies/ microscopic field						Mean
		15	20	25	30	35	40	
1	KAMP-1	4.00	9.00	21.33	45.00	24.33	1.67	17.56
2	KAMP-2	3.00	23.00	36.33	50.33	13.67	11.00	22.89
3	KAMP-3	1.33	31.33	53.33	44.33	31.33	3.00	27.44
4	KAMP-4	2.00	11.33	26.33	69.00	31.33	1.33	23.56
5	KAMP-5	3.00	17.67	33.33	64.33	28.33	11.67	26.39
6	APMP-6	1.33	3.33	15.33	66.33	9.00	11.00	17.72
7	APMP-7	2.00	32.33	16.67	31.00	42.67	3.00	21.28
8	APMP-8	2.67	16.00	31.33	60.67	20.33	5.00	22.67
9	APMP-9	1.67	3.00	27.33	54.67	6.33	2.67	15.94
10	TNMP-10	1.00	3.33	34.67	47.00	23.00	2.00	18.50
11	TNMP-11	2.67	4.67	22.00	32.67	17.00	3.67	13.78
12	MHMP-12	2.33	25.67	14.00	77.00	36.33	11.00	27.72
13	MPMP-13	3.00	5.33	40.67	37.67	47.33	6.67	23.44
14	PUMP-14	3.67	2.00	32.00	29.67	19.33	6.00	15.44
15	UPMP-15	1.67	3.00	42.33	45.33	29.33	4.33	21.00
16	UPMP-16	0.33	1.67	36.67	23.00	41.67	5.00	18.06
17	UPMP-17	1.00	7.00	34.00	44.00	19.00	5.33	18.39
18	UKMP-18	4.33	28.67	41.00	45.67	13.33	2.00	22.50
19	WBMP-19	12.33	25.00	6.00	36.67	33.00	12.67	20.94
20	HPMP-20	5.67	26.00	17.00	25.33	17.67	2.00	15.61
MEAN		2.95	13.97	29.08	46.48	25.21	5.55	20.54
		Isolates (A)		Temperature (B)		Interaction (AXB)		
S.Em ±		1.72		0.94		4.21		
CD @ 1 %		4.46		2.44		10.93		

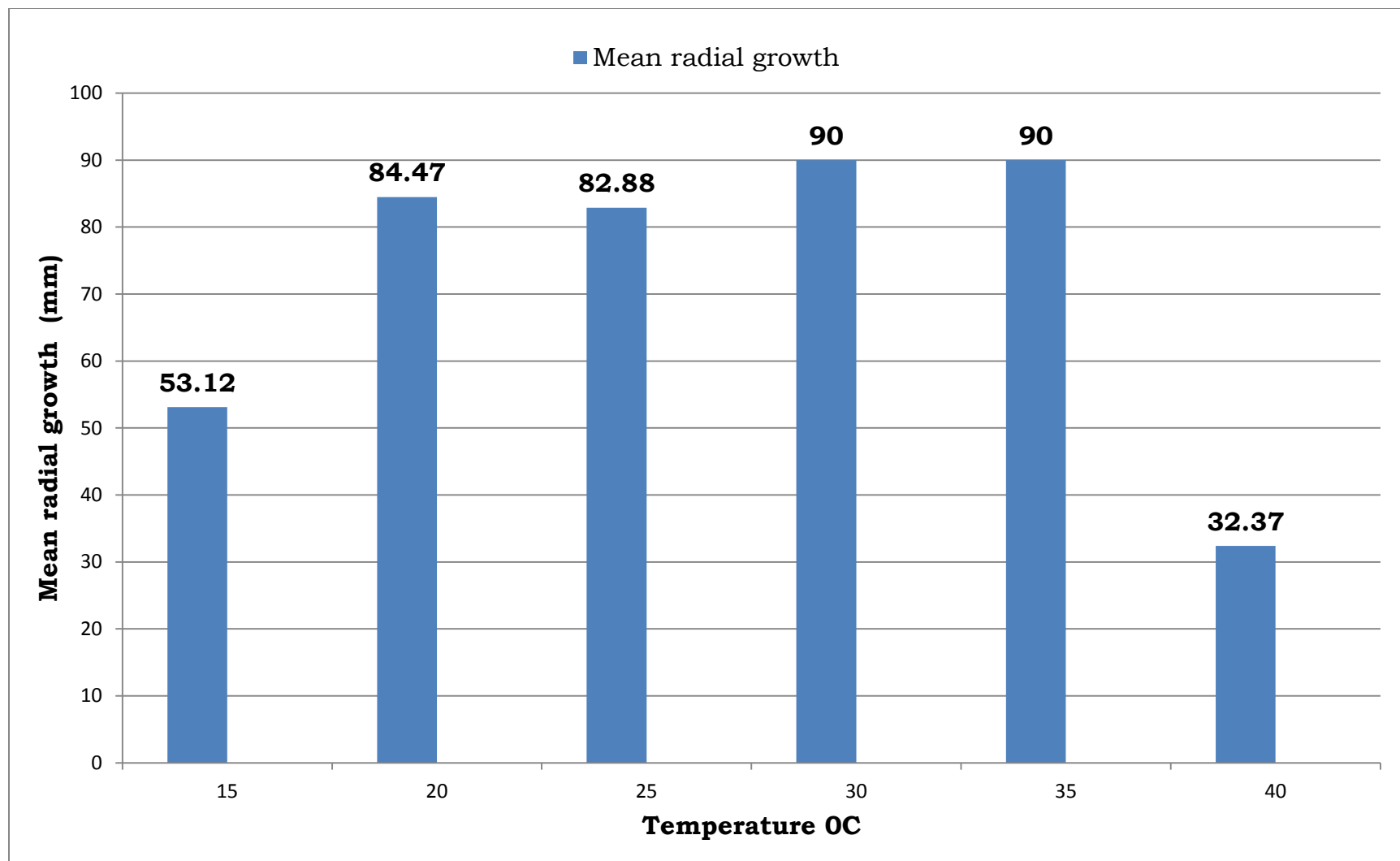


Fig. 3. Effect of temperature levels on growth of *M. phaseolina* isolates

bodies (13.78) were observed in the isolate TNMP-11 and it was on par with APMP-6, KAMP-1, APMP-9, HPMP-20, PUMP-14 and TNMP-11.

Temperature of 30°C favored more production of microsclerotial bodies of 46.48 per microscopic field followed by 40°C with a mean microsclerotial bodies of 32.37 per microscopic field and the least of that was observed at the temperature of 15°C (2.95). The change in optimum temperature from 30°C has adversely affected the production of microsclerotial bodies. Thus the temperature 30°C was found to be excellent for production of microsclerotial bodies in all the isolates of *M. phaseolina*. So, the effect of temperature on microsclerotia production by isolates and interaction were found to be highly significant (Fig 4).

4.5.3.2 Effect of pH levels on growth of *M. phaseolina* isolates

The hydrogen ion concentration (pH) of the medium influenced the growth of *M. phaseolina*. Every organism had its minimum, maximum and optimum pH for growth.

The effect of pH concentrations on growth of isolates and their interactions were found to be highly significant. The isolate MPMP-13 produced maximum mean growth with 579.44 mg of dry mycelial weight and it was on par with KAMP-2, KAMP-1 and UKMP-18. pH has no influence on growth of isolate PUMP-14 which has recorded least dry mycelial weight of 183.33 mg (Table 8).

In general, pH 5.0 was found to be better for majority of *M. phaseolina* isolates with mean maximum dry mycelial weight of 482.33 mg followed by pH 6.0 with 365.55 mg respectively (Fig 5). pH 4, pH 7 and pH 8 were on par with respect to each other. It could be clearly seen that there was a wide variation in growth with respect to pH among the isolates of different origin.

Table 8. Effect of hydrogen ion concentration (pH) levels on growth *M. phaseolina* isolates

Sl. No.	Isolates	pH levels and dry mycelial weight (mg)						Mean
		pH 4.0	pH 5.0	pH 6.0	pH 7.0	pH 8.0	pH 9.0	
1	KAMP-1	273.33	420.00	366.67	626.67	613.33	340.00	440.00
2	KAMP-2	236.67	653.33	456.67	560.00	566.67	330.00	467.22
3	KAMP-3	423.33	533.33	456.67	263.33	266.67	176.67	353.33
4	KAMP-4	320.00	350.00	306.67	343.33	263.33	216.67	300.00
5	KAMP-5	216.67	550.00	266.67	353.33	420.00	280.00	347.78
6	APMP-6	373.33	436.67	180.00	436.67	510.00	300.00	372.78
7	APMP-7	246.67	550.00	536.67	333.33	340.00	263.33	378.33
8	APMP-8	330.00	626.67	293.33	163.33	136.67	126.67	279.44
9	APMP-9	126.67	416.67	330.00	183.33	146.67	103.33	217.78
10	TNMP-10	203.33	433.33	226.67	226.67	356.67	226.67	278.89
11	TNMP-11	213.33	346.67	433.33	146.67	246.67	156.67	257.22
12	MHMP-12	390.00	606.67	640.00	256.67	160.00	150.00	367.22
13	MPMP-13	336.67	653.33	770.00	660.00	523.33	533.33	579.44
14	PUMP-14	206.67	236.67	173.33	230.00	100.00	153.33	183.33
15	UPMP-15	620.00	426.67	183.33	253.33	326.67	226.67	339.44
16	UPMP-16	153.33	533.33	196.67	133.33	100.00	133.33	208.33
17	UPMP-17	230.00	370.00	233.33	250.00	160.00	123.33	227.78
18	UKMP-18	316.67	680.00	526.67	353.33	396.67	233.33	417.78
19	WBMP-19	543.33	473.33	380.00	156.67	130.00	246.67	321.67
20	HPMP-20	636.67	350.00	353.33	253.33	350.00	213.33	359.44
MEAN		319.83	482.33	365.50	309.17	305.67	226.67	334.86
		Isolates (A)		pH (B)		Interaction (AXB)		
S.Em ±		20.21		11.06		49.50		
CD @ 1 %		52.47		28.74		128.53		

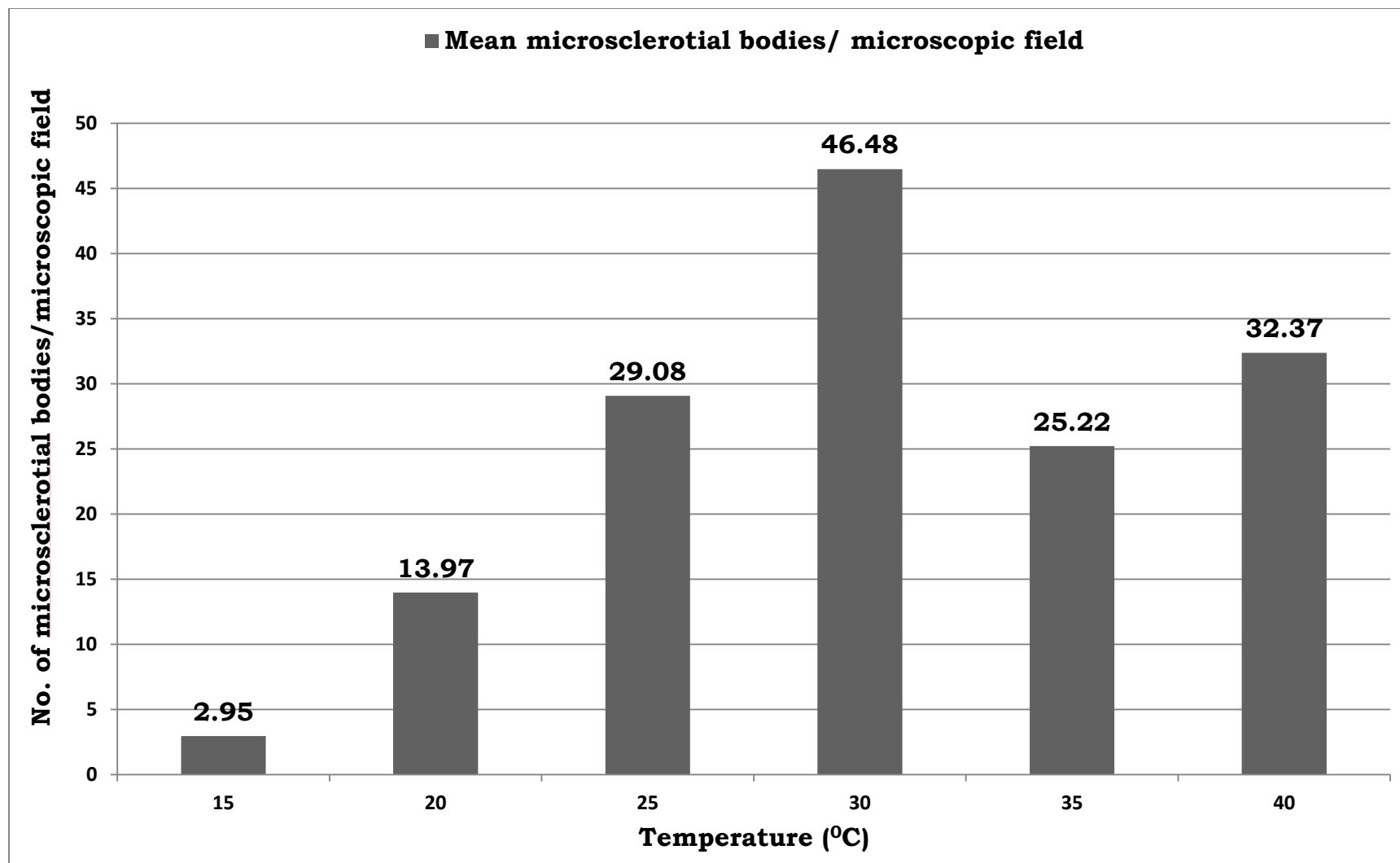


Fig. 4. Effect of temperature levels on microscerotia production by *M. phaseolina* isolates

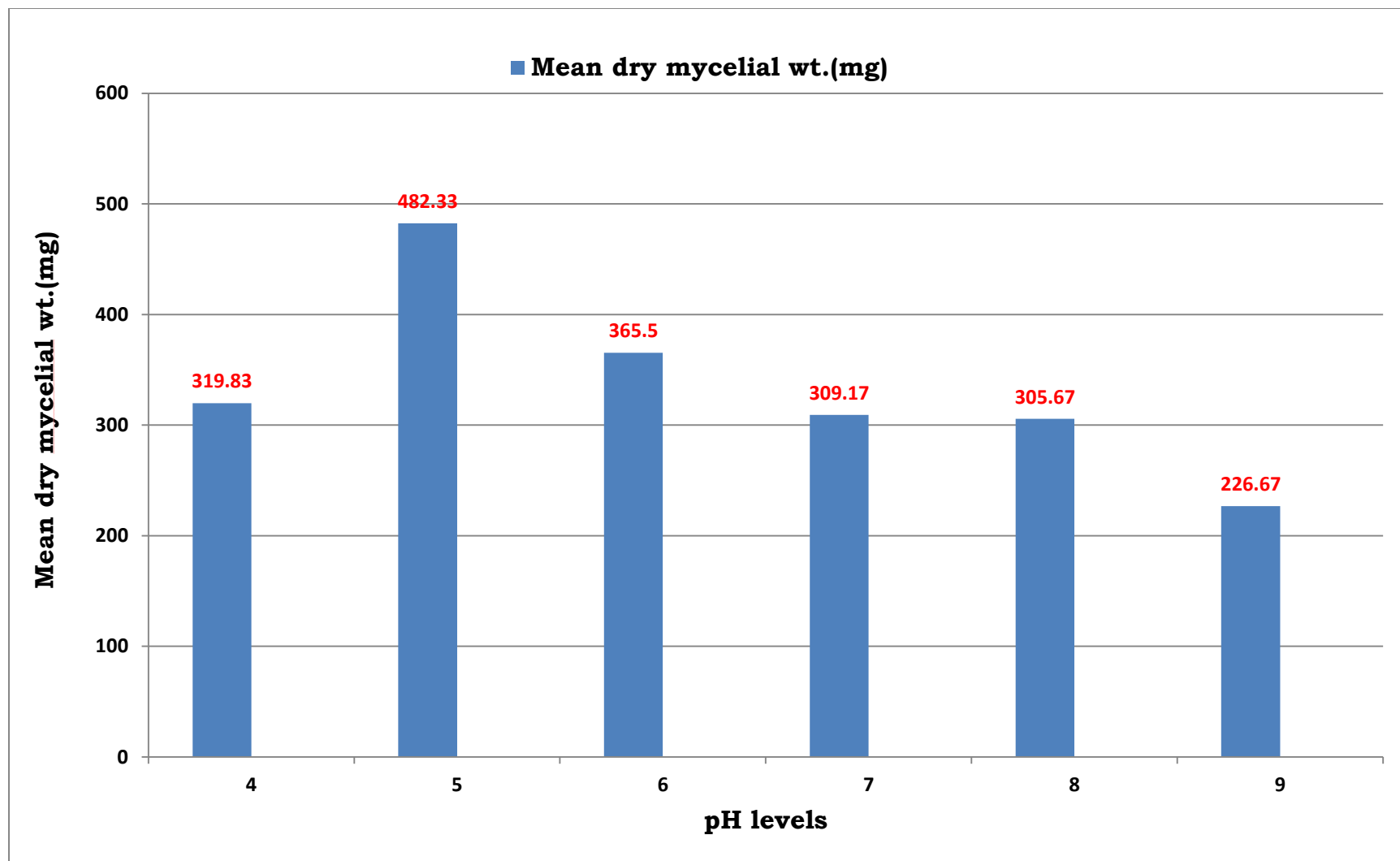


Fig. 5. Effect of pH levels on growth of *M. phaseolina* isolates

4.5.3.3 Effect of light on growth and microsclerotia production of *M. phaseolina* isolates

The effect of light on growth and production of microsclerotial bodies is presented in the Table 9a and 9b.

The mean colony diameter of all the isolates were maximum (84.43 mm) when cultures were exposed to alternate cycles of light and darkness which is on par with continuous dark (83.18 mm) and the least growth (71.36 mm) was observed when cultures were exposed to continuous light. However, the mean colony diameter was maximum (84.11 mm) in KAMP-4 isolate is on par with APMP-6, APMP-7, APMP-9, MHMP-12, APMP-8, UPMP-17, UKMP-18, PUMP-14, TNMP-10, KAMP-5, MPMP-13, HPMP-20 and UPMP-15 and the least colony diameter of 72.67 mm was observed in the isolate KAMP-3 which is significantly different from other isolates (Fig 6).

Alternate cycles of light and darkness supported the maximum mean microsclerotial production (44.73) by *M. phaseolina* followed by continuous dark (26.87) and continuous light (19.92). However, among all isolates tested, the mean sclerotia were higher in KAMP-5 isolate (42.00) which is on par with KAMP-2, APMP-8, MPMP-13, UPMP-17 and KAMP-4 and least microsclerotial production was recorded in the isolates TNMP-11 and UPMP-16 of 21.00. So, the effect of light on growth, microsclerotial production and their interaction was found to be significant.

The number of days taken for microsclerotial bodies production (Table 9b) was less in alternate cycles of light and darkness i.e., 2.2 days followed by continuous darkness (2.95 days) and continuous light (3.15 days) (Fig 7).

Table 9a. Effect of light on radial growth of *M. phaseolina* isolates

Sl. No.	Isolates	Light/ growth (mm)			Mean
		CL*	CD*	A.L/D*	
1	KAMP-1	70.00	80.67	82.50	77.72
2	KAMP-2	67.33	83.33	85.00	78.56
3	KAMP-3	46.67	84.67	86.67	72.67
4	KAMP-4	81.67	85.67	85.00	84.11
5	KAMP-5	75.83	83.00	82.00	80.28
6	APMP-6	73.33	84.33	88.00	81.89
7	APMP-7	75.00	85.00	85.00	81.67
8	APMP-8	75.33	82.33	85.33	81.00
9	APMP-9	73.50	83.00	87.33	81.28
10	TNMP-10	71.50	82.67	86.83	80.33
11	TNMP-11	70.00	80.00	81.67	77.22
12	MHMP-12	72.83	86.00	85.00	81.28
13	MPMP-13	71.33	83.00	85.33	79.89
14	PUMP-14	75.33	81.67	85.00	80.67
15	UPMP-15	69.33	81.67	85.00	78.67
16	UPMP-16	72.67	83.67	74.00	76.78
17	UPMP-17	74.00	83.00	85.67	80.89
18	UKMP-18	74.67	83.33	84.50	80.83
19	WBMP-19	72.00	82.00	79.17	77.72
20	HPMP-20	64.83	84.67	89.67	79.72
Mean		71.36	83.18	84.43	79.66
		Isolates (A)	Light (B)	Interaction (AXB)	
S.Em ±		2.19	0.85	3.80	
CD @ 1 %		5.75	2.22	9.96	

*CL-Continuous Light

CD-Continuous Dark

A.L/D- Alternate Dark and Light

Table 9b. Effect of light on microsclerotial production by *M. phaseolina* isolates

Sl. No.	Isolates	Light, number of microsclerotia/ microscopic field and days taken						Mean of microsclerotia
		CL*	Days	CD*	Days	A.L/D*	days	
1	KAMP-1	20.33	2	23.00	3	42.33	2	28.56
2	KAMP-2	35.00	3	35.67	3	48.67	2	39.78
3	KAMP-3	27.33	2	33.00	3	38.33	2	32.89
4	KAMP-4	25.67	2	26.67	3	53.00	2	35.11
5	KAMP-5	24.00	3	35.33	3	66.67	2	42.00
6	APMP-6	11.67	3	11.00	4	63.33	2	28.67
7	APMP-7	25.00	3	33.67	3	34.00	2	30.89
8	APMP-8	31.67	2	36.67	2	50.00	2	39.44
9	APMP-9	15.00	3	16.33	4	43.33	3	24.89
10	TNMP-10	14.67	3	17.33	3	46.67	3	26.22
11	TNMP-11	16.67	3	10.67	4	35.67	2	21.00
12	MHMP-12	5.67	4	26.00	3	67.67	2	33.11
13	MPMP-13	41.33	3	36.33	3	35.67	2	37.78
14	PUMP-14	13.67	3	26.00	3	31.67	3	23.78
15	UPMP-15	13.67	3	16.67	3	50.00	2	26.78
16	UPMP-16	16.00	3	21.33	3	25.67	2	21.00
17	UPMP-17	23.33	3	38.33	3	46.33	2	36.00
18	UKMP-18	6.00	4	26.67	3	47.33	2	26.67
19	WBMP-19	17.33	3	32.33	3	40.67	3	30.11
20	HPMP-20	14.33	4	34.33	4	27.67	2	25.44
Mean		19.92	3.15	26.87	2.95	44.73	2.2	30.51
		Isolates (A)		Light (B)		Interaction (AXB)		
S.Em ±		3.30		1.28		5.72		
CD @ 1 %		8.65		3.35		14.98		

*CL-Continuous Light

CD-Continuous Dark

A.L/D- Alternate Dark and Light

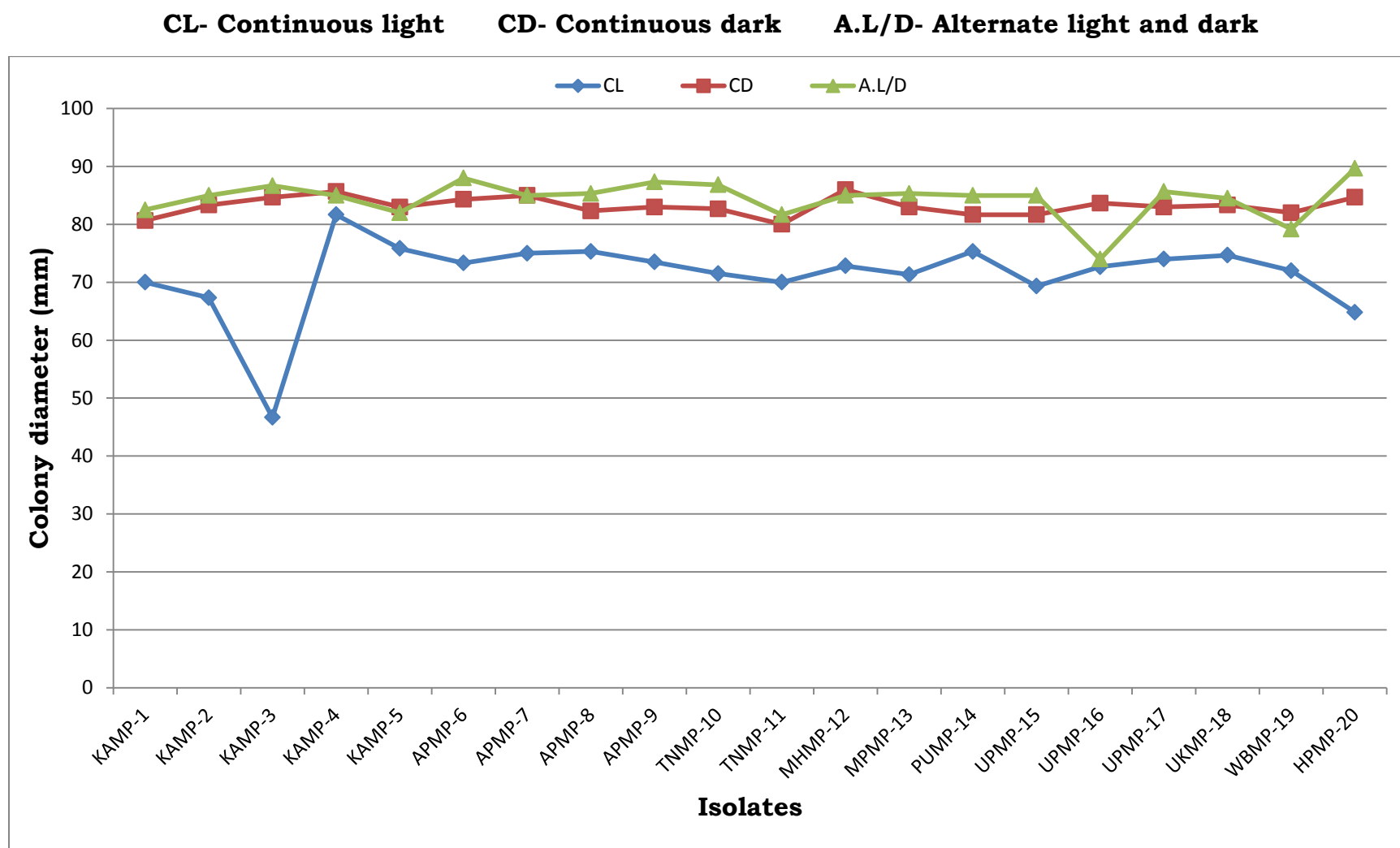


Fig. 6. Effect of light on radial growth of *M. phaseolina* isolates

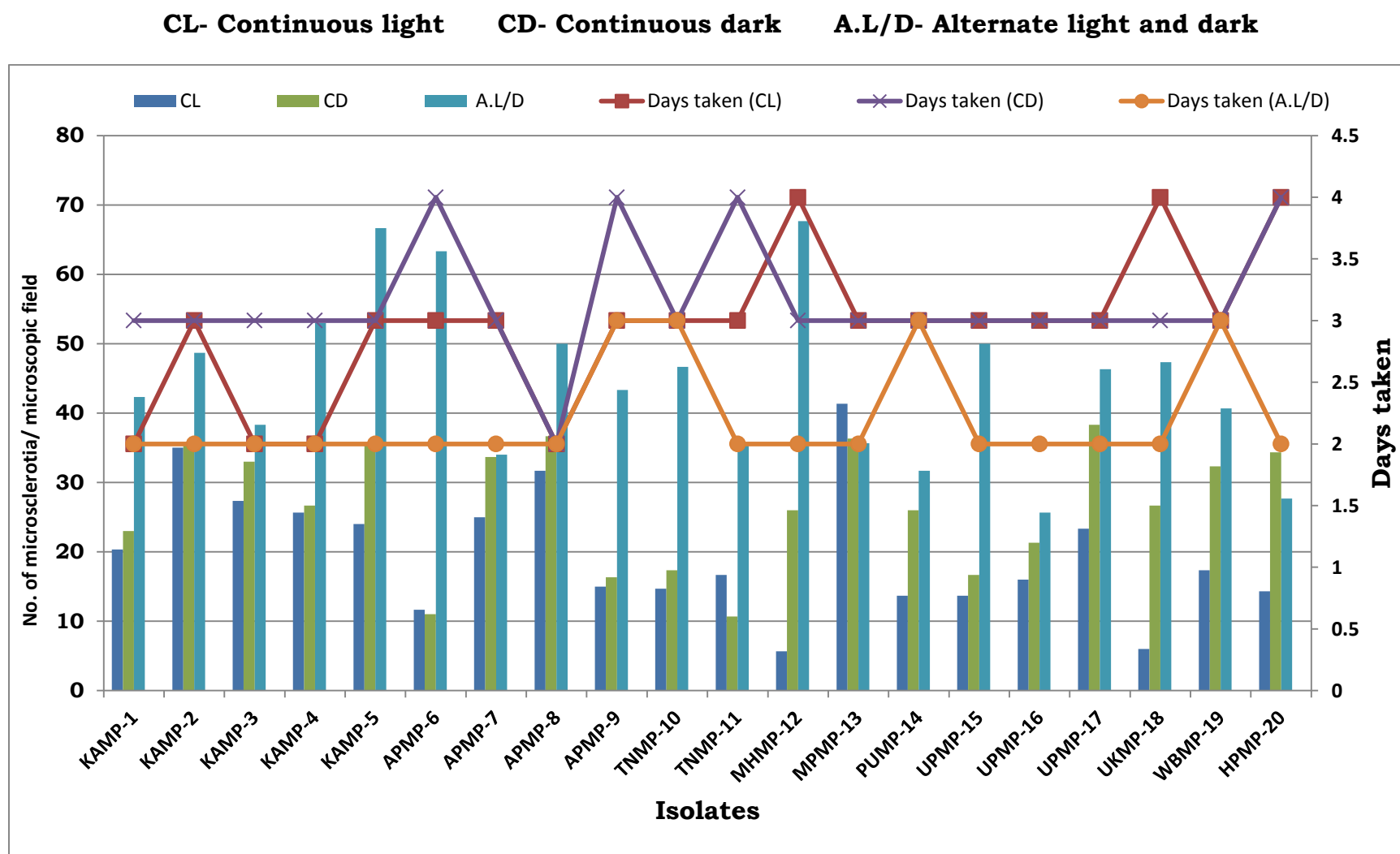


Fig. 7. Effect of light on microsclerotia production and days taken by *M. Phaseolina* isolates

4.6 Molecular characterization of *M. phaseolina* isolates

Twenty *M. phaseolina* isolates collected from different locations of India were used to characterize, with respect to molecular identification of isolates by using species specific primers and studying the molecular variability among the isolates by using PCR-RAPD and ITS sequence.

4.6.1 Molecular identification of *M. phaseolina* isolates

The total genomic DNA of twenty *M. phaseolina* isolates were amplified using species specific primers MpKFI (5' CCGCCAGAGGACTATCAAAC-3') and MpKRI (5' CGTCCGAAGCGAGGTGTATT-3') with a PCR. The primers yielded single amplified product of 350 bp with all the isolates tested. The amplification of this band confirms that all the twenty isolates belonged to *M. phaseolina* (Plate 8).

4.6.2 Molecular variability among *M. phaseolina* isolates

Molecular variability of *M. phaseolina* isolates was studied by using PCR-RAPD and ITS sequence

4.6.2.1 PCR-RAPD (Restriction Amplified Polymorphic DNA)

Genetic variation was detected among the 20 isolates of *M. phaseolina* using 16 RAPD markers, which produced scorable amplicons (Plate 9). Out of the 16 primers, OPA-1, OPA-5 and OPA-9 showed 100 per cent polymorphism among isolates. OPA-2, OPA-3, OPA-12, OPA-15 and OPA-17 showed more than 80 per cent polymorphism among isolates, remaining primers viz., OPA-4, OPA-7, OPA-8, OPA-10, OPA-13, OPA-14, OPA-16 and OPA-20 showed more than 50 per cent polymorphism. A total of 132 amplicon levels resulted from 16 primers and were available for analysis (Table 10). Information on banding pattern for all the primers were used to determine genetic distance

Table 10. Scorable bands generated with 16 RAPD primers in 20 *M. phaseolina* isolates

Sl. No.	Primer	Total number of bands amplified	Number of polymorphic bands	Per cent polymorphism
1	OPA-1	5	5	100.00
2	OPA-2	5	4	80.00
3	OPA-3	6	5	83.33
4	OPA-4	8	6	75.00
5	OPA-5	12	12	100.00
6	OPA-7	8	6	75.00
7	OPA-8	8	6	75.00
8	OPA-9	9	9	100.00
9	OPA-10	7	5	71.43
10	OPA-12	11	10	90.91
11	OPA-13	8	6	75.00
12	OPA-14	8	5	62.50
13	OPA-15	9	8	88.89
14	OPA-16	9	7	77.78
15	OPA-17	8	7	87.50
16	OPA-20	11	6	54.55
	Total	132		

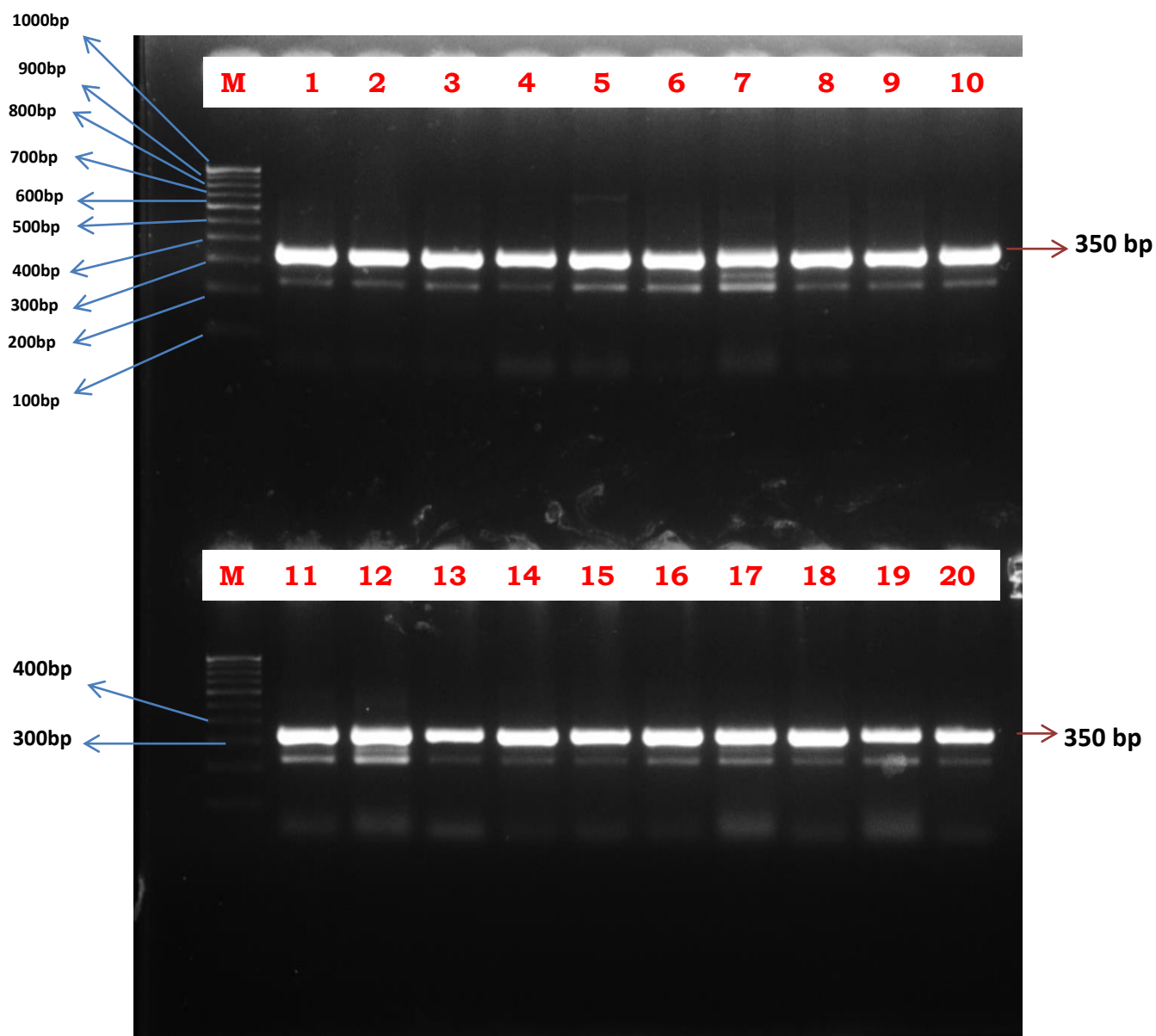


Plate 8. PCR amplification of *M. phaseolina* with species specific primers MpKF1 and MpKR1

M=100 bp ladder , 1-20 *M. phaseolina* isolates

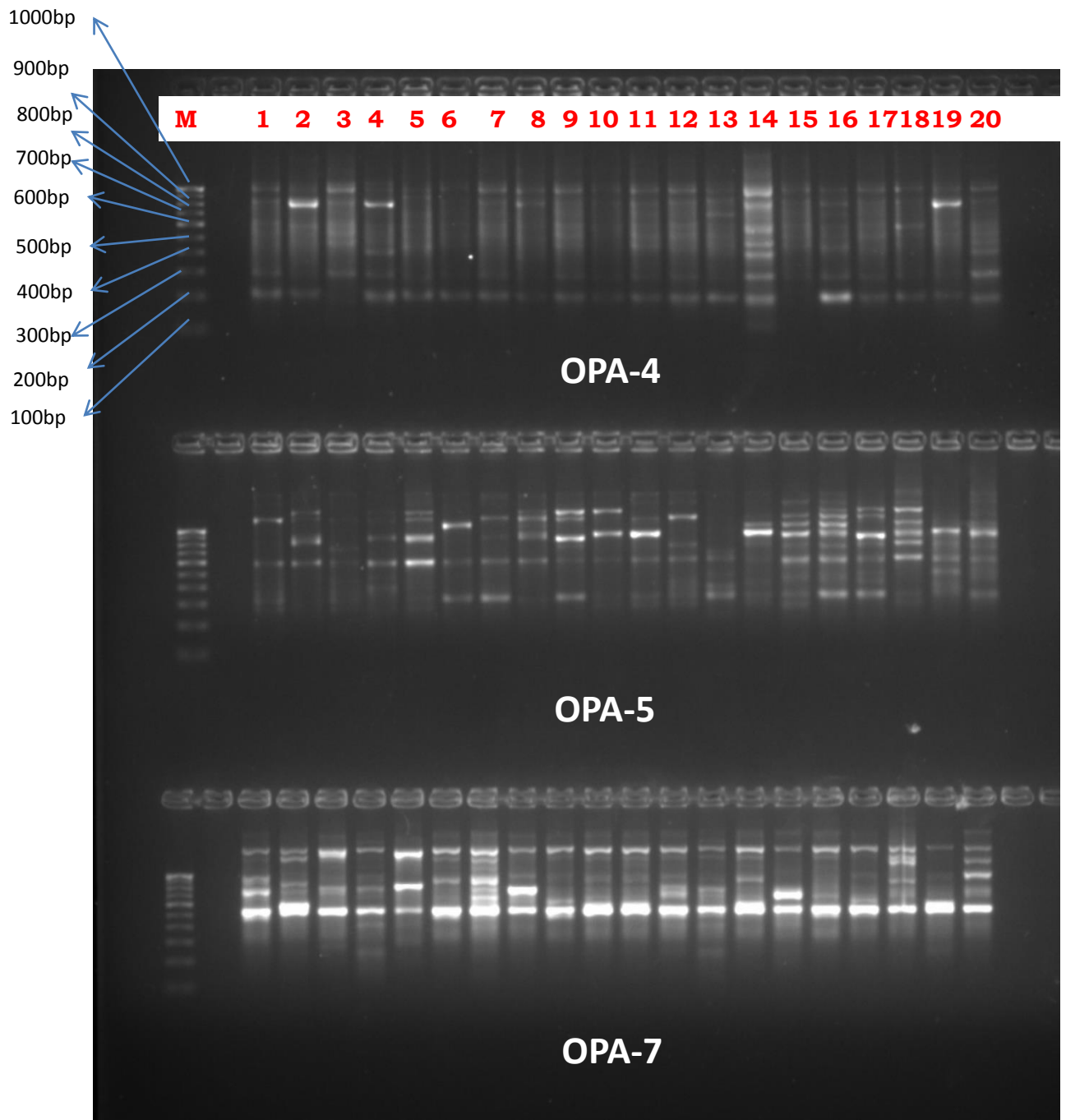


Plate 9. Banding pattern of Random amplified polymorphic DNA (RAPD) obtained from 20 *M. phaseolina* isolates

M=100 bp ladder , 1-20 *M. phaseolina* isolates

between isolates and to construct a dendrogram. Based on simple matching co-efficient a genetic similarity matrix was constructed to assess the genetic relatedness among the isolates of *M. phaseolina*. Genetic similarity based on RAPD analysis of 20 isolates of *M. phaseolina* is given in Table 11. Similarity coefficient ranged from 61 to 86 per cent. The maximum genetic similarity of 86 was between APMP-6 and APMP-7 between MHMP-12 and APMP-7 whereas least genetic similarity (61 per cent) was observed between UPMP-15 and KAMP-1. TNMP-11 and KAMP-4. The suitability of RAPD used was to detect the variations among the isolates of *M. phaseolina*.

In the present investigation, 16 OPA series primers were used to determine genetic distance between isolates and to construct a dendrogram. The dendrogram by RAPD data revealed that, the twenty isolates were differentiated into two major clusters A and B. The cluster A includes four isolates such as KAMP-1, MAMP-13, WBMP-19 and KAMP-4 and the remaining sixteen isolates *viz.*, KAMP-2, KAMP-3, KAMP-5, APMP-6, APMP-7, APMP-8, APMP-9, TNMP-10, TNMP-11, MHMP-12, PUMP-14, UPMP-15, UPMP-16, UPMP-17, UKMP-18, HPMP-20 form cluster B (Fig 8).

4.6.2.2 Molecular variability among *M. phaseolina* isolates by using Internal Transcribed Spacer (ITS) region

ITS region of rDNA was amplified using ITS1 and ITS4 primers for sequencing of its 18S rDNA region. The PCR amplified 18S rDNA region using ITS primers produced a band size of about 600bp (Plate 10). The sequence obtained upon sequencing of the ITS region from 20 different isolates were subjected to homology study using Mega 6 software along with sequences downloaded from NCBI database.

Table 11. Similarity coefficients of *M. phaseolina* isolates based on RAPD analysis

	KAMP-1	KAMP-2	KAMP-3	KAMP-4	KAMP-5	APMP-6	APMP-7	APMP-8	APMP-9	TNMP-10	TNMP-11	MHMP-12	MPMP-13	PUMP-14	UPMP-15	UPMP-16	UPMP-17	UKMP-18	WBMP-19	HPMP-20
KAMP-1	1.00																			
KAMP-2	0.80	1.00																		
KAMP-3	0.80	0.80	1.00																	
KAMP-4	0.71	0.69	0.70	1.00																
KAMP-5	0.72	0.82	0.77	0.72	1.00															
APMP-6	0.78	0.76	0.81	0.69	0.83	1.00														
APMP-7	0.78	0.76	0.84	0.69	0.79	0.86	1.00													
APMP-8	0.67	0.75	0.71	0.67	0.77	0.75	0.75	1.00												
APMP-9	0.68	0.78	0.76	0.64	0.83	0.80	0.81	0.76	1.00											
TNMP-10	0.69	0.73	0.75	0.64	0.74	0.76	0.76	0.72	0.78	1.00										
TNMP-11	0.71	0.77	0.77	0.61	0.77	0.84	0.83	0.70	0.82	0.81	1.00									
MHMP-12	0.77	0.80	0.83	0.67	0.78	0.83	0.86	0.76	0.79	0.78	0.85	1.00								
MPMP-13	0.73	0.70	0.73	0.75	0.70	0.73	0.73	0.66	0.69	0.74	0.73	0.75	1.00							
PUMP-14	0.69	0.74	0.73	0.66	0.71	0.77	0.77	0.75	0.73	0.73	0.78	0.83	0.74	1.00						
UPMP-15	0.61	0.71	0.69	0.66	0.74	0.73	0.71	0.72	0.77	0.67	0.75	0.73	0.67	0.74	1.00					
UPMP-16	0.70	0.77	0.76	0.68	0.75	0.78	0.78	0.76	0.79	0.77	0.82	0.83	0.77	0.84	0.81	1.00				
UPMP-17	0.67	0.73	0.77	0.65	0.78	0.72	0.75	0.71	0.82	0.78	0.74	0.74	0.77	0.75	0.78	0.82	1.00			
UKMP-18	0.67	0.76	0.70	0.66	0.79	0.70	0.73	0.69	0.75	0.70	0.70	0.75	0.65	0.67	0.73	0.75	0.75	1.00		
WBMP-19	0.74	0.73	0.73	0.68	0.67	0.69	0.64	0.64	0.65	0.69	0.71	0.73	0.75	0.70	0.67	0.74	0.70	0.67	1.00	
HPMP-20	0.66	0.73	0.67	0.63	0.74	0.73	0.74	0.67	0.72	0.68	0.75	0.78	0.68	0.76	0.76	0.80	0.73	0.77	0.73	1.00

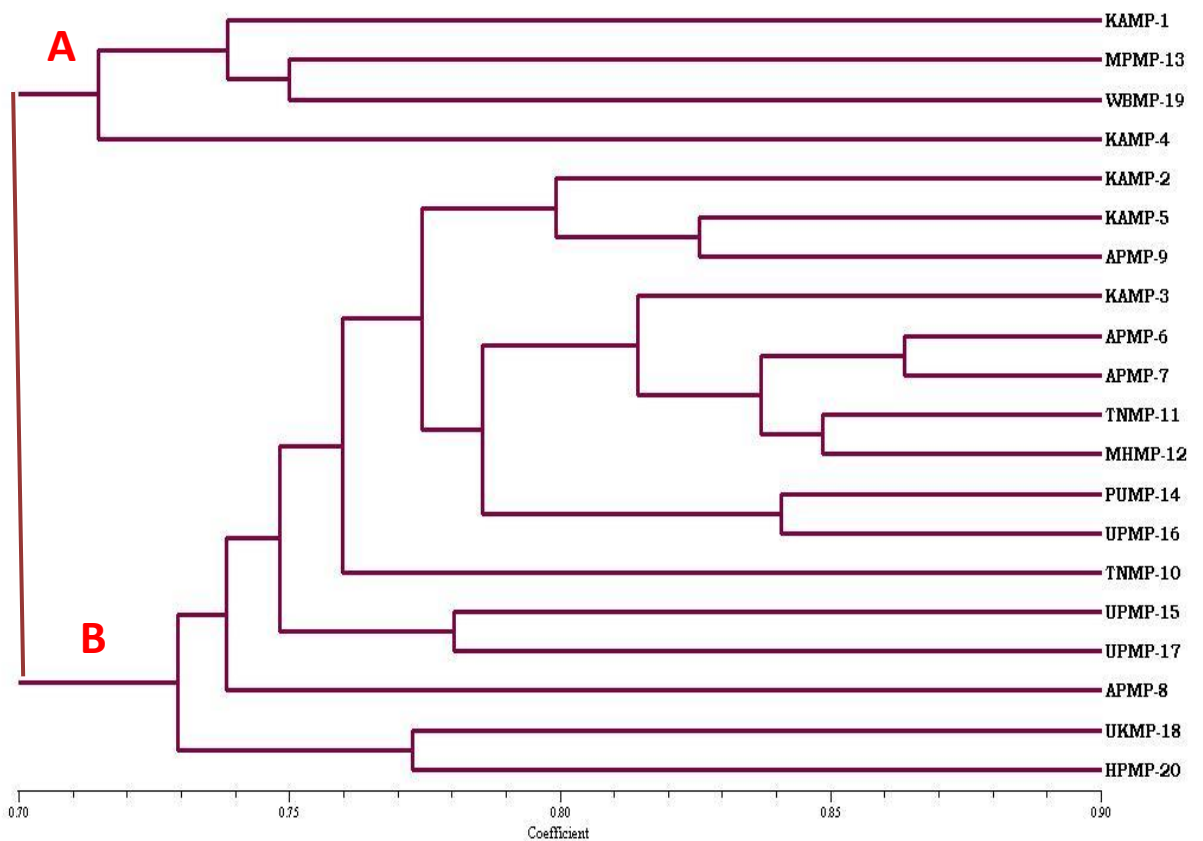


Fig. 8. Dendrogram constructed with UPGMA clustering method among 20 isolates of *M. phaseolina* based on polymorphism

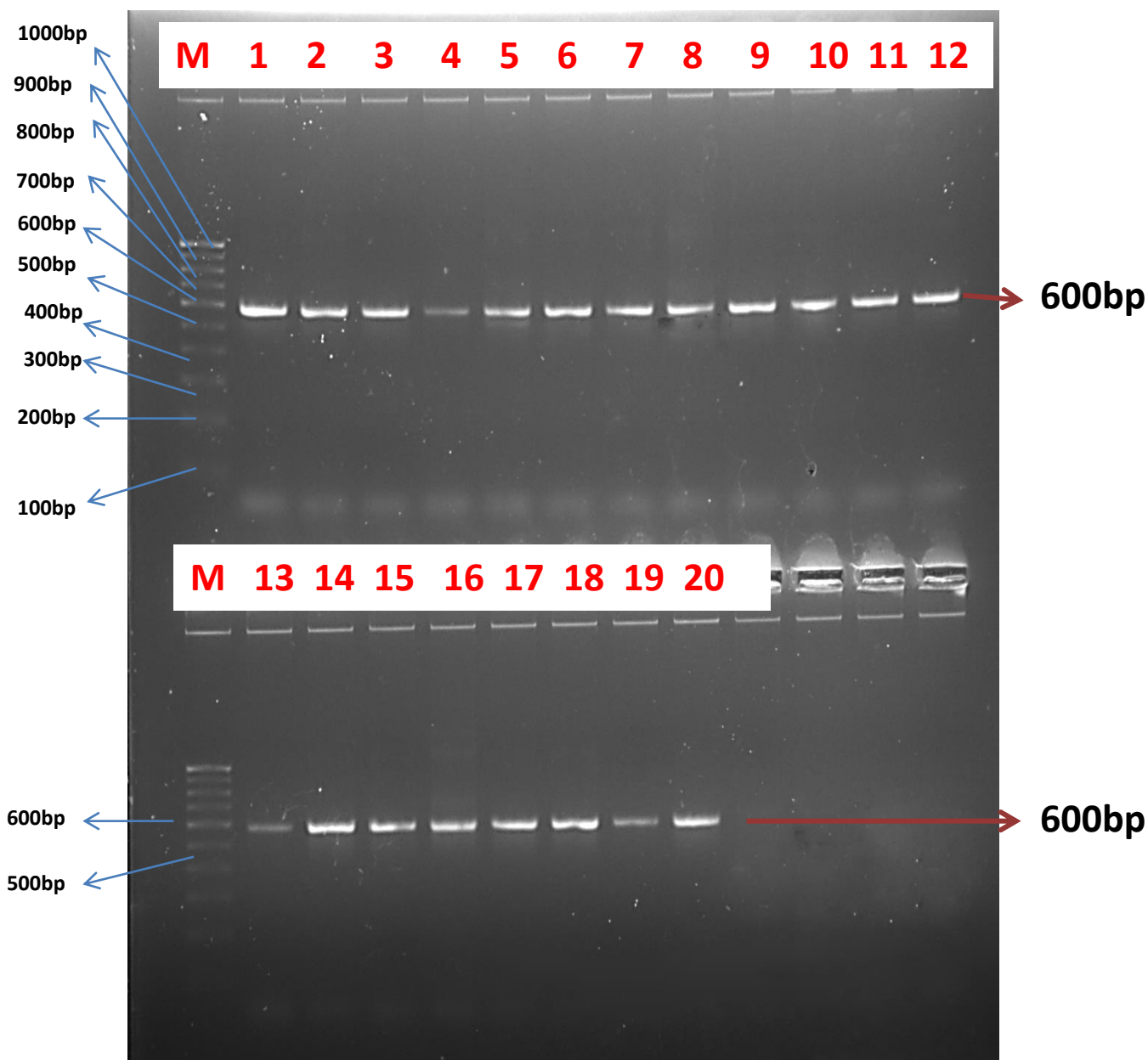


Plate10. PCR amplification of *M. phaseolina* isolates with ITS 1 and ITS 4 primers

M=100 bp ladder , 1-20 *M. phaseolina* isolates

The homology studies involving isolates of *M. phaseolina* revealed that the partial sequence of ITS1- 5.8S-ITS4 rRNA gene is as variable as rDNA regions. The sequence information was then analysed through BLASTn program which indicated that the sequences contain the genetic information of internal transcribed spacer region of rDNA gene of *M. phaseolina* with nearly 100% similarity.

Multiple sequence alignment revealed that there were quite a number of gaps introduced in the alignment within the ITS region which were closely related. Multiple and pair wise sequence alignment were generated and used to calculate evolutionary distances and percent homology of sequence and to construct a phylogenetic tree.

The phylogenetic tree constructed by using ITS sequence from 20 different *M. phaseolina* isolates along with 11 sequences of *M. phaseolina*, one *Rhizoctonia solani*, one *Athelia rolfsii* sequence obtained from NCBI gene Bank Data base generated a phylogenetic tree with two clusters A and B. All 31 sequences of *M. phaseolina* were grouped in cluster A with two sub clusters. The only one isolate KAMP-2 diverged to form single subcluster, whereas all isolates formed another sub cluster. While two sequences of *R. solani* and *A. rolfsii* were grouped in cluster B (Fig 9) which indicated that all the isolates tested were identified as *M. phaseolina* (Table 12).

All the isolates showed about 97 to 100 per cent similarity with already deposited *M. phaseolina* sequences in NCBI data base whereas, the lowest similarity (32 to 37 %) was observed between the isolates of other genus plant pathogens such as *Rhizoctonia solani* and *Athelia rolfsii*. Among the 20 isolates used in the present study the highest similarity (100 per cent) was observed between most of the isolates while the lowest similarity (94 per cent) was observed between the isolates KAMP-2 and PUMP-14 (Table 12).

Table 12. Nucleotide identities for ITS region sequences of *M. phaseolina* isolates with other selected sequences from GenBank

[illegible]

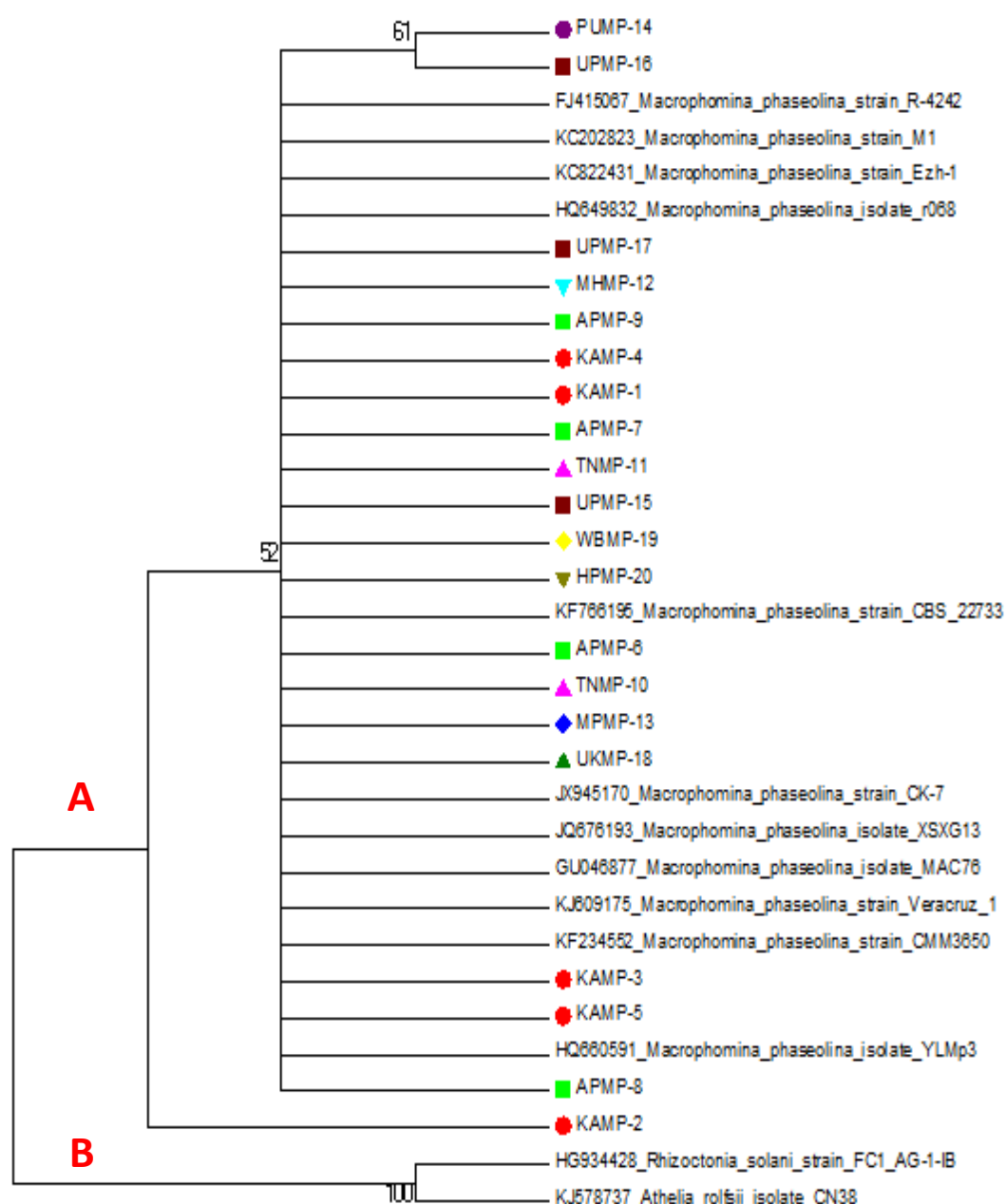


Fig. 9. Phylogenetic tree showing relationships among the *M. phaseolina* isolates based on their ITS sequences.

4.7 Evaluation of chickpea genotypes for identification of host plant resistance

Different chickpea entries were screened for dry root rot disease in dry root rot sick plot maintained at All India Co-ordinated Research Project on chickpea, ZARS, GKVK, Bengaluru during 2012-13 *Rabi* and 2013-14 *Rabi* seasons (Plate 11). *In vitro* screening was done in blotter paper method as described in Material and methods (Plate 12 & 13). The observations were recorded at around flowering to pod formation stage. The entries were categorized into disease reaction categories based on the disease incidence.

4.7.1 Screening of desi entries for dry root rot

One hundred and thirteen desi entries were screened for dry root rot under field conditions and blotter method. The results obtained are presented in Table 13a.

Field screening

Among one hundred and thirteen desi entries screened for dry root rot, none of the entry showed resistant reaction, whereas nine entries *viz.*, GNG 1581, PBG 5, JAKI 9218, CSJ 697, IPC 08-69, RSG 888 (2), RSG 931, Vijay and RKG 160 showed moderately resistant reactions with 11-20 per cent disease incidence, remaining twenty four entries showed moderately susceptible reaction, fifty six entries showed susceptible reaction and twenty three entries showed highly susceptible reaction (Table 13b).

Blotter paper method

Among one hundred and thirteen desi entries screened against dry root rot *in vitro* none of the entries showed resistant reaction, while one entry PBG 5 showed moderate resistant reaction with a score of 3,

Table 13a. Screening of desi entries for dry root rot under field and blotter method (2012-13 Rabi)

Sl. No.	Entries	Dry root rot			
		Field		Blotter paper method	
		Disease incidence (%)	Disease reaction	Disease score	Disease reaction
1.	GNG 1581	14.76	MR	5	MS
2.	GJG 0809	30.51	S	7	S
3.	PBG 5	16.27	MR	3	MR
4.	GNG 469	32.05	S	5	MS
5.	GL 27104	24.76	MS	7	S
6.	GPF 2	30.13	S	7	S
7.	GNG 1958	25.00	MS	5	MS
8.	Vishal	30.56	S	7	S
9.	PhuleG 0204-4	34.06	S	9	HS
10.	GCP 101	42.56	S	7	S
11.	GJG 0906	27.69	MS	7	S
12.	RSG 888 (1)	57.58	HS	9	HS
13.	GJG 0907	35.49	S	7	S
14.	JAKI 9218	18.38	MR	9	HS
15.	NDG 11-11	75.21	HS	7	S
16.	BGD 1073	47.62	S	9	HS
17.	NBeG 165	54.58	HS	9	HS
18.	NDG 11-12	38.48	S	9	HS
19.	PG 0120	29.66	MS	7	S
20.	RKG 201-95	75.00	HS	7	S
21.	CSJ 563	49.41	S	5	MS
22.	BGD 1074	31.88	S	7	S
23.	RKG 202-22	57.73	HS	5	MS
24.	CSJ 724	58.18	HS	9	HS
25.	H 09-96	60.39	HS	9	HS
26.	Phule G 0302-10	45.83	S	7	S
27.	GNG 2124	51.46	HS	7	S
28.	IPC 08-57	36.67	S	7	S
29.	PhuleG 08108	40.57	S	9	HS
30.	GNG 2171	43.67	S	7	S
31.	GJG 1001	51.59	HS	7	S
32.	IPC 06-126	58.08	HS	5	MS
33.	GL 28295	31.67	S	9	HS
34.	AKG 1108	47.22	S	5	MS
35.	GL 28297	44.45	S	7	S
36.	JG 29	58.08	HS	7	S

Sl. No.	Entries	Dry root rot			
		Field		Blotter paper method	
		Disease incidence (%)	Disease reaction	Disease score	Disease reaction
37.	GJG 1010	43.08	HS	7	S
38.	BG 3033	37.73	S	9	HS
39.	DKG 972	25.16	MS	7	S
40.	JG 30	29.64	MS	9	HS
41.	BG 3034	40.59	S	7	S
42.	DKG 1030	28.14	MS	7	S
43.	H 09-90	33.97	S	9	HS
44.	CSJ 697	14.97	MR	5	MS
45.	NDG 11-5	81.82	HS	7	S
46.	BGD 1071	32.64	S	9	HS
47.	GJG 0921	31.37	S	7	S
48.	IPC 06-77	37.65	S	7	S
49.	H 08-75	20.26	MS	7	S
50.	BG 3032	35.00	S	9	HS
51.	Phule G 0204-16	39.17	S	5	MS
52.	GJG 0922	41.63	S	9	HS
53.	JG 14	38.14	S	5	MS
54.	NDG 11-21	22.50	MS	7	S
55.	GL 29390	26.04	MS	7	S
56.	IPC 07-13	32.38	S	9	HS
57.	NDG 11-24	70.00	HS	7	S
58.	BG 3037	41.67	S	9	HS
59.	PG 0100	27.19	MS	7	S
60.	CSJ 513	42.48	S	7	S
61.	H 09-53	20.51	MS	7	S
62.	IPC 08-69	46.97	S	5	MS
63.	CSJ 692	11.11	MR	7	S
64.	BG 3038	62.02	HS	9	HS
65.	GJG 1004	48.18	S	7	S
66.	RSG 963	30.48	MS	9	HS
67.	JG 31	25.00	MS	7	S
68.	NBeG 3	36.36	S	9	HS
69.	GNG 2127	50.00	S	7	S
70.	BGD 1075	36.46	S	9	HS
71.	GJG 1013	33.27	S	7	S
72.	GNG 2144	33.70	S	9	HS
73.	H 09-65	25.90	MS	9	HS
74.	PG 099	56.86	HS	9	HS
75.	GNG 1995 (R)	54.38	HS	7	S

Sl. No.	Entries	Dry root rot			
		Field		Blotter paper method	
		Disease incidence (%)	Disease reaction	Disease score	Disease reaction
76.	JG 32	63.74	HS	9	HS
77.	GL 29389	58.89	HS	9	HS
78.	RKG 207-61	44.32	S	7	S
79.	RKG 206-31	41.43	S	7	S
80.	GJG 0814	45.83	S	7	S
81.	CSJ 515 (AVT-2)	31.34	S	5	MS
82.	RSG 888 (2)	19.43	MR	7	S
83.	H 08-71	40.14	S	9	HS
84.	Vijay	57.31	HS	5	MS
85.	JG 25	39.78	S	5	MS
86.	GJG 0904	51.40	HS	5	MS
87.	RSG 931	15.77	MR	5	MS
88.	CSJ 730	43.96	S	5	MS
89.	IPC 07-09	24.40	MS	5	MS
90.	NDG 11-41	37.50	S	7	S
91.	GNG 2146	31.58	S	9	HS
92.	IPC 2009-21	28.48	MS	9	HS
93.	CSJ 739	36.41	S	7	S
94.	GJG 1012	32.78	S	7	S
95.	GL 28186	34.52	S	7	S
96.	JG 33	30.97	S	9	HS
97.	CSJ 741	28.13	MS	7	S
98.	PhuleG 0752	43.91	S	9	HS
99.	H 09-54	25.00	MS	7	S
100.	RSG 888	30.91	S	7	S
101.	GJG 1003	27.72	MS	7	S
102.	BG 3035	45.63	S	9	HS
103.	JG 34	55.90	HS	7	S
104.	RSG 931	55.13	HS	9	HS
105.	IPC 2008-76	22.98	MS	9	HS
106.	BG 3036	35.86	S	9	HS
107.	RKG11-301	39.76	S	9	HS
108.	GNG 2145	23.55	MS	9	HS
109.	Phule G 0305-3	30.30	MS	7	S
110.	H 09-23	22.21	MS	7	S
111.	NBeG 47-1	34.09	S	7	S
112.	Vijay	18.00	MR	5	MS
113.	RKG 160	12.92	MR	7	S
114.	L-550 (Check)	59.02	HS	9	HS

Table 13b. Disease reaction of desi entries for dry root rot under field conditions

Sl. No.	Reaction	Entries	No. of entries
1	Resistant (0-10%)	Nil	0
2	Moderately resistant (11-20 %)	GNG 1581, PBG 5, JAKI 9218, CSJ 697, CSJ 692, RSG 888 (2), RSG 931, Vijay and RKG 160	9
3	Moderately susceptible (21-30%)	GL 27104, GJG 0906, PG 0120, DKG 972, JG 30, DKG 1030, H 08-75, NDG 11-21, GL 29390, PG 0100, BG 3038, BGD 1075, GJG 1013, PG 099, GNG 1995 (R), IPC 07-09, IPC 2009-21, CSJ 741, H 09-54, GJG 1003, IPC 2008-76, GNG 2145, Phule G 0305-3 and H 09-23	24
4	Susceptible (31-50%)	GJG 0809, GNG 469 ,GPF 2,Vishal, Phule G 0204-4, GCP 101, GJG 0907, BGD 1073, NDG 11-12, CSJ 563, BGD 1074, Phule G 0302-10, IPC 08-57, Phule G 08108, GNG 2171, GL 28295, AKG 1108, GL 28297, BG 3033, BG 3034, BGD 1071, GJG 0921, IPC 06-77, BG 3032, Phule G 0204-16, GJG 0922, JG 14, IPC 07-13, BG 3037, CSJ 513, IPC 08-69, GJG 1004, NBeG 3, GNG 2127, BGD 1075, GJG 1013, GNG 2144, RKG 207-61, RKG 206-31, GJG 0814, CSJ 515 (AVT-2), H 08-71, JG 25, CSJ 730, NDG 11-41, GNG 2146, CSJ 739, GJG 1012, GL 28186, JG 33, Phule G 0752, RSG 888, BG 3035, BG 3036, RKG 11-301 and NBeG 47-1	57
5	Highly susceptible (51-100%)	RSG 888 (1), NDG 11-11, NBeG 165, RKG 201-95, RKG 202-22, CSJ 724, H 09-96, GNG 2124, GJG 1001, IPC 06-126, JG 29, GJG 1010, NDG 11-5, NDG 11-24, BG 3038, PG 099, GNG 1995 (R), JG 32, GL 29389, Vijay, GJG 0904, JG 34,RSG 931 and L-550	24
Total No. of genotypes			114



Plate 11. General view of dry root rot screening plot



***M. phaseolina* inoculum**



Dipping of roots in the inoculum



Arrangement of seedlings on blotter paper



Incubation

Plate 12. Screening of chickpea genotypes by blotter paper technique



Resistant (1)



Moderately resistant (3)



Moderately susceptible (5)



Susceptible (7)



Highly susceptible (9)

Plate 13. Dry root rot disease reaction under blotter paper technique (Scale 1-9)

eighteen entries *viz.*, GNG 1581, GNG 469, GNG 1958, CSJ 563, RKG 202-22, IPC 06-126, AKG 1108, CSJ 697, Phule G 0204-16, JG 14, IPC 08-69, CSJ 515 (AVT-2), Vijay, JG 25, GJG 0904, RSG 931, CSJ 730 and IPC 07-09 showed moderately susceptible reaction with a score of 5, whereas fifty five entries showed susceptible reaction (score of 7) and thirty eight entries showed highly susceptible reactions (score of 9) (Table 13c).

In both the methods of screening, none of the entry showed resistant reaction. However, nine entries *viz.*, GNG 1581, PBG 5, JAKI 9218, CSJ 697, IPC 08-69, RSG 888, RSG 931, Vijay and RKG 160 under field conditions and one entry PBG 5 in blotter method showed moderately resistant reaction, while two entries *viz.*, GNG 1958 and IPC 07-09 showed moderately susceptible reaction, twenty five entries *viz.*, GJG 0809, GPF 2, Vishal, GCP 101, GJG 0907, Phule G 0302-10, IPC 08-57, GNG 2171, GL 28297, BG 3034, GJG 0921, IPC 06-77, CSJ 513, GJG 1004, GNG 2127, GJG 1013, RKG 207-61, RKG 206-31, GJG 0814, NDG 11-41, CSJ 739, GJG 1012, GL 28186, RSG 888 (2) and NBeG 47-1 showed susceptible reaction, nine entries *viz.*, RSG 888 (1), NBeG 165, CSJ 724, H 09-96, BG 3038, PG 099, JG 32, GL 29389 and RSG 931 showed susceptible reaction in both field and blotter methods.

4.7.2 Screening of Kabuli entries against dry root rot

Sixty one Kabuli entries were screened against dry root rot in sick plot and blotter method and the results are presented in Table 14a.

Field Screening

Among sixty one entries screened in field none of the entries showed resistant reactions. However, six entries *viz.*, Phule G 09316, GNG 2182, CSJK 72, IPCK 2008-108, Phule G 10404 and IPCK 2008-109 showed moderately resistant reaction with 11-20 per cent disease

Table 13c. Disease reaction of desi entries for dry root rot under blotter paper method

Rating	Reaction	Entries	No. of entries
1	Resistant	Nil	0
3	Moderately resistant	PBG 5	1
5	Moderately susceptible	GNG 1581, GNG 469, GNG 1958, CSJ 563, RKG 202-22, IPC 06-126, AKG 1108, CSJ 697, Phule G 0204-16, JG 14, IPC 08-69, CSJ 515 (AVT-2), Vijay, JG 25, GJG 0904, RSG 931, CSJ 730 and IPC 07-09	18
7	Susceptible	GJG 0809, GL 27104, GPF 2, Vishal, GCP 101, GJG 0906, GJG 0907, NDG 11-11, PG 0120, RKG 201-95, BGD 1074, Phule G 0302-10, GNG 2124, IPC 08-57, GNG 2171, GJG 1001, GL 28297, JG 29, GJG 1010, DKG 972, BG 3034, DKG 1030, NDG 11-5, GJG 0921, IPC 06-77, H 08-75, NDG 11-21, GL 29390, NDG 11-24, PG 0100, CSJ 513, H 09-53, CSJ 692, GJG 1004, JG 31, GNG 2127, GJG 1013, GNG 1995 (R), RKG 207-61, RKG 206-31, GJG 0814, RSG 888, NDG 11-41, CSJ 739, GJG 1012, GL 28186, CSJ 741, H 09-54, RSG 888, GJG 1003, JG 34, Phule G 0305-3, H 09-23, NBeG 47-1 and RKG 160	55
9	Highly susceptible	Phule G 0204-4, RSG 888 , JAKI 9218, BGD 1073, NBeG 165, NDG 11-12, CSJ 724, H 09-96, PhuleG 08108, GL 28295, BG 3033, JG 30, H 09-90, BGD 1071, BG 3032, GJG 0922, IPC 07-13, BG 3037, BG 3038, RSG 963, NBeG 3, BGD 1075, GNG 2144, H 09-65, PG 099, JG 32, GL 29389, H 08-71, GNG 2146, IPC 2009-21, JG 33, PhuleG 0752, BG 3035, RSG 931, IPC 2008-76, BG 3036, RKG11-301, GNG 2145and L-550	40
Total No. of genotypes			114

Table 14a. Screening of Kabuli entries for dry root rot under field and blotter paper method (2012-13 Rabi)

Sl. No.	Entries	Dry root rot			
		Field		Blotter paper method	
		Disease incidence (%)	Disease reaction	Disease score	Disease reaction
1.	CSJK 54	22.25	MS	7	S
2.	BG 3026	33.99	S	9	HS
3.	BDNGK 798	45.00	S	7	S
4.	CSJK 74	28.18	MS	7	S
5.	IPCK 06-143	33.82	S	7	S
6.	HK 08-231	53.85	HS	7	S
7.	GNG 2104 (AVT 2)	67.86	HS	5	MS
8.	PhuleG 09316	11.44	MR	7	S
9.	JGK 2	27.27	MS	9	HS
10.	BG 3025	40.00	S	7	S
11.	HK 06-171	70.24	HS	7	S
12.	JGK-1	76.19	HS	9	HS
13.	CSJK 27	69.05	HS	7	S
14.	NDGK 11-31	47.92	S	7	S
15.	JGK 20	31.67	S	7	S
16.	BG 3040	51.90	HS	7	S
17.	NDGK 11-32	32.50	S	7	S
18.	HK 2	30.95	S	9	HS
19.	IPCK 2009-164	31.60	S	7	HS
20.	GNG 2182	18.21	MR	7	S
21.	NBeG 72	33.64	S	9	HS
22.	BGD 1076	46.03	S	9	HS
23.	GNG 2196	75.00	HS	5	MS
24.	Phule G 10406	53.75	HS	5	MS
25.	HK 09-219	70.24	HS	5	MS
26.	GLK 27211	20.94	MS	7	S
27.	BGD 1077	31.11	S	5	MS
28.	IPCK 2009-47	44.20	S	7	S
29.	GLK 28331	75.00	HS	7	S
30.	JGK 19	26.11	MS	9	HS

Sl. No.	Entries	Dry root rot			
		Field		Blotter paper method	
		Disease incidence (%)	Disease reaction	Disease score	Disease reaction
31.	HK 04	75.00	HS	9	HS
32.	BG 3039	30.59	S	7	S
33.	BDNG 799	33.33	S	7	S
34.	HK 09-202	28.57	MS	7	S
35.	Phule G 10306	34.19	S	7	S
36.	SKUA-C-23311	75.00	HS	9	HS
37.	JGK 18	60.00	HS	9	HS
38.	CSJK 70	40.00	S	7	S
39.	JGK 13	46.43	S	7	S
40.	GLK 28127 (AVT 2)	66.67	HS	7	S
41.	IPCK 08-120	41.67	S	9	HS
42.	BG 3022	56.25	HS	9	HS
43.	JGK 17	23.81	MS	7	S
44.	BG 3028	50.00	S	9	HS
45.	Kripa	65.93	HS	7	S
46.	CSJK 69	75.00	HS	7	S
47.	BGD 1079	37.14	S	7	S
48.	NBeG 119	24.29	MS	7	S
49.	IPCK 2008-108	12.50	MR	5	MS
50.	CSJK 72	20.24	MR	5	MS
51.	Phule G 10404	16.67	MR	5	MS
52.	JGK 21	29.29	MS	5	MS
53.	BG 3041	23.38	MS	5	MS
54.	Kripa	63.33	HS	7	S
55.	HK 09-201	66.67	HS	7	S
56.	IPCK 2008-109	17.05	MR	9	HS
57.	BG 3042	52.78	HS	7	S
58.	PKV 4	58.33	HS	9	HS
59.	HK 09-206	83.33	HS	7	S
60.	JGK 22	45.96	S	9	HS
61.	BGD 1078	43.75	S	7	S
62.	L-550(Check)	77.98	HS	9	HS

incidence, ten entries *viz.*, CSJK 54, CSJK 74, JGK 2, GLK 27211, JGK 19, HK 09-202, JGK 17, NBeG119, JGK 21 and BG 3041 showed moderately susceptible reaction (20-30 %) twenty three entries showed susceptible reaction and twenty two entries showed highly susceptible reaction (Table 14b).

In *in vitro* screening out of sixty one genotypes screened for dry root rot, none of the entries showed resistant reaction and moderate resistant reaction, while ten entries *viz.*, GNG 2104 (AVT 2), GNG 2196, Phule G 10406, HK 09-219, BGD 1077, IPCK 2008-108, CSJK 72, Phule G 10404, JGK 21 and BG 3041 showed moderately susceptible reaction with a score of 5, thirty four entries showed susceptible reaction and seventeen entries showed highly susceptible reactions (Table 14c).

Among two methods of screening, none of the entries showed resistant reaction. However, six entries *viz.*, Phule G 09316, GNG 2182, CSJK 72, IPCK 2008-108, Phule G 10404 and IPCK 2008-109 showed moderately resistant reactions in field but not in blotter methods, two entries *viz.*, JGK 21 and BG 3041 showed moderately susceptible in both field and blotter method, fourteen entries *viz.*, BDNGK 798, IPCK 06-143, BG 3025, NDGK 11-31, JGK 20, NDGK 11-32, IPCK 2009-47, BG 3039, BDNG 799, Phule G 10306, CSJK 70, JGK 13, BGD 1079 and BGD 1078 showed susceptible reaction and six entries *viz.*, JGK-1, HK 04, SKUA-C-23311, JGK 18, BG 3022 and PKV 4 showed highly susceptible reaction in both field and blotter method.

4.7.3 Screening of local germplasm and advance breeding material (which include CWRRN entries)

Forty one chickpea entries, which include ten entries of advanced breeding and germplasm material and thirty one entries from CWRRN

Table 14b. Disease reaction of Kabuli entries for dry root rot under field conditions

Sl. No.	Reaction	Entries	No. of entries
1	Resistant (0-10%)	Nil	0
2	Moderately resistant (11-20 %)	Phule G 09316, GNG 2182, CSJK 72, IPCK 2008-108, Phule G 10404 and IPCK 2008-109	6
3	Moderately susceptible (21-30%)	CSJK 54, CSJK 74, JGK 2, GLK 27211, JGK 19, HK 09-202, JGK 17, NBeG119, JGK 21 and BG 3041	10
4	Susceptible (31-50%)	BG 3026, BDNGK 798, IPCK 06-143, BG 3025, NDGK 11-31, JGK 20, NDGK 11-32, HK 2, IPCK 2009-164, NBeG 72, BGD 1076, BGD 1077, IPCK 2009-47, BG 3039, BDNG 799, Phule G 10306, CSJK 70, JGK 13, BG 3028, IPCK 08-120, BGD 1079, JGK 22 and BGD 1078,	23
5	Highly susceptible (51-100%)	HK 08-231, GNG 2104 (AVT 2), HK 06-171, JGK-1, CSJK 27, BG 3040, GNG 2196, Phule G 10406, HK 09-219, GLK 28331, HK 04, SKUA-C-23311, JGK 18, GLK 28127 (AVT 2), BG 3022, BG 3028, CSJK 69, Kripa, HK 09-201, BG 3042, PKV 4, HK 09-206 and L-550	23
Total No. of genotypes			62

Table 14c. Disease reaction of Kabuli entries for dry root rot under blotter method

Rating	Reaction	Entries	No. of entries
1	Resistant	Nil	0
3	Moderately resistant	Nil	0
5	Moderately susceptible	GNG 2104 (AVT 2), GNG 2196, Phule G 10406, HK 09-219, BGD 1077, IPCK 2008-108, CSJK 72, Phule G 10404JGK 21 and BG 3041	10
7	Susceptible	CSJK 54, BDNGK 798, CSJK 74, IPCK 06-143, HK 08-231, Phule G 09316, BG 3025, HK 06-171, CSJK 27, NDGK 11-31, JGK 20, BG 3040, NDGK 11-32, GNG 2182, GLK 27211, IPCK 2009-47, GLK 28331, BG 3039, BDNG 799, HK 09-202, Phule G 10306, CSJK 70, JGK 13, GLK 28127 (AVT 2), JGK 17, Kripa (178), CSJK 69, BGD 1079, NBeG 119, Kripa (189) HK 09-201, BG 3042, HK 09-206 and BGD 1078	34
9	Highly susceptible	BG 3026, JGK 2, JGK-1, HK 2, IPCK 2009-164, NBeG 72, BGD 1076, JGK 19, HK 04, SKUA-C-23311, JGK 18, IPCK 08-120, BG 3022, BG 3028, IPCK 2008-109, PKV 4, JGK 22 and L-550	19
Total No. of genotypes			62

(ICRISAT) were screened against dry root rot in sick plot and blotter method. The results of this experiments are presented in Table 15a.

Field Screening

Among forty one entries screened for dry root rot, two entries ICCV 04514 and ICCV 96854 showed resistant reaction with 0-10 per cent disease incidence, six entries *viz.*, ICCV 08124, ICCV 93706, ICC 4951, BBG 2, HIR 70 and HIR 60 showed moderately resistant reactions with 11-20 per cent disease incidence, seven entries *viz.*, ICCV 07306, ICCV 08125, ICCV 08321, ICCV 08323, ICCV 98505, ICC 11322 and Vishal showed moderately susceptible reaction with 21-30 per cent disease incidence, thirteen entries *viz.*, ICCV 07107, ICCV 07111, ICCV 07309, ICCV 073111, ICCV 08117, ICCV 08120, ICCV 08310, ICCV 08315, ICC 5003, A1, BBG 1, BG 256 and HIR 55 showed susceptible reaction with 31-50 per cent disease incidence, thirteen entries *viz.*, ICCV 07105, ICCV 07118, ICCV 07304, ICCV 07305, ICCV 08113, ICCV 08116, ICCV 08123, ICCV 08305, ICCV 08311, ICCV 08317, ICCV 08319, KAK 2 and JG-62 showed highly susceptible reaction with 51-100 per cent disease incidence (Table 15b).

Blotter paper method

Out of forty one entries screened under *in vitro* conditions none of the entries showed resistant and moderately resistant reaction, while two entries *viz.*, ICCV 07111 and ICCV 96854 showed moderately susceptible reaction with a score of 5, twenty one entries *viz.*, ICCV 07105, ICCV 07107, ICCV 07304, ICCV 07305, ICCV 07306, ICCV 07309, ICCV 073111, ICCV 08113, ICCV 08116, ICCV 08117, ICCV 08120, ICCV 08125, ICCV 08305, ICCV 08315, ICCV 93706, ICC 11322, ICC 4951, ICC 5003, Vishal, KAK 1 and JG-62 showed susceptible reaction with a score of 7 and eighteen entries *viz.*, ICCV 04514, ICCV 07118, ICCV 08123, ICCV 08124, ICCV 08310, ICCV 08311, ICCV 08317, ICCV

Table 15a. Screening of local germplasm and advance breeding entries for dry root rot under field and blotter paper method (2012-13 Rabi)

Sl. No.	Entries	Dry root rot			
		Field		Blotter paper method	
		Disease incidence (%)	Disease reaction	Disease score	Disease reaction
1.	ICCV 04514	3.57	R	9	HS
2.	ICCV 07105	51.67	HS	7	S
3.	ICCV 07107	33.03	S	7	S
4.	ICCV 07111	50.00	S	5	MS
5.	ICCV 07118	75.00	HS	9	HS
6.	ICCV 07304	58.33	HS	7	S
7.	ICCV 07305	58.57	HS	7	S
8.	ICCV 07306	28.57	MS	7	S
9.	ICCV 07309	33.33	S	7	S
10.	ICCV 073111	35.88	S	7	S
11.	ICCV 08113	50.00	HS	7	S
12.	ICCV 08116	57.78	HS	7	S
13.	ICCV 08117	40.18	S	7	S
14.	ICCV 08120	30.84	S	7	S
15.	ICCV 08123	62.14	HS	9	HS
16.	ICCV 08124	18.06	MR	9	HS
17.	ICCV 08125	20.83	MS	7	S
18.	ICCV 08305	62.86	HS	7	S
19.	ICCV 08310	48.33	S	9	HS
20.	ICCV 08311	87.50	HS	9	HS
21.	ICCV 08315	37.50	S	7	S
22.	ICCV 08317	58.33	HS	9	HS
23.	ICCV 08319	80.00	HS	9	HS
24.	ICCV 08321	22.62	MS	9	HS
25.	ICCV 08323	27.68	MS	9	HS
26.	ICCV 93706	18.33	MR	7	S
27.	ICCV 96854	7.14	R	5	MS
28.	ICCV 98505	26.25	MS	9	HS
29.	ICC 11322	24.29	MS	7	S
30.	ICC 4951	13.66	MR	7	S
31.	ICC 5003	35.00	S	7	S
32.	A1	49.50	S	9	HS
33.	BBG 1	41.56	S	9	HS
34.	BBG 2	17.15	MR	9	HS
35.	BG 256	48.57	S	9	HS
36.	Vishal	28.21	MS	7	S
37.	HIR 70	13.70	MR	9	HS
38.	HIR 55	33.09	S	9	HS
39.	KAK 2	65.00	HS	7	S
40.	HIR 60	16.67	MR	9	HS
41.	JG-62	53.14	HS	7	S
42.	L-550 (Check)	77.98	HS	9	HS

Table 15b. Disease reaction of local germplasm and advance breeding entries for dry root rot under field

Sl. No.	Reaction	Entries	No. of entries
1	Resistant (0-10%)	ICCV 04514 and ICCV 96854	2
2	Moderately resistant (11-20 %)	ICCV 08124, ICCV 93706 ,ICC 4951, BBG 2, HIR 70 and HIR 60	6
3	Moderately susceptible (21-30%)	ICCV 07306, ICCV 08125, ICCV 08321, ICCV 08323, ICCV 98505, ICC 11322 and vishal	7
4	Susceptible (31-50%)	ICCV 07107, ICCV 07111,ICCV 07309, ICCV 073111, ICCV 08117, ICCV 08120, ICCV 08310, ICCV 08315, ICC 5003A1, BBG 1, BG 256 and HIR 55	13
5	Highly susceptible (51-100%)	ICCV 07105, ICCV 07118, ICCV 07304, ICCV 07305, ICCV 08113, ICCV 08116, ICCV 08123, ICCV 08305, ICCV 08311, ICCV 08317, ICCV 08319 L-550, KAK 2 and JG-62	14
Total No. of genotypes			42

08319, ICCV 08321, ICCV 08323, ICCV 98505, A1, BBG 1, BBG 2, BG 256, HIR 70, HIR 55 and HIR 60 showed highly susceptible reaction with a score of 9 (Table 15c).

Among both the methods two entries *viz.*, ICCV 04514 and ICCV 96854 showed resistant in field but not in blotter method, three entries *viz.*, ICCV 08124, ICCV 93706 and ICC 4951 showed moderate resistance in field but not *in vitro*, seven entries *viz.*, ICCV 07107, ICCV 07309, ICCV 073111, ICCV 08117, ICCV 08120, ICCV 08315 and ICC 5003 found susceptible in both field and blotter method, five entries *viz.*, ICCV 07118, ICCV 08123, ICCV 08311, ICCV 08317 and ICCV 08319 showed highly susceptible reaction in both field and blotter method.

4.7.4 Screening of promising germplasm

Out of two hundred and fifteen chickpea entries including desi, Kabuli, local germplasm and advanced breeding material screened at sick plot in the year 2012-13 *Rabi* only two entries *viz.*, ICCV 04514 and ICCV 96854 were found to be resistant with a 0-10 per cent disease incidence and twenty one entries *viz.*, GNG 1581, PBG 5, JAKI 9218, CSJ 697, CSJ 692, IPC 08-69, RSG 888 (2), RSG 931, Vijay, RKG 160, GNG 2182, IPCK 2008-108, CSJK 72, Phule G 10404, IPCK 2008-109, BBG 2, HIR 70, HIR 60, ICCV 08124, ICCV 93706 and ICC 4951 showed moderate resistance with a 11-20 per cent disease incidence and all these 23 entries were once again screened for disease resistance during the year 2013-14 *Rabi* (Table 16a).

Among the twenty three entries screened with L-550 susceptible check for dry root rot, only two entries *viz.*, RSG 888 and ICCV 93706 were showed moderately resistant reaction, three entries *viz.*, PBG 5, CSJ 692 and ICCV 04514 showed susceptible reaction and eighteen entries *viz.*, GNG 1581, JAKI 9218, CSJ 697, Phule G 09316, RSG 931,

Table 15c. Disease reaction of local germplasm and advance breeding entries for dry root rot under blotter paper method

Rating	Reaction	Entries	No. of entries
1	Resistant	Nil	0
3	Moderately resistant	Nil	0
5	Moderately susceptible	ICCV 07111 and ICCV 96854	2
7	Susceptible	ICCV 07105, ICCV 07107, ICCV 07304, ICCV 07305, ICCV 07306, ICCV 07309, ICCV 073111, ICCV 08113, ICCV 08116, ICCV 08117, ICCV 08120, ICCV 08125, ICCV 08305, ICCV 08315, ICCV 93706, ICC 11322, ICC 4951 ICC 5003, Vishal, KAK 2 and JG-62	21
9	Highly susceptible	ICCV 04514, ICCV 07118, ICCV 08123, ICCV 08124, ICCV 08310, ICCV 08311, ICCV 08317, ICCV 08319, ICCV 08321, ICCV 08323, ICCV 98505, A1, BBG 1, BBG 2, BG 256, HIR 70, HIR 55, HIR 60 and L-550	20
Total No. of genotypes			42

Table 16a. Screening of promising entries for dry root rot under field 2013-14 (*Rabi*)

Sl. No.	Entries	Per cent disease (<i>Rabi</i>)		Disease reaction (<i>Rabi</i>)	
		2012-13	2013-14	2012-13	2013-14
1	GNG 1581	14.76	90.91	MR	HS
2	PBG 5	16.27	39.58	MR	S
3	JAKI 9218	18.38	81.82	MR	HS
4	CSJ 697	14.97	95.83	MR	HS
5	CSJ 692	11.11	46.15	MR	S
6	Phule G 09316	11.14	53.85	MR	HS
7	RSG 888 (2)	19.43	18.18	MR	MR
8	RSG 931	15.77	91.67	MR	HS
9	Vijay	18.00	100.00	MR	HS
10	RKG 160	12.92	71.43	MR	HS
11	GNG 2182	18.21	92.86	MR	HS
12	IPCK 2008-108	12.50	62.50	MR	HS
13	CSJK 72	20.24	53.85	MR	HS
14	Phule G 10404	16.67	100.00	MR	HS
15	IPCK 2008-109	17.05	53.33	MR	HS
16	BBG 2	17.85	72.50	MR	HS
17	HIR 70	13.70	62.50	MR	HS
18	HIR 60	16.67	68.18	MR	HS
19	ICCV 08124	18.06	73.68	MR	HS
20	ICCV 93706	18.33	15.38	MR	MR
21	ICC 4951	13.50	58.82	MR	HS
22	ICCV 04514	3.17	50.00	R	S
23	ICCV 96854	7.14	88.89	R	HS
24	L-550 (Check)	57.58	53.14	HS	HS

Vijay, RKG 160 , GNG 2182, IPCK 2008-108, CSJK 72, Phule G 10404, IPCK 2008-109, BBG 2, HIR 70, HIR 60, ICCV 08124, ICC 4951 and ICCV 96854 showed highly susceptible reaction in the year 2013-14 (*Rabi*) (Table 16b).

4.8 Management of dry root rot and root rot complex in chickpea

***Kharif* 2013-14 at GKVK**

Results of trials conducted during the year 2013-14 in *Kharif* season by employing fourteen treatments are presented in the Table 17 and Plate 14. Among them, T10: Seed Treatment (ST) with tebuconazole (2 g/kg of seed) recorded lowest root rot incidence (25.98 %) with highest yield of 666.57 kg/ha, followed by T11: ST with tebuconazole (2 g/kg) + carbendazim (1 g/kg) recorded 36.49 per cent incidence with yield of 564.43 kg/ha, and are found significantly superior when compared to control which recorded 99.04 per cent disease incidence with yield of 310.55 kg/ha.

The treatment T10: ST with tebuconazole (2 g/kg of seed) recorded the lowest collar rot, wilt and dry root rot incidence as 5.13, 9.08 and 11.77 respectively with a yield of 666.57 kg/ha, is at par with T11: ST with tebuconazole (2 g/kg) + carbendazim (1 g/kg) which recorded 8.04, 13.36 and 15.09 per cent disease incidence respectively with a yield of 564.43 kg/ha and these are found significantly superior over control which recorded highest collar rot, wilt and dry root rot incidence as 21.44, 35.73 and 42.87 respectively with a lowest yield of 310.55 kg/ha.

***Rabi* 2013-14 at GKVK**

Results obtained from management of root rot complex conducted during the year 2013-14 in *Rabi* season by imposing fourteen treatments are presented in the Table 18. Among them, T10: ST with tebuconazole (2 g/kg of seed) recorded lowest root rot complex per cent disease incidence

Table 16b. Disease reaction of promising chickpea entries for dry root rot under field condition

Sl. No.	Reaction	Entries	No. of entries
1	Resistant (0-10%)	Nil	0
2	Moderately resistant (11-20 %)	RSG 888 and ICCV 93706	2
3	Moderately susceptible (21-30%)	Nil	0
4	Susceptible (31-50%)	PBG 5 , CSJ 692 and ICCV 04514	3
5	Highly susceptible (51-100%)	GNG 1581, JAKI 9218, CSJ 697, Phule G 09316, RSG 931, Vijay, RKG 160, GNG 2182, IPCK 2008-108, CSJK 72, Phule G 10404, IPCK 2008-109, BBG 2, HIR 70, HIR 60, ICCV 08124, ICC 4951, ICCV 96854 and L-550	19
Total No. of genotypes			24

Table 17. Management of root rot complex in chickpea during *Kharif* 2013-14 at GKVK, Bangalore

Sl. No.	Treatments	Per cent disease incidence				Yield (kg/ha.)
		Collar rot	Wilt	Dry root rot	Root rot complex	
1	T1: ST* with <i>T. viridae</i> @ 5 g/kg of seed	16.31(23.67) ^{bc}	25.79(30.48) ^{bcde}	30.34(33.12) ^{cd}	72.44(58.31) ^g	368.48 ^{abcd}
2	T2: ST with <i>T. viridae</i> @ 10 g/kg of seed	13.25(21.05) ^{bc}	18.35 (25.22) ^{abcd}	28.14(31.61) ^{bcd}	59.74(50.60) ^{de}	437.55 ^{ef}
3	T3: ST with <i>P. fluorescens</i> @ 5 g/kg of seed	20.65 (26.41) ^c	34.27(35.15) ^{de}	38.03(37.98) ^d	92.95(74.57) ^h	345.32 ^{abc}
4	T4: ST with <i>P. fluorescens</i> @ 10 g/kg of seed	19.27(25.91) ^c	31.86(33.73) ^{cde}	39.57 (38.88) ^d	90.70(72.21) ^h	380.42 ^{bcde}
5	T5: Application of <i>T. viridae</i> @ 5 kg in 50 kg of FYM	15.90(23.26) ^{bc}	27.86 (31.83) ^{cde}	28.15(32.03) ^{bcd}	71.92(57.97) ^g	406.98 ^{cdef}
6	T6: Application of <i>T. viridae</i> @ 10 kg in 50 kg of FYM	10.32(18.73) ^{ab}	23.97(29.23) ^{bcde}	26.81 (30.40) ^{bcd}	61.10(51.39) ^{def}	360.38 ^{abcd}
7	T7: ST with carbendazim @ 2 g + thiram @ 1 g/kg of seed	9.68 (17.93) ^{ab}	19.81(26.14) ^{abcd}	18.32 (24.82) ^{abc}	47.81(43.73) ^c	552.25 ^g
8	T8: ST with carbendazim @ 2 g + captan @ 1 g/kg of seed	10.48(18.56) ^{ab}	20.82(26.87) ^{abcd}	21.17 (26.58) ^{abc}	52.47(46.40) ^{cd}	453.02 ^f
9	T9: ST with carbendazim + mancozeb @ 2 g/kg of seed	13.88(21.71) ^{bc}	23.18(28.59) ^{bcde}	27.60(31.43) ^{bcd}	64.66 (53.50) ^{efg}	337.54 ^{ab}
10	T10: ST with tebuconazole @ 2 g/kg	5.13 (13.05) ^a	9.08 (17.50) ^a	11.77 (19.84) ^a	25.98 (30.63) ^a	666.57 ^g
11	T11: ST with tebuconazole 2 g + carbendazim @ 1 g/kg of seed	8.04 (16.27) ^{ab}	13.36 (21.18) ^{ab}	15.09 (22.55) ^{ab}	36.49 (37.15) ^b	564.43 ^g
12	T12: ST with carboxin 37.5 % + thiram 37.5 % (Vitavax power) @ 2 g/kg seed	12.90(20.68) ^{bc}	15.41 (22.97) ^{ab}	28.52 (32.22) ^{bcd}	56.83 (48.91) ^{cde}	421.98 ^{def}
13	T13: ST with TV @ 5 g + carboxin 37.5 % + thiram 37.5 % (Vitavax power) @ 2 g/kg seed	15.04 (22.60) ^{bc}	17.70 (24.64) ^{abc}	36.34 (37.01) ^d	69.09(56.20) ^{fg}	417.92 ^{def}
14	T14: Control	21.44 (27.38) ^c	35.73 (36.54) ^c	41.87(40.26) ^d	99.04(84.35) ⁱ	310.55 ^a
	S.Em±	3.48	4.93	4.97	1.71	33.05
	CD @ 5 %	7.16	10.14	10.22	3.52	67.94
	CV	20.10	21.68	19.43	7.75	11.85

*Figures in the parenthesis are arcsine transformed values

*ST-Seed treatment

Table 18. Management of root rot complex in chickpea during Rabi 2013-14 at GKV, Bangalore

Sl. No.	Treatments	Per cent disease incidence				Yield (kg/ha.)
		Collar rot	Wilt	Dry root rot	Root rot complex	
1	T1: ST with <i>T. viridae</i> @ 5 g/kg of seed	27.82 (31.53) ^{cdef}	12.19 (20.21) ^{bc}	27.26(31.17) ^{cd}	67.27 (55.08) ^e	427.38 ^a
2	T2: ST with <i>T. viridae</i> @ 10 g/kg of seed	21.85 (27.63) ^{bcd}	9.43 (17.86) ^{abc}	21.08 (27.24) ^{cd}	52.36 (46.34) ^{cd}	468.49 ^{ab}
3	T3: ST with <i>P. fluorescens</i> @ 5 g/kg of seed	33.70 (35.40) ^{ef}	11.07(19.38) ^{abc}	35.86 (36.72) ^e	80.63 (63.86) ^g	672.38 ^b
4	T4: ST with <i>P. fluorescens</i> @ 10 g/kg of seed	30.06 (32.79) ^{def}	13.13(20.80) ^{cd}	31.77(34.19) ^d	74.96 (59.95) ^f	555.71 ^{abc}
5	T5: Application of <i>T. viridae</i> @ 5 kg in 50 kg of FYM	27.52 (31.49) ^{cdef}	12.38 (20.45) ^{bc}	24.16 (29.38) ^{cd}	64.07(53.15) ^e	452.93 ^{ab}
6	T6: Application of <i>T. viridae</i> @ 10 kg in 50 kg of FYM	23.41 (28.03) ^{bcd}	9.12 (17.11) ^{abc}	20.27 (26.38) ^{bcd}	52.81 (46.59) ^{cd}	573.49 ^{abc}
7	T7: ST with carbendazim 2 g + thiram 1 g/kg of seed	18.25(25.22) ^{abcde}	10.03(18.41) ^{abc}	16.94 (24.17) ^{abc}	45.22 (42.24) ^{bc}	466.26 ^{ab}
8	T8: ST with carbendazim 2 g + captan 1 g/kg of seed	22.00 (27.75) ^{bcd}	8.83 (17.19) ^{abc}	9.24 (16.88) ^{ab}	40.07 (39.26) ^{ab}	586.82 ^{abc}
9	T9: ST with carbendazim + mancozeb @ 2 g/kg of seed	24.00 (29.12) ^{bcd}	13.56 (21.46) ^{cd}	19.83 (26.18) ^{bcd}	57.39 (49.23) ^{de}	474.04 ^{ab}
10	T10: ST with tebuconazole 2 g/kg	13.65 (21.51) ^{abc}	6.49 (14.73) ^a	7.08 (15.19) ^a	27.23 (31.44) ^a	779.04 ^c
11	T11: ST with tebuconazole 2 g + carbendazim 1 g/kg of seed	15.59 (22.67) ^{abcd}	10.05(18.33) ^{abc}	15.05 (22.48) ^{abc}	40.69 (39.62) ^b	629.04 ^{abc}
12	T12: ST with carboxin 37.5 % + thiram 37.5 % (Vitavax power) @ 2 g/kg seed	7.08 (15.20) ^a	7.47 (15.76) ^{ab}	23.93 (28.59) ^{cd}	38.48 (38.32) ^{ab}	729.04 ^c
13	T13: ST with TV 5 g + carboxin 37.5 % + thiram 37.5 % (Vitavax power) @ 2 g/kg seed	11.32 (19.36) ^{ab}	8.44 (16.71) ^{abc}	21.40 (27.28) ^{cd}	41.16 (39.89) ^b	673.49 ^{bc}
14	T14: Control	35.41 (36.29) ^f	18.77 (25.53) ^d	38.33 (37.97) ^f	92.50 (74.08) ^h	413.49 ^a
	S.Em±	5.15	2.41	4.83	2.89	113.70
	CD @ 5 %	10.59	4.96	9.93	5.94	233.72
	CV	23.01	10.77	21.59	12.91	29.88

*Figures in the parenthesis are arcsine transformed values

*ST-Seed treatment

(27.23) with highest yield of 779.04 kg/ha, followed by T12: ST with carboxin 37.5 % + thiram 37.5 % (Vitavax power) @ 2 g/kg seed which recorded 38.48 per cent incidence with yield of 729.04 kg/ha and these treatments are found significantly superior over control with 92.50 per cent disease incidence and the lowest yield of 413.49 kg/ha.

Individually the treatment T12: ST with carboxin 37.5 % + thiram 37.5 % (Vitavax power) 2 g/kg seed) recorded the lowest collar rot per cent disease incidence of 7.08, followed by T13: ST with TV 5g+carboxin 37.5 % + thiram 37.5 % (Vitavax power) @ 2 g/kg seed which recorded 11.32 per cent disease incidence.

The lowest wilt incidence was observed in the treatment T10: ST with tebuconazole @ 2 g/kg which recorded 6.49 per cent disease incidence, followed by T12: ST with carboxin 37.5 % + thiram 37.5 % (Vitavax power) @ 2 g/kg seed with 7.47 per cent disease incidence and the least dry root rot incidence was recorded in the treatment T10: ST with tebuconazole @ 2 g/kg with 7.08 per cent disease incidence followed by T8: ST with carbendazim 2 g + captan @ 1 g/ kg of seed which recorded 9.24 per cent disease incidence.

4.8.1 Pooled data of treatments employed to manage root rot complex of chickpea at GKV over two seasons

The data obtained from two seasons at GKV was pooled and analyzed statistically to know the best treatments in both the seasons and represented in the table 19 and Fig 10.

In both the seasons the treatment T10: ST with tebuconazole @ 2 g/kg recorded lowest per cent disease incidence of 26.61, with a highest yield of 722.81 kg/ha, which is at par with T11: ST with tebuconazole 2 g + carbendazim 1 g/kg which recorded 38.59 per cent disease incidence with a yield of 596.74 kg/ha followed by T8: ST with carbendazim 2 g +

Table 19. Management of root rot complex in chickpea during 2013-14

Sl. No.	Treatments	Root rot complex (Per cent disease incidence)			Yield (kg/ha.)		
		2013-14 (<i>Khari</i> f)	2013-14 (<i>Rabi</i>)	Mean	2013-14 (<i>Khari</i> f)	2013-14 (<i>Rabi</i>)	Mean
1	T1: ST with <i>T. viridae</i> @ 5 g/kg of seed	72.44	67.27	69.85 (56.68) ^f	368.48	427.38	397.93 ^{ab}
2	T2: ST with <i>T. viridae</i> @ 10 g/kg of seed	59.74	52.36	56.05 (48.46) ^{cde}	437.55	468.49	453.02 ^{abcd}
3	T3: ST with <i>P. fluorescens</i> @ 5 g/kg of seed	92.95	80.63	86.79 (68.66) ^g	345.32	672.38	508.85 ^{abcd}
4	T4: ST with <i>P. fluorescens</i> @ 10 g/kg of seed	90.70	74.96	82.83 (65.49) ^g	380.42	555.71	468.06 ^{abcd}
5	T5: Application of <i>T. viridae</i> @ 5 kg in 50 kg of FYM	71.92	64.07	67.99 (55.52) ^{ef}	406.98	452.93	429.96 ^{abcd}
6	T6: Application of <i>T. viridae</i> @ 10 kg in 50 kg of FYM	61.10	52.81	56.95 (48.98) ^{cdef}	360.38	573.49	466.94 ^{abcd}
7	T7: ST with carbendazim 2 g + thiram 1 g/kg of seed	47.81	45.22	46.51 (42.98) ^{bc}	552.25	466.26	509.26 ^{abcd}
8	T8: ST with carbendazim 2 g + captan 1 g/kg of seed	52.47	40.07	46.27 (42.85) ^{bc}	453.02	586.82	519.92 ^{abcd}
9	T9: ST with carbendazim + mancozeb @ 2 g/kg of seed	64.66	57.39	61.03 (51.35) ^{def}	337.54	474.04	405.79 ^{abc}
10	T10: ST with tebuconazole 2 g/kg	25.98	27.23	26.61 (31.04) ^a	666.57	779.04	722.81 ^e
11	T11: ST with tebuconazole 2 g + carbendazim 1 g/kg of seed	36.49	40.69	38.59 (38.39) ^{ab}	564.43	629.04	596.74 ^{de}
12	T12: ST with carboxin 37.5 % + thiram 37.5 % (Vitavax power) @ 2 g/kg seed	56.83	38.48	47.65 (43.64) ^{bcd}	421.98	729.04	575.51 ^{cde}
13	T13: ST with TV 5 g + carboxin 37.5 % + thiram 37.5 % (Vitavax power) @ 2 g/kg seed	69.09	41.16	55.13 (47.92) ^{cde}	417.92	673.49	545.70 ^{bcd}
14	T14: Control	99.04	92.50	95.77 (78.11) ^h	310.55	413.49	362.02 ^a
	S.Em±	3.73			80.53		
	CD @ 5%	8.07			173.98		
	CV	7.08			19.95		

*Figures in the parenthesis are arcsine transformed values

*ST-Seed Treatment

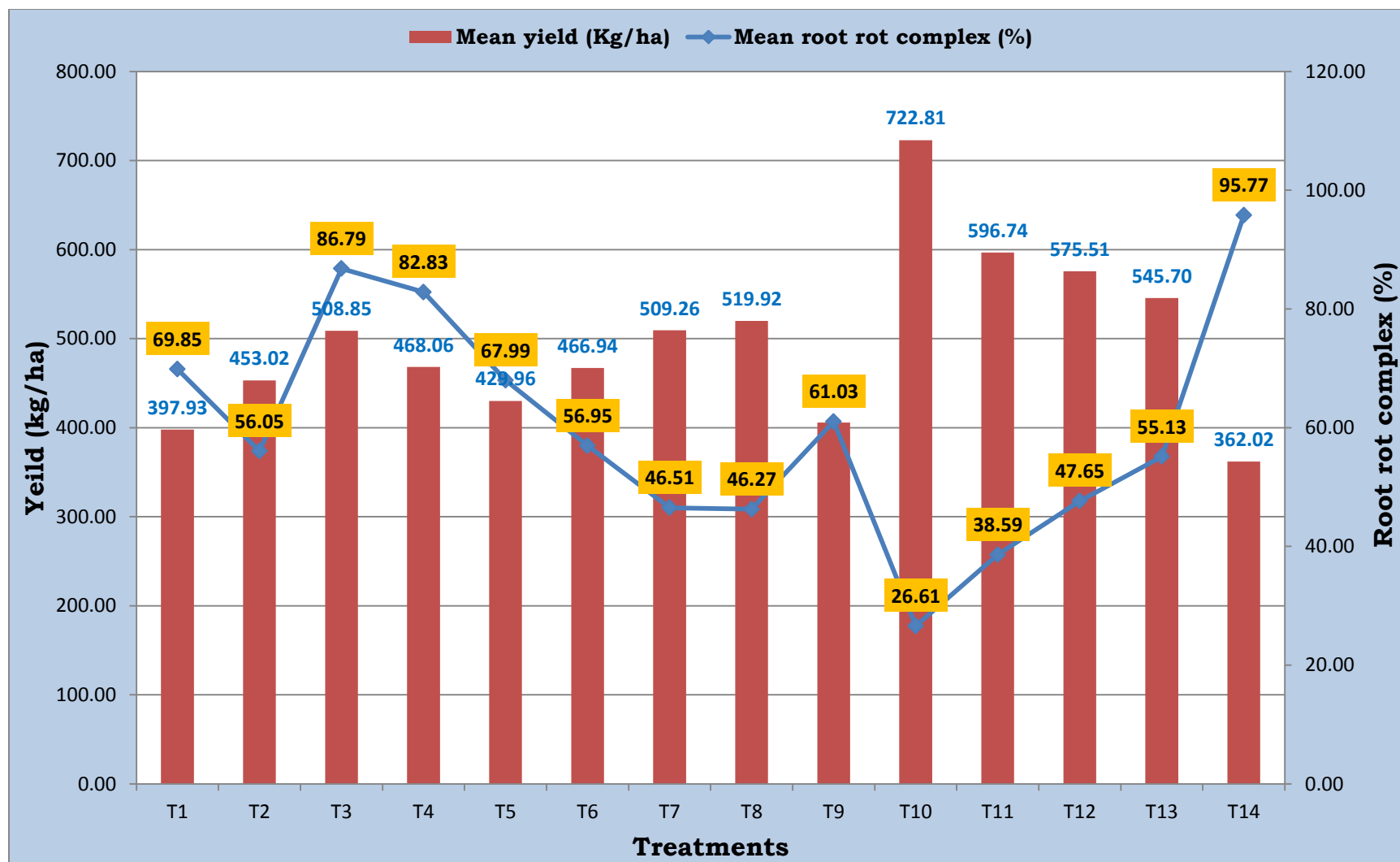


Fig. 10. Management of root rot complex in chickpea during the year 2013-14 (Pooled data)

captan 1 g/kg of seed which recorded 46.27 per cent disease incidence with a mean yield of 519.92 kg/ha. These treatments are significantly superior to control which recorded 95.77 per cent disease incidence with a lowest mean yield of 362.02 kg/ha.

4.8.2 Root rot complex management during the year 2013-14 Rabi

At Hassan

The results pertaining to management of root rot complex of chickpea conducted during the year 2013-14 *Rabi* season by employing fourteen treatments are presented in the Table 20 and Plate 15. Among the treatments employed, T10: Seed treatment with tebuconazole (2 g/kg of seed) recorded lowest root rot complex per cent disease incidence (5.57) with a yield of 2000 kg/ha, which is at par with T3: ST with *P. fluorescens* @ 5 g/kg of seed and T5: Application of *T. viride* @ 5 kg in 50 kg of FYM where the disease incidence of 5.70 and 5.93 with a yield of 2710 and 1830 kg/ha respectively were recorded followed by T2: ST with *T. viride* @ 10 g/kg of seed with a disease incidence of 7.83 with a yield of 1900 kg/ha. The treatments were found significantly superior over control with 21.80 per cent disease incidence and the lowest yield of 1520 kg/ha.

Table 20. Management of root rot complex in chickpea during Rabi 2013-14, Hassan

Sl. No.	Treatments	Root rot complex (per cent disease)	Yield (kg/ha.)
1	T1: ST with <i>T. viridae</i> @ 5 g/kg of seed	11.17 (19.50) ^d	1630 ^{ab}
2	T2: ST with <i>T. viridae</i> @ 10 g/kg of seed	7.83 (16.21) ^b	1900 ^{def}
3	T3: ST with <i>P. fluorescens</i> @ 5 g/kg of seed	5.70 (13.77) ^a	2710 ^h
4	T4: ST with <i>P. fluorescens</i> @ 10 g/kg of seed	8.97 (17.41) ^{bc}	2200 ^g
5	T5: Application of <i>T. viridae</i> @ 5 kg in 50 kg of FYM	5.93 (14.08) ^a	1830 ^{cde}
6	T6: Application of <i>T. viridae</i> @ 10 kg in 50 kg of FYM	8.30 (16.74) ^b	2080 ^f
7	T7: ST with carbendazim 2 g + thiram 1 g/kg of seed	8.76 (17.21) ^{bc}	1620 ^{ab}
8	T8: ST with carbendazim 2 g + captan 1 g/kg of seed	13.17 (21.24) ^d	1640 ^{abc}
9	T9: ST with carbendazim + mancozeb @ 2 g/kg of seed	8.92 (17.37) ^{bc}	1680 ^{abc}
10	T10: ST with tebuconazole 2 g/kg	5.57 (13.64) ^a	2000 ^{ef}
11	T11: ST with tebuconazole 2 g + carbendazim 1 g/kg of seed	10.07 (18.45) ^{cd}	1740 ^{bcd}
12	T12: ST with carboxin 37.5 % + thiram 37.5 % (Vitavax power) @ 2 g/kg seed	8.60 (17.03) ^b	2010 ^{ef}
13	T13: ST with TV 5 g + carboxin 37.5 % + thiram 37.5 % (Vitavax power) @ 2 g/kg seed	15.17 (22.91) ^e	1800 ^{bcd}
14	T14: Control	21.80 (27.81) ^f	1520 ^a
	S.Em±	0.67	94.80
	CD @ 5 %	1.38	194.86
	CV	4.54	6.16

*Figures in the parenthesis are arcsine transformed values

*ST-Seed Treatment



Plate 14. General view of chickpea root rot complex management field at GKVK



Plate 15. General view of chickpea root rot complex management field at Hassan



Discussion

V. DISCUSSION

Chickpea (*Cicer arietinum* L.) is one of the major grain legumes grown worldwide. It is a rich source of protein (20 to 25%) and also enriches soil fertility by biological nitrogen fixation (Zia-Ul-Haq *et al.*, 2012). The crop is susceptible to many diseases (Nene *et al.*, 1984) and among them dry root rot caused by *Macrophomina phaseolina* (Tassi) Goid. is one of the major limitations in chickpea production causing 10 to 20% annual loss (Dhar and Chaudhary, 2001).

The dry root rot pathogen *M. phaseolina* as presently understood, is an important soil borne fungus causing different types of diseases like root rot, charcoal rot, damping-off and stem rot *etc.*, on a wide variety of plants, over a large part of the world and under more diverse environmental conditions, than any other plant pathogenic species (Smith and Carvil, 1977).

The literature reviewed reflected that there is need to take up the investigations on certain aspects of chickpea dry root rot. Wide gaps still exist in areas relating to occurrence of variability in the pathogen and its management. Due to its economic importance a detailed investigation was carried out including collection of disease samples and isolation of pathogen from different geographical regions across India, identification of pathogen by morphological, cultural and molecular approaches and evaluation of chickpea entries for host plant resistance and finally management of disease by imposing various biological and chemical treatments. The results obtained are discussed in this chapter.

The causal organism of dry root rot pathogen was isolated from diseased plants of chickpea by following standard tissue isolation method. The pure culture obtained was maintained on potato dextrose agar and stored in refrigerator for further studies. The work was

inconformity with earlier workers Shyla (1998), Bobby and Bagyaraj (2003), Sachidananda (2005) and Manjunatha (2009) who also collected disease specimens and isolated pathogen from different locations.

Affected plants showed various types of symptoms *viz.*, yellowing, drooping, drying and shredding of leaves. The root system of infected plant was dark with extensive rotting. Roots became brittle, lateral roots were destroyed and bark was peeled off easily, because *M. phaseolina* generally affects the fibrovascular system of the roots and basal internodes, impedes the transport of nutrients and water to the upper parts of the plant, which causes progressive wilting, premature dying, loss of vigor, and reduced yield. The diseased plants could be very easily pulled out from the soil and showed discoloration of roots by the presence of black colored microsclerotia on roots. Similar symptoms were recorded by several earlier workers *viz.*, Byadgi (1979), Sinclair (1982), Yang and Owen (1982), Hoes (1985), Kolte (1985), Singh (1989), Veerendra Kumar (2004) and Manjunatha (2009), Gowdra Nagamma and Saifulla (2013).

Virulence of all the twenty isolates was analyzed in sick pot culture technique by using three susceptible chickpea entries H 09-23, L-550 and BG-212. Dry root rot symptoms were observed after 15- 30 days of sowing. Out of twenty isolates, five isolates *viz.*, KAMP-2, KAMP-3, KAMP-4, TNMP-10 and PUMP-14 caused cent per cent dry root rot incidence on all three cultivars, four isolates *viz.*, KAMP-1, MHMP-12, UPMP-16 and UPMP-17 caused > 90 per cent incidence, six isolates *viz.*, KAMP-5, APMP-6, APMP-7, MPMP-13, UKMP-18 and WBMP-19 caused > 80 per cent incidence and the remaining three isolates APMP-8, APMP-9 and TNMP-11 caused > 70 per cent incidence.

The isolates KAMP-2, KAMP-3, KAMP-4, TNMP-10 and PUMP-14 had fast growth with larger sclerotia and were found to be highly

pathogenic to all the cultivars tested, whereas the three isolates APMP-8, APMP-9 and TNMP-11 produced small sclerotia and had slow growth. Therefore, the size of sclerotia and growth were linked to the virulence of *M. phaseolina*. Similar results were reported by Hooda and Grover (1982) wherein sclerotial production and virulence or pathogenicity of isolates had direct correlation. According to Aghakhani and Dubey (2009a) the most virulent isolate (RBI from Bengaluru, Karnataka) was fast growing and produced largest sclerotia. A set of cultivars was proposed for the first time for differentiating the pathotypes of *R. bataticola* causing dry root rot of chickpea. In contrast to the present results, Fernandiz *et al.* (2006) observed that pathogenic specialization of *M. phaseolina* was related to the geographic origin of the isolates.

Symptoms consisted of seed rot, yellowing, drooping and drying of leaves, dark root system of infected plant and destruction of lateral roots due to extensive rotting. Symptoms due to dry root rot in potted plants inoculated with *M. phaseolina* were similar to that observed in the field. Re-isolation of the pathogen from infected root portion was made and pathogen cultures obtained were compared with original culture of *M. phaseolina*. The re-isolated cultures were found to be similar in the morphology with the original culture. The symptoms obtained in the present study were in conformity with those reported by Devesa *et al.* (2009) who conducted a study by seedling inoculation with *M. phaseolina* in beans.

In nature, new strains may arise by mutation, hybridization, differential cytoplasmic inheritance (Hughes, 1956), heterokaryosis (Buxton, 1954) and by parasexual life cycle (Pontecarvo, 1949). Study of pathogenic variability is essential for breeding disease resistance in a crop improvement programme.

A potential pathogen is often blessed with biodiversity within its population. Basically, variation in pathogen is desirable trait for its existence in nature. This variability among the pathogens underlies their diverse nature and ability to withstand the host environment. Variability of pathogens studied by cultural and morphological characters, physiological, molecular variability and pathogenic behavior focused the existence of variation in *M. phaseolina* isolates collected from different locations.

With reference to the cultural and morphological characters observed in the present study, the mycelium was pale white in color in the initial stages of the growth but later turned to dark brown to black as and when microsclerotial formation started. Right angled branching of the mycelium was also observed. The microsclerotia varied in shape and size with scattered growth. These observations are in line with the descriptions made by Butler (1925) for pycnidial stage *M. phaseolina* and other characters described by Ashby (1927), Manjunatha (2009) and Gowdra Nagamma and Saifulla (2013).

Variation in cultural and morphological characters observed in the first cultures nevertheless is important from the point of view of the biology of the fungus as it occurs in nature. Since it is closely linked to the question of physiological races of the pathogens, so there may be a chances of arising new races of the pathogen.

Variation was observed in all the isolates with respect to colony appearance, colony diameter, number and days taken to produce microsclerotia in culture plate, size and shape of microsclerotial bodies. However there was a negligible variation in cultural characters *viz.*, mycelia colour, growth pattern mycelial branching and there was wide variation noticed with respect to morphological characters *viz.*, size and shape of microsclerotial bodies. The rapid growth or spread of mycelia

and the abundant occurrence of sclerotia due to conducive environmental conditions may also have caused variation. This fact has been observed by earlier workers (Jimenez *et al.*, 1983, Waseer *et al.* (1990), Ali and Dennis (1992), Vishwadar and Sarabhay, 1993, Shekhar *et al.* 2006a).

Cultural and morphological characters of different isolates collected from distinct chickpea growing agro-ecological zones of India was also reported by Sharma *et al.* (2004) and results revealed that the morphological characters of *R. bataticola* vary with the isolate and age of the culture. The cultural and morphological variations in *R. bataticola* (*M. phaseolina*) in different hosts such as cowpea, sunflower, groundnut, pearl millet and bean has also been reported by several workers (Rantoo *et al.*, 1997; Atiq *et al.*, 2001; Okwulehie, 2001; Suriachandraselvan and Seetharaman, 2000; Fernandez *et al.*, 2006; Ndiaye, 2007).

The influence of temperature on the rate of growth, metabolism and morphological characters of fungi has been well documented. The temperatures of 30°C and 35°C was found best for the growth of all the twenty *M. phaseolina* isolates (90 mm). The temperature 35°C is on par with 30°C and 30°C is highly significant with the remaining temperature, where as temperatures below 30°C and above 35°C reduced the growth of isolates with least growth recorded at 40°C and 15°C in all the isolates. Temperature of 30°C favored more production of microsclerotial bodies of 46.48 per microscopic field followed by 40°C with a mean microsclerotial bodies of 32.37 per microscopic field and the least of that was observed at the temperature of 15°C (2.95). In general this is in line with other findings reporting an optimal growth rate between 30 and 35°C (Dhingra and Sinclair, 1978) but this may vary depending on the geographical region from where the fungus was isolated. Manici *et al.* (1995) concluded that the isolates from various

climatic regions in Italy grew best at temperatures close to those in the soils from which they were isolated. They also suggest that high soil temperatures may lead to the selection of hyphae, microsclerotia and spores that are adapted to better growth at higher or lower temperatures (Manici *et al.*, 1995). The results of the present study are in conformity with reports of Raut and Bhombe (1976), Singh and Mehrotra (1980) and Mishra and Sinha (1982) who also reported that the temperatures of 30 and 35°C favored maximum growth of *M. phaseolina*.

The metabolic event like transformation of a substrate into product is carried out with the help of biological catalysts *i.e.* enzymes. Various fungal enzymes have been seen to be active at a particular pH range. The pH of the medium has profound effect on the rate and amount of growth and other life process of the fungus (Lilly and Barnett, 1951).

In the present study, variation in *M. phaseolina* due to change in hydrogen ion concentration (pH) was recorded. Highest mean growth was observed in pH 5.0 with 482.33 mg mean mycelial weight closely followed by 365.50 mg dry mycelial weight at pH 6.0 indicating that the preferential range is between pH 5.0 and 6.0.

The hydrogen ion concentration of the medium and growth of the isolates are interdependent. Every organism requires optimum pH for its better growth and sporulation. Bai *et al.* (1991) reported 6.7 to 7.5 to be optimum pH for the growth of *Fusarium* sp. isolated from maize plants in Northern China.

According to the present results, the optimal pH for the pathogen is between 5.0 and 6.0. However, some of the earlier studies showed that the optimal pH ranged from 3.6 to 5.0 (Bruton *et al.*, 1995). In the present investigation pH 3.0, 7.0 and 8.0 also provided satisfactory growing condition. This is in line with the reports in which good growth

was found between pH 5.0 and 8.0 (Dhingra and Sinclair, 1978), while others concluded that the best pH value was between 4.0 and 6.0 (Nischwitz *et al.*, 2004). It can thus be concluded that the growth rate of *M. phaseolina* at different pH values depends mostly on the isolate and the conditions of origin.

Uppal *et al.* (1936) while studying variation among *M. phaseolina* observed the isolates sustaining wide range of pH between 3.4 to 6.4 and concluded that the optimum pH range varies with media composition. In support of the present investigation, Khare *et al.* (1970) observed difference in growth pattern and sclerotial size of *R. bataticola* isolates from urd bean at pH 6.5 and attributed the variation to changes in biochemical processes in the plant parts from which they were isolated.

Kulkarni (2000) has studied the effect of different pH levels on 13 isolates of *M. phaseolina* isolated from maize. He observed that the highest growth was observed at pH 7.0 and second best was at 6.5. These 13 isolates were categorized into three groups based on their dry mycelial weight.

Light is a very important signal for every living cell and since optimal adaptation to both the beneficial and harmful effects of light significantly enhances fitness of an organism. The ability to sense light is essential to recognize and anticipate conditions unfavorable for vegetative growth, such as loss of water and deprivation of nutrient on the soil surface.

Upon illumination of fungal mycelia, the cell wall is the first site of impact of photons. Given the harmful effects of light, it is not surprising that several studies revealed an altered composition of the cell wall in response to light. Transient alterations in cell wall structure were observed in *P. blakesleeana* (Herrera-Estrella and Ruiz-Herrera, 1983),

and the content of chitin in the cell wall of *A. giganteus* is doubled if the mycelium is cultivated in light as compared to darkness (Fiema, 1983).

Light has a profound influence on growth and sclerotial production of the fungus. It influences the melanization process in the mycelial cells that leads to formation of microsclerotial bodies. In agreement with these findings Nemcovic and Farkas (2001) reported that illumination of *Trichoderma harzianum* with light affects enzymes involved in biosynthesis of cell wall components: Accordingly, after 10 min of illumination, specific activity of beta-1,3- glucan synthase increases by about 130%, and specific activity of chitin synthase decreases by 50% compared to dark controls. Correspondingly, while the chitin content remains essentially constant, the beta-1,3-glucan content in illuminated mycelia is elevated by up to 50%.

In the present investigation, significantly higher mycelial growth (84.43 mm) and sclerotia production (44.73 per microscopic field) of *M. phaseolina* isolates was noticed when culture was exposed to alternate cycles of 12 h light and 12 h darkness followed by continuous dark. On the contrary continuous light recorded least growth (71.36 mm) and sclerotia production (19.92). These findings are supported by Muthukrishnan *et al.* (1995) who noticed that continuous light or light alternated with dark favored the growth of *Rhizoctonia* from pulses *in vitro* on Czapeck's Dox liquid medium. Veerendra Kumar (2004) also showed the effect of alternate light and darkness on good growth and sclerotia formation as compared to continuous light and continuous darkness.

The traditional method of identification of *M. phaseolina* is usually based on morphological characters, but due to wide variations in the phenotype of the isolates these criteria are often not reliable. Pathogenic variability among the isolates or sensitivity to chlorate are some other

characters utilized to categorize the isolates (Byadgi and Hegde, 1985; Cloud and Rupe, 1991; Khan *et al.*, 1976; Pearson *et al.*, 1987). Molecular methods involving the use of the PCR have been recently described to resolve genetic variation among the *M. phaseolina* isolates (Das *et al.*, 2006, Jana *et al.*, 2005a, b). Moreover, these techniques are laborious, time consuming, not reliable and useful for grouping rather than identification. Therefore, a rapid diagnostic test that can exactly identify and detect *M. phaseolina* isolates, both *in vitro* and *in vivo* are required.

Nuclear rDNAs, particularly in the internal transcribed spacer (ITS) region, are good targets for phylogenetic analysis in fungi (Bruns *et al.*, 1991). Oligonucleotide specific primers or probes targeting the ITS region have been demonstrated to selectively detect several agriculturally important fungi like *Trichoderma*, *Hypocrea* (Irina *et al.*, 2005), *Fusarium* (Edel *et al.*, 2000), *Verticillium* (Nazar *et al.*, 1991), *Phytophthora* (Lee, *et al.*, 1993) and *Mycosphaerella* (Johanson *et al.*, 1994) etc. However, no such markers are available for specific detection of *M. phaseolina*. Babu *et al.* (2010) designed two species-specific primers MpKFI and MpKRI from the conserved region, adjacent to ribosomal 5.8 S gene. The designed primers yielded single amplified product of 350 bp with all the *M. phaseolina* isolates tested.

In this study the twenty isolates of *M. phaseolina* were identified by using species specific primers developed by Babu *et al.* (2010) yielded single amplified product of 350 bp with all the isolates tested. The results were in conformity with Mahdizadeh *et al.* (2011) who also identified fifty two *M. phaseolina* isolates that were recovered from 24 host plant species through the 14 Iranian provinces using species-specific primers MpKFI and MpKRI.

In the present investigation, 16 OPA series primers were used to determine genetic distance between isolates and to construct a dendrogram. The dendrogram constructed by RAPD data revealed that, the twenty isolates were differentiated into two major clusters A and B and the results revealed that the isolates collected from different geographical locations showed genetic variability. While *M. phaseolina* is a generalist pathogen with clonal reproduction affinity (Anderson and Kohn, 1995), the results indicate the coexistence of different haplotypes in India. Besides the geographical dominance of a given haplotype, the possibility of a closer genetic relationship between spatially distinct haplo groups is suggested. Despite the asexual nature of this pathogen (Dhingra and Sinclair, 1978; Crous *et al.*, 2006) genetic diversity levels were responsible for cluster formation. This may indicate that genetic variability between isolates of *M. phaseolina* is due to the fusion of vegetative cells, favoring heterokaryons or parasexual recombination between nuclear genes, as suggested by previous genetic studies (Saleh *et al.*, 2010, Almeida *et al.*, 2008). Although no teleomorph for *M. phaseolina* is known, Baird *et al.* (2010) detected heterogeneity in solely asexually reproducing populations from the USA using SSR markers.

The results are also in confirmation with the studies conducted by Shekhar *et al.* (2006b) wherein they analyzed seven isolates of *M. phaseolina*, incitant of maize charcoal rot through RAPD marker for genetic diversity. They observed that the most closely related isolates were Hyderabad and Delhi with an affinity percentage of 75.5 followed by Udaipur and Bangalore isolates with 62.9 per cent similarity. Rajkumar and Kuruvinashetti (2007) studied the genetic variability in ten isolates of *M. phaseolina* using PCR- RAPD markers. UPGMA clustering of data indicated that the isolates shared varied levels of genetic similarity within a range of 0.14 to 0.72 similarity coefficient index and it was suggestive that grouping of isolates was not related to sampling location in anyway.

Monga *et al.* (2007) made studies on 25 isolates of *R. bataticola* and obtained similar results. From their observations, it is amply clear that all the isolates belonging to one geographical location did not come under a single group, reflecting the fact that the variation is independent of geographical nearness of agroclimatic zone. These findings indicated a clear variation among the isolates and strongly support the possibility of existence of pathotypes in *M. phaseolina*.

Amplification of target DNA through PCR with sequence specific primers is potentially more sensitive and rapid than microbiological techniques, as a number of constraints are removed. Unlike culture, PCR does not require the presence of viable organisms for success and may be performed even when sample volumes are small. Differences in the nucleotide composition of the variable ITS region have been successfully employed to design specific primer sets that amplify DNA selectively among and within species of plant pathogens (Nazar *et al.*, 1991; Moukahmedov *et al.*, 1994; Schilling *et al.*, 1996; Moricca *et al.*, 1998). In the broader context, taxon-selective amplification of ITS regions is likely to become a common approach in molecular identification strategies. ITS regions have been used successfully to generate specific primers capable of differentiating closely related fungal species (Bryan *et al.*, 1995). These rDNA are highly stable and exhibit a mosaic of conserved and diverse regions within the genome (Hibbett, 1992). They also occur in multiple copies with up to 200 per haploid genome (Bruns *et al.*, 1991) arranged in tandem repeats with each repeat consisting of the 18s small subunit (SSU), the 5.8s and the 28s large subunit (LSU) genes.

In the present study, ITS regions of ribosomal genes for the construction of primers were used to identify and to study variability among *M. phaseolina* isolates. ITS region of rDNA was amplified using

genus specific ITS-1 and ITS-4 primers for *M. phaseolina*. Amplified products of size in the range of 600bp was produced by the primer pairs.

Amplification of DNA fragments of *M. phaseolina* with primers indicate the usefulness of molecular technique for detection, identification and variability studies. Using the primers ITS 1 and ITS 4, only a single band of 600bp was generated in the amplification pattern of all the isolates. *M. phaseolina* as first described by Pearson *et al.* (1986) suggested that isolates from one specific host are more suited to colonize it. Later, differences in pathogenicity among the isolates of soybean and sorghum have been observed (Cloud and Rupe, 1991). This has been further confirmed with isolates from soybean, sorghum and cotton (Su *et al.*, 2001).

The ITS sequence of all the isolates showed about 97 to 100 per cent similarity with already deposited *M. phaseolina* sequences in NCBI data base, the lowest similarity (32 to 37 %) was observed between the isolates of other genus plant pathogens like *Rhizoctonia solani* and *Athelia rolfsii*. Among the 20 isolates used in the present study the highest similarity (100 %) was observed between in most of the isolates while the lowest similarity (94 %) was observed between the isolates KAMP-2 and PUMP-14.

The phylogenetic tree obtained from ITS sequence of 20 *M. phaseolina* isolates along with 11 sequences of *M. phaseolina* of other hosts, one *Rhizoctonia solani*, one *Athelia rolfsii* sequence generated a tree with two clusters A and B. All 31 sequences of *M. phaseolina* were grouped in cluster A with two sub clusters. The only one isolate KAMP-2 diverged to form single sub cluster, whereas all isolates formed another sub cluster. While two sequences of *R. solani* and *A. rolfsii* were grouped in cluster B. The present findings are partially in accordance with the earlier results that the *M. phaseolina* isolates of same host were

genetically similar and differed distinctly from the isolates of other hosts (Purkayastha *et al.*, 2006; Su *et al.*, 2001). The existence of genetic diversity among the isolates from the same provinces might be due to the movement of *M. phaseolina* through infested seeds and soil (Aghakhani and Dubey, 2009b). Hence it is concluded that ITS sequence of *M. phaseolina* and bioinformatics tools can be used for both rapid identification and analysis of genetic variability among the isolates collected from different places.

Chemical control of dry root rot is not effective as *M. phaseolina* has a broad host range and survives in soil for longer periods in the form of microsclerotia. The microsclerotial bodies can survive up to 10 months even in the absence of the host plants and under prevailing dry soil conditions. Use of host plant resistance is the most economical approach for the management of dry root in chickpea. A few chickpea lines with field tolerance to dry root rot have been identified, but high levels of resistance are scarce in cultivated genotypes (Reddy *et al.*, 1991). Keeping this in view the present investigations were undertaken to evaluate performance of chickpea genotypes against dry root rot disease under sick plot condition and blotter paper technique during the year 2011-12.

Out of two hundred and fifteen chickpea entries including desi, Kabuli, local germplasm and advanced breeding material screened at sick plot in the year 2012-13 *Rabi* only two entries viz., ICCV 04514 and ICCV 96854 were found to be resistant and twenty one entries viz., GNG 1581, PBG 5, JAKI 9218, CSJ 697, CSJ 692, IPC 08-69, RSG 888 (2), RSG 931, Vijay, RKG 160, GNG 2182, IPCK 2008-108, CSJK 72, Phule G 10404, IPCK 2008-109, BBG 2, HIR 70, HIR 60, ICCV 08124, ICCV 93706 and ICC 4951 showed moderate resistant and among two hundred and fifteen entries screened in blotter paper method only one

entry PBG 5 was found to be moderately resistant. Twenty three promising entries obtained in the year 2012-13 were screened in sick plot with L-550 as checks in the year 2013-14 only two entries *viz.*, RSG 888 and ICCV 93706 were showed moderately resistant.

The only entry PBG-5 was shown moderately resistant in both sick plot and blotter paper method in the year 2012-13 also found to be susceptible in the sick plot during the year 2013-14. Since, *M. phaseolina* is having wide host range, prolonged saprophytic survival ability and the sclerotia can survive upto 10 months even in the absence of the host can make any resistant cultivar to become susceptible with in a less time. Therefore, development of resistant cultivars for dry root rot disease is becoming challenge for both breeders and pathologists.

The present findings are well supported by the field and blotter screening. Similarly, Saifulla *et al.* (2011a) screened chickpea entries for dry root rot pathogen under field condition and found that 21 entries *viz.*, GNG 1861, Phule G 07112, BGD1056(R), RSG 931, Phule G 07101, Vijay, RSG 888, GJG 0825, PG064, GNG 1947, JG 1307, GBC 6 (AVT-1), GNG 2002, GNG 1936, CSJ 313, JG 18, HIR 60 and JG 11, ICCV 07107 and ICCV 08323 were resistant and 36 entries were moderately resistant to dry root rot out of 196 genotypes. Whereas three germplasm lines showed resistant, 22 were moderately resistant, 19 susceptible and three were highly susceptible under blotter paper technique (Pande *et al.*, 2004).

Nagamma and Saifulla (2012) screened sixty four Kabuli genotypes against dry root rot, out of which six genotypes *viz.*, Phule G 04305, IPCK 07-62, RVSSG 12, HK 08-212, Phule G 09305 and AKG 2002-1K showed resistant reaction, nine genotypes *viz.*, HK 94-134, GNG 1888, HK 06-152, GNG 2112, HK 06-171, Phule G 09316, CSJK 74, JGK 13 and CSJK 70 showed moderately resistant reaction.

Similarly Nene *et al.* (1989), Oad *et al.* (1995), Mayek-Perez *et al.* (2002), Gangwar *et al.* (2002), Om Gupta and Anita Babbar (2006) and Al Shareef *et al.* (2009) also reported resistant genotypes, moderate resistant genotypes, moderate susceptible genotypes, susceptible genotypes and highly susceptible genotypes for dry root rot.

Management of soil borne diseases only with host plant resistance and cultural methods is difficult in chickpea. Thus there is an urgent need to formulate effective disease management module by using chemicals as a first barrier for plant disease development by treating the seed with chemicals while sowing. Therefore, experiments were conducted to find out the best chemical and biocontrol agents for effective management of root rot complex of chickpea caused by *Sclerotium rolfsii*, *Fusarium oxysporum* f.sp. *ciceri* and *Macrophomina phaseolina*. Since, three pathogens responsible for yield loss in the field conditions.

To manage the root rot complex of chickpea fourteen treatments were imposed during *Kharif* 2013-14. Among the treatments employed, seed treatment with tebuconazole (2 g/kg of seed) recorded significantly lowest root rot complex (25.98 %) with highest yield of 666.57 kg/ha, followed by seed treatment with tebuconazole (2 g/kg) + carbendazim (1 g/kg) recorded 36.49 per cent disease incidence with a yield of 564.43 kg/ha. Seed treatment with tebuconazole (2 g/kg of seed) which recorded the lowest color rot, wilt and dry root rot incidence as 5.13, 9.08 and 11.77 respectively, which is at par with seed treatment with tebuconazole (2 g/kg) + carbendazim (1 g/kg) with 8.04, 13.36 and 15.09 per cent disease incidence respectively.

Among the treatments employed during the year 2013-14 *Rabi*, seed treatment with tebuconazole (2 g/kg of seed) recorded significantly lowest root rot complex (27.23%) with highest yield of 779.04 kg/ha,

followed by seed treatment with carboxin 37.5 % + thiram 37.5 % (Vitavax power) 2 g/kg seed that recorded 38.48 per cent disease incidence with a yield of 729.04 kg/ha.

Individually seed treatment with carboxin 37.5 % + thiram 37.5 % (Vitavax power) @ 2 g/kg seed) recorded the lowest collar rot incidence 7.08 (%), followed by seed treatment with TV 5 g + carboxin 37.5 % + thiram 37.5 % (Vitavax power) 2 g/kg seed which recorded 11.32 % disease incidence, the lowest wilt 6.49 % disease incidence was observed in the treatment seed treatment with tebuconazole 2 g/kg, followed by T12: ST with carboxin 37.5 % + thiram 37.5 % (Vitavax power) 2 g/kg seed recorded 7.47 % disease incidence and the least dry root rot incidence was recorded in the seed treatment with tebuconazole 2 g/kg with 7.08 % disease incidence followed by T8: seed treatment with carbendazim 2 g + captan 1 g/ kg of seed which recorded 9.24 % disease incidence.

In both the seasons seed treatment with tebuconazole 2 g/kg recorded lowest disease incidence of 26.61 per cent, with highest yield of 722.81 kg/ha, followed by seed treatment with tebuconazole 2 g + carbendazim 1 g/kg which recorded 38.59 per cent disease incidence with a yield of 596.74 kg/ha. Both the disease incidence and yield were significantly higher compare to control having 95.77 per cent disease incidence with a mean yield of 362.02 kg/ha.

Management of root rot complex experiment was conducted at College of Agriculture, Hassan. Among the fourteen treatments employed seed treatment with tebuconazole (2 g/kg of seed) recorded lowest root rot complex incidence (5.57 %) with a yield of 2000 kg/ha, which is on par with seed treatment with *P. fluorescens* @ 5 g/kg of seed with 5.70 % incidence and a highest yield of 2710 kg/ha. The treatments

were significantly superior to control which recorded 21.80 % disease incidence and lowest yield of 1520 kg/ha.

It has been shown that tebuconazole ((RS)-1-(4-chlorophenyl)-4,4-dimethyl-3-(1H-1,2,4-triazole-1-ylmethyl)-pentan-3-alcohol) was highly effective against *Fusarium graminearum* (Spolti *et al.*, 2012). It is a systemic fungicide of the triazole group, and the primary mode of action is the inhibition of ergosterol biosynthesis in fungi (Hewitt, 1998). Even though different triazole fungicides have a similar mechanism of action, they may show marked differences in their activity against different fungal pathogens (Buchenauer, 1987; Scheinpflug and Kuck, 1987). This broad-spectrum, relatively new triazole fungicide is being used for its effectiveness against soil-borne and foliar fungal diseases in nut, fruit, cereal and vegetable crops worldwide (Munoz-Leoz *et al.*, 2011). Results of the present study are in agreement with Brenneman *et al.* (1991) who reported the antifungal activity against *S. rolfsii* and *R. solani* *in vitro*. Kanwal *et al.* (2012) evaluated different concentrations *viz.* 35, 70, 105 and 140 ppm of tebuconazole against three problematic soil-borne fungal phytopathogens namely *M. phaseolina*, *F. oxysporum* f. sp. *lycopersici* and *S. rolfsii* using food poisoning technique, all the concentrations of tebuconazole completely arrested the growth of all the three fungal species.

The development of biological techniques using PGPR amended with suitable bioformulations is an emerging trend in plant protection to reduce the plant diseases. Production of antibiotics *viz.*, HCN, pyrrolnitrin, phenazine and 2,4-diacetyl phloroglucinol and lytic enzymes by *P. fluorescens* against fungal pathogens were reported by many workers (Vivekananthan *et al.*, 2004; Sendhilvel *et al.*, 2005; Saravanakumar *et al.*, 2007). The above facts suggest that the inhibition of root rot pathogen, *M. phaseolina* by *P. fluorescens* may be due to the

production of antibiotics, siderophore mediated competition and lytic enzymes, *viz.*, chitinase, β -1,3-glucanase which degraded the fungal cell wall and restricted the growth of the fungus under *in vitro*. In the present investigation seed treatment with *P. fluorescens* @ 5 g/kg of seed recorded 5.70 per cent disease incidence with a highest yield of 2710 kg/ha at Hassan. Similar results were obtained by Jayasree *et al.* (2000) who investigated that the *P. fluorescens* strain pf1 effectively inhibited the mycelia growth of *M. phaseolina*, the pathogen causing dry root rot in blackgram and sesame. Among the various methods of pf1 application, seed treatment + soil application recorded maximum yield in blackgram (1238 kg/ha) and sesame (1200 kg/ha) followed by seed treatment + soil drenching of carbendazim, which recorded grain yield to the extent of 1205 kg/ha in blackgram and 820 kg/ha in sesame.

Future line of work

- Identification of stable resistant varieties by evaluating over years in sick plots
- Effect of climate change on disease development
- Identification of potential bio-control agents for management of soil borne diseases



Summary

VI. SUMMARY

Macrophomina phaseolina has long been recognized as a destructive pathogen on a wide variety of crops throughout the world. Dry root rot of chickpea caused by *M. phaseolina* once considered a minor disease has become a major problem in successful production of chickpea in South India. The present study comprised collection of pathogenic isolates from different geographical regions across India for exploring cultural, morphological and molecular diversity analysis, screening of chickpea entries to identify host plant resistance and management of dry root rot under field condition using seed treatment and soil application of chemicals and bio-agents. Results of the investigations are summarized here under.

Twenty dry root rot infected chickpea samples were collected from major chickpea growing states of India viz., Andhra Pradesh, Himachal Pradesh, Karnataka, Madhya Pradesh, Maharashtra, Punjab, Tamil Nadu, Uttarakhand, Uttar Pradesh and West Bengal during *Rabi* 2012-13.

Pure culture of dry root rot pathogen, *M. phaseolina* was obtained by tissue isolation and further pathogenicity test was done with susceptible chickpea varieties. Out of twenty isolates, five isolates KAMP-2, KAMP-3, KAMP-4, TNMP-10 and PUMP-14 were most virulent and caused cent per cent dry root rot incidence on all three the cultivars.

Based on cultural characters like growth, the isolates were categorized into three groups viz., group A (80-90 mm), group B (70-80 mm) and group C (60-70 mm). Based on colony appearance twenty isolates were categorized into three groups viz., fluffy, partially appressed and appressed. Based on colony color the isolates categorized into four

groups *viz.*, Group I-Grayish white, Group II- Blackish gray, Group III- Cottony white and Group IV-Deep black.

Based on morphological characters like mean diameter of microsclerotial bodies, the isolates were categorized into four groups *viz.*, group I (30 -50 μm), group II (50- 100 μm), group III (100-150 μm) and group IV (> 150 μm) and based on shape of microsclerotia the isolates were categorized into two groups like round and oblong.

The number of microsclerotial bodies produced per microscopic field was minimum (25.33) by isolate UPMP-16 and the isolate MHMP-12 produced maximum microsclerotial bodies (77) under 10x magnification. All the isolates took two days to form microsclerotial bodies in the culture except four isolates *viz.*, APMP-9, TNMP-10, PUMP-14 and WBMP-19 which took three days in culture after inoculation. The minimum length and breadth of microsclerotia 37.53 x36.43 μm was observed in isolate TNMP-11, while the maximum length and breadth of about 183.63x137.83 μm was noticed in KAMP-4 isolate.

The cultural characters and morphological characters suggested that there is existence of variability among the isolates.

Among physiological requirements for growth, temperatures 30°C and 35°C favored good growth and production of microsclerotial bodies of the fungus, pH 5 and 6 were found to be best for growth and alternate light and darkness favored growth and microsclerotial production.

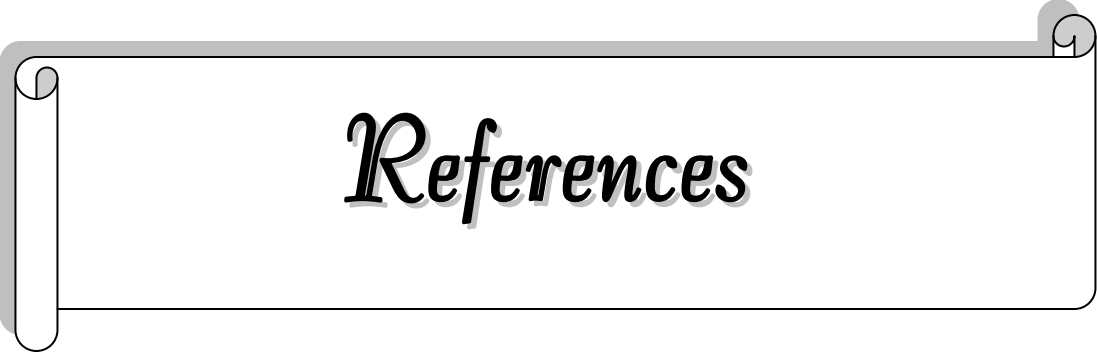
Molecular characterization of twenty isolates includes identification with species specific primers MpKFI (5'-CCGCCAGAGGACTATCAAAC-3') and MpKRI (5' CGTCCGAAGCGAGGTGTATT-3') confirms that all isolates were belongs to *M. phaseolina* and molecular variability was studied by employing PCR-RAPD revealed that the twenty isolates were

differentiated into two major clusters A and B and ITS sequence generated a phylogenetic tree with two clusters A and B. All 31 sequences of *M. phaseolina* were grouped in cluster A with two sub clusters. The only one isolate KAMP-2 diverged to form single subcluster, whereas all isolates formed another sub cluster. Among both the markers used for the study RAPD helps to know only variability but ITS region sequence was found best to identify the pathogen and also to study molecular variability among the twenty isolates.

A total of two hundred and fifteen chickpea entries including desi, Kabuli and other local and advance breeding germplasm were screened for host plant resistance both in sick plot and blotter paper method. Out of those only two entries *viz.*, ICCV 04514 and ICCV 96854 showed resistant and twenty one entries *viz.*, GNG 1581, PBG 5, JAKI 9218, CSJ 697, CSJ 692, Phule G 09316, RSG 888 (2), RSG 931, Vijay, RKG 160, GNG 2182, IPCK 2008-108, CSJK 72, Phule G 10404, IPCK 2008-109, BBG 2, HIR 70, HIR 60, ICCV 08124, ICCV 93706 and ICC 4951 showed moderate resistance reaction in field. The only entry PBG-5 was shown moderately resistant in both sick plot and blotter paper method in the year 2012-13 also found to be susceptible in the sick plot during the year 2013-14. Among 23 resistant and moderately resistant entries selected during previous screening year were once again tested for their resistance during the current year 2013-14. None of the entries showed resistant reaction, while only two entries *viz.*, RSG 888 (2) and ICCV 93706 showed moderately resistant reaction during the current year of screening.

In management of chickpea root rot complex among the fourteen treatments imposed in both *Kharif* and *Rabi* at ZARS, GKVK and College of agriculture Hassan, seed treatment with tebuconazole 2 g/kg was found to be the best to manage the dry root rot and root rot complex

diseases in chickpea followed by seed treatment with tebuconazole @ 2 g + carbendazim 1 g/kg at GKVK and seed treatment with *P. fluorescens* @ 5 g/kg of seed at Hassan.



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VII. REFERENCES

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